

Phil 102 Logic

Numbered Transcript

PHIL 102 • Dr. Stephen R. C. Hicks

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Lecture 1: Why Logic Matters

At a glance — Hicks introduces why logic matters — theory and application.

- 1.1 Logic, a lot of fun, but also a lot of work. Applied, can be very theoretical and abstract, too. I want to start with the question of why bother with dealing with all the hard abstractions, doing all of the sometimes difficult work involved in logic, mastering it and then being able to apply it? I'll make a claim that, for human beings, logic is the most practical thing one can master. That's a very strong claim, and it's a controversial claim, even within pro-logic circles, but I'm going to make it.
- 1.2 Some of the reason for this is for thinking about what makes it possible for us human beings even to survive, but beyond mere survival, flourish and to accomplish wonderful things that human beings can accomplish. It has a lot, of course, to do with our bodies, learning how to use our bodies well. But for human beings, what matters more is how we use our brains, how we use our minds, and that brain-mind combination, however we sort out what that is.
- 1.3 We are not particularly fast in the animal kingdom. We're not particularly hardy in the animal kingdom. Our range of comfort and temperature is pretty narrow. But we do have extraordinarily powerful brains, extraordinarily powerful minds, compared to others. The amount of information that we can take in and our capacities for sorting, differentiating, noting similarities, noting differences, repackaging information, drawing inferences, building enormously complex understandings of the world makes us amazingly powerful human beings.
- 1.4 Now, that takes work. And part of the work is that our brains and our minds, they come as a set of capacities. But a big part of what we need to do is become self-conscious about our capacities and learn how to use them—what they can do, what they can't do, and how to use them properly and improperly. This is going to be primarily a course on logic methods, logic techniques, increasing our sophistication, and being able to understand and evaluate arguments.
- 1.5 At the end of each lecture, where we go over a certain body of important material, I want you to step back for a few minutes and be philosophical and identify either a point of controversy; because while this is a course that the premise of it is that logic is great and wonderful, not everyone believes that, and there are, in some cases, important controversies within the philosophical traditions of logic or people who are outside of that tradition. So I want to flag, in each case, what those are, or give credit to the giants in the history of logic who developed the important things.
- 1.6 On that latter score, I want to turn for a few minutes, rather, to Aristotle, who is almost always the person who gets the lion's share of the credit for founding the discipline of logic. This is 2400 years ago in ancient Greece, ancient Greece being an amazing place

where all sorts of things were happening in the development of early sciences, literature, political developments, artistic developments, and so forth.

- 1.7 Aristotle was a student of Plato who was in turn a student of Socrates, and these individuals had already elevated thinking, and to some extent, thinking about thinking, to a higher art. But typically, the lion's share of the credit for founding logic goes to Aristotle to be the one who is making it most explicit. This is what we are doing, developing axiomatic or foundational principles for logic, why logic is the right way for us to use our thinking. Then over the course of several books, developing an impressive array of logical techniques.
- 1.8 Collectively, those books that Aristotle wrote that are devoted to logic and logical techniques are called the Organon. And as a Greek word, this means a tool. So these are tools of thinking and different tools for different kinds of circumstances. So the first passage I want to draw our attention to is this question of, why is logic so fundamental to human thinking? Our answer can't just be that, "Well, I happen to like being logical." The answer has to be, from this perspective, that logic is what keeps our minds connected to reality, on track with reality, no matter how complicated our thinking is.
- 1.9 If we push that, this is where it gets controversial, it's the only way that we can be sure, particularly in our highly abstract, highly complicated thinking, that we are being connected to reality. If our goal is to know the real world, know being, know what exists, then we need to have a certain set of techniques, and that needs to be justified. An important historical passage is from one of the books that's typically in the Organon, called the Categories. The statement might not be too impressive to us, but there's a lot built into this.
- 1.10 "What is makes statements true or false." The idea of an axiom as a founding principle. Where do we start? Where we start on this Aristotelian phrase is with what is, a recognition of reality, a recognition of existence. We might emphasize it, what is, is. Reality is real. Facts are facts. Existence exists. That's where we start. And if we're interested in truth and falsehood, we are making statements and claims and abstract propositions and so forth, we know sometimes those are true and sometimes those are false. It is reality that is the determiner of truth and falsity.
- 1.11 So statements have to be connected to reality; that makes them true. If we can't connect them to reality, that's a mark of their being false. So, start with what is, and our job is to make all of our statements connected to, or in reflection to that. And that's a very pithy statement, but a lot built into it. As an axiom, we have one that is called the principle of non-contradiction, as it comes through. We have our minds, it gets very complicated, and the rule of logic, the operative rule, is don't contradict yourself.
- 1.12 If you have a contradiction, you've made a mistake, and that means you need to go back and do a whole lot more work of whatever it is. Aristotle argues that this principle of non-contradiction is also axiomatic in our thinking. Then as mature thinkers, we

want to bring it to our conscious, explicit attention. I'm not going to contradict myself in my thinking. The idea is that what is, is. Reality is what it is. It works the way that it does, and that there's some consistent causality.

- 1.13** Things have an identity, or things have a nature, or certain patterns emerge as a result of reality doing whatever it is that reality is doing. But reality doesn't contradict itself. It is what it is. And that means, in our thinking, we can't come up with contradictions; or we can come up with contradictions, but that means we have made a mistake. So here is Aristotle's formulation of this, for the historical record, from his *Metaphysics*.
- 1.14** "Such a principle is the most certain of all; which principle is this? Let us proceed to say. It is that the same attribute cannot at the same time belong and not belong to the same subject in the same respect." So whatever we want to say about reality, reality is X, then we are committed to saying it is X, but it's not also not X. And if I find myself ever saying, well, this thing is X, and then a little bit later, it's also not X, that's where the mistake has happened. Non-contradiction. I need to go back. Is it X, or is it not X?
- 1.15** And that's the point of the third one, drawing our attention to this principle. Sometimes it's called a principle of either-or. "To hold equally to the opinions and seemings of those disputing against one another is foolish, for clearly one of them is mistaken." So it might be in your own thinking, or it might be in the thinking of two people who are disputing with each other. But if they have a dispute, there's a contradiction, this person says it is, this person says it is not, this person says it is X, this person says it is not X, they can't both be true. It's either this one, or it's that one. And this is the either-or principle.
- 1.16** So what we have then is Aristotle enunciating what you might think of as four foundational principles or four axioms for logic. First, what is. Start there. Reality comes first, and that's a commitment. It's a fact and it's a commitment. Reality comes first. Our minds, our statements about it, what we come to believe is true, what we come to believe is false, takes its bearings from and has to be connected to what is. That's number two. Non-contradiction. Things are not both having a certain property and not having that property. Then number four, either-or.
- 1.17** Now, this is historically foundational, and this puts Aristotle, what we call philosophically the small-O objectivist tradition. Objects are what they are or objective reality is what it is. The commitment to logic is the commitment then to being objective. My thinking as a subject is trying to stay in connection with objects or objective reality as it is. And logic is the set of tools increasingly that enables me to do so.
- 1.18** Now, in the philosophical tradition, there are various challenges attempted to these as axiomatic. If you are interested in the history of logic, that can also be a lot of fun and rewarding reading. There is an analogy here. We are partly physical, partly mental. There's a mind-body integrate of a sorts. But if you think about logic by means of this

analogy to physical fitness, one healthy human goal is to have the best physical body you possibly can. And a part of that then means you start to pay attention to your body and you learn its capacities.

- 1.19 And then at various points, you decide, "I'm going to work out." And what we then do when we are working out is, we're using our body for day-to-day life, but we are self-consciously setting the project of attending to our bodies. And we say, "I need to work on this, that, or the other thing." I want, for example, to have, say, more strength of body. Then I do exercises that will build my strength. I want to have more flexibility in my body. We do exercises to increase the flexibility of our bodies.
- 1.20 I want to have more endurance to be able to do things over the long term. I do exercises to develop the endurance of my body. By analogy, and slightly metaphorically, the same holds for the use of our minds. We want to have strength of mind. And what that means is to be able to take up difficult problems, ones that require heavy lifting, intellectually hard challenges, not easy problems to solve or things to deal with.
- 1.21 I also want to have flexibility of mind because one of the things we know is, things change. Sometimes I need to think and rethink and be able to think quickly in a different way and look at things from different perspectives. So the flexibility of mind is also something that we need to practice and train. Then there's an endurance issue as well, so that carries over. Sometimes we don't need to think about something for very long. We can make a decision and act on it. Other things require a lot of thought.
- 1.22 They require that I have developed my mind so that I can stay with a problem, stay with a challenge, or stay with whatever it is that I'm thinking about for minutes, hours, sometimes days, weeks, before I've done enough thinking to be able to figure out what it is that I want to do. So just as I might say I want to have a lean, mean, powerful body and design workouts and commit myself to a life-long working out to keep my body as healthy as I possibly can, the same thing holds for the mind, and that's what logic primarily is—self-consciously attending to the development of one's mind's capacities over the course of one's life.
- 1.23 Now, the analogy breaks down a little bit because a lot of what we do physically, in terms of internal processes, is done automatically. I take some stuff into my mouth, I swallow it, it goes into my stomach, hopefully it's mostly nutritious stuff. But at that point, my stomach and the rest of my physiology more or less automatically sorts out what's good for me and what's bad for me. It sorts the good stuff, uses it in various ways automatically, and then automatically gets rid of the stuff that we don't need.
- 1.24 When we take information in, observations, ideas, suggestions, and we have other things kicking around in our minds, it's not automatic that we are going to sort what's good from what's bad. We have to pay attention to it. One of the things that we do know is sometimes we make a lot of mistakes. We let too much false stuff in, or we can

take perfectly good information but combine it in weird ways. And so we end up, as a result of our thinking, not in a place where we want to be. We are fallible in our thinking.

- 1.25** And this is where logic, in addition to being an enormously important set of exercises, needs to be self-consciously directed. And there's great value to sorting out and learning explicitly what kinds of thinking processes work to preserve truth, to extend our understanding of truth, sometimes to enable us to tease out new ideas and new truths that are not very obvious, and distinguish those from the ways of using our minds. Sometimes they're pretty easy and seductive that lead us astray, so that we can then self-consciously apply our mind.
- 1.26** Let me give you a couple of examples here. We'll start abstractly, just to make this point here. We are concrete thinkers, but we're also capable of abstraction, and logic is a matter often of shuttling back and forth between. But suppose I say, all of the P's are Q. Now, already this is an abstraction. This is just a letter, but it's standing for something or other. We don't know what that P is. It might be a person, it might be Paul, it might be penguins, whatever it is that we are talking about.
- 1.27** We're saying that all of them, there's a quantity term, are, and then there's some other trait or property or action, Q, there. I say all of the Q's are M. Then I say, so, pretend like we're doing a little bit of math. If you take this and you take this, it follows that, or we can conclude that, all of the P's are M. So all of the P's are Q's, all of the Q's are M, so all of the P's must be M's. So we've made a logical move, taken this piece of information, this piece of information, and then made an inference.
- 1.28** Now, this is not the best word, but how does that feel to you? Because you don't know what the P's and the Q's and the M's are, but does it make sense? We start to nod. We feel pretty comfortable about that one. Suppose I had done this. All of the P's are Q. All of the M's are Q. I say, well, therefore, or it seems to follow, is that it must be that all of the P's are M. All of the P's are Q's, and all of the M's are Q's, so all of the P's are M's.
- 1.29** And again, how does that feel? Does that seem sensible to you? And I'm not getting any noddings of the heads right at this moment. I'm getting nothing, and that's a good sign that you're uncertain about that one. And part of the uncertainty is, you might be getting set up for a trick, and you are being set up for a trick, because, yes, your sense that this is good and that this is not so good is exactly right. But it's hard sometimes, especially at the beginning, when you're just dealing with abstract variables here, to tell one way or the other.
- 1.30** So, shuttling from the abstract to the particular, one thing we might start to do is say, well, look, we have in this case, we have a P and we have a P. There's two P's. There's also two Q's, and there's two Q's here, and then there's two M's. Then the same thing over here. We have P, we have P, we have two M's, and we have two Q's. But abstractly, there's something different. It's not that the content is any different. It's just

All's and R's, and then the same three big letters, capital letters, the variables. But structurally, they are different.

- 1.31** The common term here is the Q in both cases. The common term over here, Q, is the same. But in this case, the Q appears in what we call the predicate position from our grammar. In this case, it appears in the subject position from our grammar. In this case, over here, both times, it's in the predicate position. That's a structural difference or a formal difference abstractly. One of the things that we will learn is, there's actually lots of ways of putting P's and Q's together, and sometimes we're talking about all things, and sometimes we're talking about no things, or some things, and so on.
- 1.32** So it gets a lot more complicated. But the more we do this sort of thing, we recognize, even if without knowing the P's and Q's, this is a bad pattern. Maybe the first lesson, of course, will be, anytime you've got the same term, the linking term, the common term, we sometimes call it the middle term, and it's in the same predicate position, a little red flag will go up in your mind. "I'm suspicious about that." And chances are good that that suspicion will bear you out. You won't buy into that argument as much.
- 1.33** So let me give you a concretization, taking this abstract pattern to show how this would follow. So as I said, this argument meant, and every time we're talking about P's, we're going to be talking about humans. So all humans are, and I'll make a claim about them, they are mammals. I'm substituting mammals for Q. I'm substituting humans for P. Then we maintain that pattern. And then I say, all mammals are what I want M to be, say, mortal. I'll make another claim about them. Then I say, well, therefore, if all humans are mammals and all mammals are mortals, I'm going to make the inference that all humans are mortals.
- 1.34** This is good information. This is good information. Then I made a logical move, and I end up with something that is good information. So the move structurally, formally, whatever I did logically, preserves the truth of the information. It doesn't lead me astray. But suppose we do the exact same stuff here and do it over here. So all P, R, Q. So all humans are mammals. Now, let me mix it up just to show. Suppose I say, all pigs are mammals. Perfectly good substitution. Q, Q, mammals, mammals. P, humans. In this case, M is pigs.
- 1.35** If I use this pattern, logically, and I then say, well, therefore, all of the P's are M, that means I'm drawing the conclusion, all humans are pigs. We're not metaphorically talking here, we're talking biology. The point of this is, and we laugh because it's a ridiculous conclusion to me, but what it tells us is, we started with perfectly good information. Humans are mammals. Pigs are mammals. That's true, that's true. But whatever manipulation, logically, we did took us astray, and we end up with a false conclusion.
- 1.36** And that tells us, this pattern of reasoning is not a good pattern of reasoning. We'll say it's an invalid pattern when we get to the terminology. This is a good pattern of

reasoning, but there are hundreds and hundreds of patterns of reasoning, and so one needs to spend some time looking at all of them and then sorting out some rules and methods for being able to tell, for the ones you're not sure about, which ones are good and which ones are not good.

- 1.37** Some of it we can do by memorizing. So, as I've indicated here, always look for duplication in the predicate. That should be a sign that maybe something is wrong, and then I'll take that one away. Now, logic is also useful in discovery mode. Sometimes we're taking existing information and just repackaging it in a way that's not particularly revelatory. None of you were particularly surprised when I said, "Oh, by the way, humans are mortal beings."
- 1.38** But logic is useful not only for tidying up our existing knowledge and structuring things in a way perhaps that's more economical, but in figuring out brand new things. And this is where one of its greatest strengths comes along. Let me give you an example from the history of science, and it will sound childish when I tell you the example initially, but it's the way we make progress.
- 1.39** So one of the questions in ancient societies was, an observant guy noticed that when people work out, we've all noticed this, their skin will get wet. We start to perspire. And this is ancient times, though. They don't have microscopes. They haven't been able to look at the skin very closely, so they're not able to see that human beings have pores through which moisture can pass. So there's a question here, perhaps even a little bit of a mystery question. Where's this water coming from?
- 1.40** It doesn't seem necessarily like, if you're in some places, as coming from the air; I didn't splash water on myself, but my skin seems like it's pretty solid. So how could water be coming from the inside of me? So question, puzzle. But then this person noticed that one night he had a bucket and he filled the bucket up with water and he would leave it outside for his animals to drink during the night. And then the next morning, he got up, and the bucket was completely empty of water, and there was just a big puddle around the empty bucket.
- 1.41** So how did the water that was in the bucket get out of the bucket when the bucket is supposed to be a solid, impermeable container? And then he takes the bucket that's now empty and he holds it up to the light and he looks really closely. You can see there are some very tiny holes at the bottom of the bucket. They're barely visible to the naked eye. And probably the water then escaped through these little tiny holes, and that's how it got out. So, aha moment.
- 1.42** This question of water on the skin, maybe it's the case that, on our skin, we have little tiny holes that we can't even see, but they are big enough for water to get through. Now I've got a scientific hypothesis. But how did I get to that scientific hypothesis? By an act of reasoning, I observed a bucket of water that had water in it, that then emptied, and I observed that it had holes. I observed that somehow water is appearing

on my skin. And so, by analogy, I'm drawing the conclusion that maybe skin also has holes.

- 1.43** So we do a lot of analogical reasoning. We have lots of information over here, lots of knowledge over here. Can we transfer that knowledge to some other issue, some other domain, and figure out new things that we otherwise would not know? Now, just for fun, though, the logical methods are often used in humor. One of the things that's delightful about logic is that the more you learn logic, the more you appreciate a wider variety of humor. Here's an example. I'll just read it to you.
- 1.44** Someone who really loves chocolate made this joke. I hope it comes across as a joke. It says, chocolate is crucial to our survival. Dinosaurs didn't have chocolate, and look what happened to them. All right, now that's kind of hilarious, but what makes it hilarious in part is that the person is reasoning analogically. In some sense, the point really just is, chocolate is really important to me, and you're exaggerating it; it's essential to my survival. Then the way you make that is by making an analogy to the dinosaurs.
- 1.45** The dinosaurs didn't survive, and why didn't the dinosaurs survive? Well, we know what the real hypotheses are there, but when we stick in this joke one, analogically, well, if I didn't have chocolate, what would happen to me? Well, that must have been what happened to the dinosaurs. And part of the fun of logic is that we've automated a certain amount of good analogical reasoning, so we can spot bad analogical reasoning when it happens, and it strikes us as funny.
- 1.46** But the point then is, then we need to study lots of analogical patterns, lots of patterns like this, to develop a good sense, sometimes an explicit sense, of what are the good ones and the bad ones. Now, there's also the endurance element, and I want to give some examples of how sometimes we get stuck on problems or we don't know what to do. And it really is a matter of sticking with the logic and being able to apply the logic and the methods that one learns. If one does so, then one sometimes unexpectedly comes across good conclusions and brand new information.
- 1.47** So let's imagine a situation, and there's going to be a logic lesson embedded in this fairy tale-type of story about discovery. Once upon a time, here's the storytelling, there was a young woman, she was a princess, and her father was the king. She got to be of marrying age, and the king decided that he wanted to get his daughter married off. So he sent the word out, and three princes showed up, all saying, "I wanted to marry the princess," for varying reasons.
- 1.48** The names of those three princes are Prince Noble, Prince Debonair, and Prince Valiant. I'm just going to abbreviate those. There's Prince Noble, there's Prince Debonair, and there's Prince Valiant. I'm putting those in that arrangement for sneaky reasons that I will reveal in just a moment. They come in, and the king has decided that the way we're going to decide who gets to marry the princess is by means of a duel. It will be a duel with pistols, equidistant.

- 1.49** So what's going to happen is, Noble will stand here, equidistant from Debonair, equidistant from Valiant. Each of them will have a pistol, and then they will take turns, taking a shot, and then at the end, whoever is left standing, that's the young man who will be able to marry the princess. This is a bit of a brutal, old-fashioned fairy tale. Now, they do some testing, and the king also is concerned; he wants to make it somewhat fair.
- 1.50** As a result of the testing, they find out actually that Noble is a crack shot. He basically never misses at this distance. I'm going to put a 100% here. He hits what he shoots at 100% of the time. Debonair is a very good shot, but he misses occasionally, and on average, he hits 80% of the time. And Valiant is a so-so shot. More often than not, he hits, but he hits his targets only 60% of the time.
- 1.51** Now, complications then set in. The princess, when she sees these young men, immediately fell in love with Valiant, and she's not particularly interested in this guy. She wants Valiant to win. Of course, Valiant wants to win, but it doesn't seem like the chances are good for him because he's not as good a shooter as the other two. Then what the king has decided, he says, "Well, the way we will handle that is, we will go in reverse order. Valiant will get the first shot. Debonair will get the second shot, and if he's still alive," it's a little dark, "Noble will get the third shot."
- 1.52** The princess really wants Valiant to win. She's smart. She's taking some logic, and so she thinks and thinks and thinks and thinks. She thinks all night before the duel. What should Valiant do to make sure that he has a better than 50% chance of winning the duel? Now, we can think about this. We're not going to think about this all night. What I want to do is then say, well, how would we start to think about this?
- 1.53** So Valiant has the first shot. I could say, here I am, Valiant. What I'm going to do is, I'm going to shoot at Noble. My reasoning might be, well, he's my most dangerous adversary. So get rid of the most dangerous one first, increase my chances that way. Now, what happens when I shoot at Noble? Well, one of two things can happen. I will kill him, and there's a 60% chance of that, or I will miss. There's a 40% chance that I will miss. What happens if I kill him? Well, my turn is over, and then it is Debonair's turn. Debonair is going to do what?
- 1.54** He's going to shoot at me, Valiant. And there's an 80% possibility, because he's an 80% shooter, that I will be dead. So, I have a 60% chance of this, but an 80% of that, 10% of that, this seems like it's going to lead me to death. It's not a very good strategy. On the other hand, I might, let's hold off on this for a moment, say, well, OK, what if I instead try to shoot at Debonair? I'm not going to shoot at Noble first. I will shoot at Debonair.
- 1.55** Well, then I say two possibilities. I might kill him, 60% chance, or I might miss him. If I kill him, what happens next? Well, he doesn't get a turn because he's dead. So the next turn is going to be Noble. So Noble is going to shoot at me. I'm the only one remaining, and I'm dead. Now, what has this revealed to us so far? I have two options, and I'm

starting to do the logical work, figuring out the implications. I have to go 1, 2, 3, 4, 5 steps into my reasoning to get to a certain conclusion. I need to keep track of all of my options, and then again, go through all of this again.

- 1.56** But at this point, you might have an idea. You say, well, look, if I try to kill him, I probably will, but then that leads to my death. If I try to kill this guy, then I'm probably going to die as well. What happens if I miss? OK, well, we have to figure out this one. 40% chance that I will miss. Then if I've missed Noble, what happens next? Well, my turn is over. It is now Debonair's turn. What's Debonair going to do?
- 1.57** He says, well, I can shoot Valiant, but he's a weak shot, or I can shoot at Noble. He's pretty dangerous. So I'm going to possibly shoot at Noble, and what is most likely to happen? Well, I can kill him, 80% chance, 20% chance that I will miss. And now we're running out of blackboard space. But if, as most likely is going to happen, Debonair shoots at Noble and kills him, 80% chance, what happens next? Well, it's my turn again. Because Noble is dead, it's now Valiant's turn, and there's only one person left, and I have a 60% chance of killing him. That's better than half.
- 1.58** The same thing would happen if we start to go along this road. So as I miss having shot at Debonair, in which case the next thing that's going to happen is, well, it will be Debonair's turn. Most likely, he's going to shoot at Noble. Most likely, he's going to kill him. And again, it's my turn, Valiant, and I will have a 60% chance of winning. Now, that's a quick run-through of the numbers. You want to check, double-check to make sure.
- 1.59** But what this indicates is, if in either of the missing cases, I suddenly have a 60% chance of winning this duel, and if I kill on my first shot, I'm almost certainly going to die, this leads to an unexpected conclusion. You're in a duel. What's your strategy? Miss. Intentionally miss your first shot. You would never come up with that idea if you did not do this somewhat hard, complicated, logical work. That's what the princess wrote in the note to Prince Valiant. "Miss your first shot." Then he thinks it through, says, "She's right," and that's what happens.
- 1.60** OK, let me give you one other example of doing some hard work. I can lead to unexpected conclusions. One of my professors in graduate school, this is now going a ways back, was a world-famous logician, but he loved doing puzzles, and he sold lots and lots of fun puzzle books. Logic puzzles, the world over. He liked to use them in his class. So when I was his teaching assistant, working with him, I started learning these, and they are fun, but they are also instructive and help to build your mind.
- 1.61** Here's another example of where the hard work can lead you to discoveries of things that you otherwise would not come across. Sometimes you might look initially at the problem and say this is an impossible problem, but it's not really. So here's the situation. Suppose you're an anthropologist and you are going to do your research work, you want to do something new and original, and you hear about this island of

people who have been studied a little bit, but not very successfully because of the language barrier.

- 1.62** The language barrier is such that we know that they have a language. We know some of their words, but we're not sure quite what those words are. We know what the sounds are. We have also learned that all of the people on these islands are either truth-tellers or they are liars. This is obviously a made-up example, but one group of people on this island, they always tell the truth. The other group of people, they always lie.
- 1.63** The other thing that we've learned is that they have two words, and those words are bal and dal. And one of those words means yes, and the other word means no, but we don't know which one it is. So all we know is, those are two words, they indicate yes or no. And the two people have these different commitments to truth, so to speak, in the language. The other thing to make this fun is, they understand English, but they don't speak English, and we don't speak their language at all.
- 1.64** So, suppose you go to the island and you say, all right, I'm going to get this figured out. Then I want to be able to study these people, to get lots of information about them to write my new anthropology book or whatever it is. So I go to the island and I get off and I start going along and I encounter a native. And this native comes up to me, and now I want to communicate with this person. So I can start asking him or her questions. Let's suppose it's a him. Start asking him questions. And sometimes he says dal and sometimes he says bal. But I don't know what that is.
- 1.65** I point to a rock. "Is that a rock?" He says, "Dal." I point to something. "Bal." Do I know what those words mean? No. I don't even know if I can trust this guy to be saying truthfully what it is. It might be that, yes, there's a certain word for rock, but he's a liar. He's trying to mislead me. He'll tell me something else. What can I do to get past this language barrier? Now suppose you're a very clever anthropologist. You've taken some logic. You sit down and you think about it, and you come up with this question that you're going to put to the native.
- 1.66** You say, "Is your word for yes, bal? Is your word for yes, bal?" And suppose, so this is you, as the anthropologist asking this question, and then the native responds, "Bal." OK. You look at that and then you say, well, he's saying that the word for yes is bal. If he's a truth-teller, it is, but if he's a liar, then it isn't. I don't know if he's a truth-teller or a liar, so I'm stuck. I don't know if I can trust this guy or not.
- 1.67** But suppose we say, let's do a little logic work on this. We say, well, here we have two things that we don't know. We don't know what bal means. Does it mean yes or no? I don't know if this guy is a liar or a truth-teller. I've got two things that I don't know, and both of them have two possibilities. I've probably taken some math classes or logic classes, and it tells you when, if you have two variables and each of which has two possibilities, there's four overall possibilities. But what are those four possibilities?

- 1.68** Now, it's just an abstract rule, right? Back from 10th grade math class, it just popped up into my mind, but let's work it through. We could say, well, I don't know if he's a truth-teller or not, but suppose he is a truth-teller. And suppose that bal in their language does, in fact, equal yes. So that's one possibility. This guy's a truth-teller, and bal does mean yes. It could also be that he's a liar and bal means yes. It could also be that the guy's a truth-teller, but bal means no, and so forth. So we figure out what the four possibilities are.
- 1.69** But this is one possible situation. The guy's a truth-teller and the word bal means yes. So when you ask him that question, "Is your word for yes, bal?", what is the true answer to that question? According to this assumption, "yes" is the right answer to that question. And this guy is a truth-teller, so he's going to tell you the right answer to the question. And in his language, the right answer is yes, and that word is bal. So he's going to say, "Bal."
- 1.70** Now, that was a lot of "if this and this that," with a couple of assumptions built into it. You have to hold all that in your mind to say, yeah, he's going to say bal. Now, on the other hand, suppose the guy is a liar. Another possibility, he's a liar. But in their language, the word bal still means yes. Now, in this case, what's going to happen? You ask this guy, "Is your word for yes, bal?" The correct answer is yes, but he's a liar. So that means he's going to say no. He wants to trick you.
- 1.71** And in their language, if bal means yes, how do you say no? You say "dal." So he's going to say "dal." Now, that's only two of the possibilities. Suppose we have truth-teller. But in fact, we don't know this guy is a truth-teller, bal means no in their language. So you ask this question, "Is your word for yes, bal?" Well, it isn't. And this guy is a truth-teller. So how is he going to say "no, it isn't," in his language? Well, if bal means no, the right answer is no. So he's going to say "bal."
- 1.72** And then this is actually the hard one. So once the guy's a liar, you don't know he's a liar yet, you ask him, "Is your word for yes, bal?" And the other assumption is, bal, in fact, means no. So, "Is your word for yes, bal?" It isn't, but he's a liar. So he's going to say it is. And how do you say it is, yes, if the word for bal is no? You will say "dal." Now, we just ran through that pretty quickly. On your own time, double-check, triple-check a few times, see if my reasoning is correct in each case. But if this is correct, and this is some work, we've learned something. What have we learned?
- 1.73** The guy said to this question, "Bal." That was the response this particular native gave to this question. And the only time someone is going to say "bal," two times that popped up, in both cases, the guy was a truth-teller. So to that question, liars are always going to say "dal," and truth-tellers are always going to say "bal." So now you know at least one thing. You know this guy is a truth-teller.
- 1.74** Now you've got something to go on. You can say, "I like you. Come with me. I want to do my anthropological research," and you can start learning from him. You can ask him

other questions, and you will know that he's a truth-teller. You also now know, since he is a truth-teller, that what he said was true, and that the word for yes is bal. So anytime he says "bal," you've got information. Anytime he says "dal," you've got information.

- 1.75** Now, this is not obvious. You're only going to get to that and break through this language barrier if you go through that process of logic. This is one of the great values of logic, is the capacity of learning more tools and learning more methods of structuring and restructuring, and in some cases torturing the information to get unexpected things to pop out. And the more of it you do, the better you become at it.
- 1.76** One of the things that's very common in social circumstances is, the people involved will all have access to the same information. But some people, partly because they've practiced a lot more, they know how to see further implications and repackage the information to get unusual ideas, creative ideas, and new information to pop out. It's enormously valuable.
- 1.77** Now, another great value for logic is not only learning methods of extracting new kinds of information, but the flexibility point. This takes a certain amount of endurance to be able to go through this and stick with it all the way through. But another thing that's important that logic helps you is learning how to adjust to new information. This is something that's very common with us as human beings. We have lots of information come in. We have a hypothesis or an idea about what it all means. Sometimes because we came up with that new idea, we want that idea to be true.
- 1.78** But then more information comes in, new information comes in, and you recognize that there's a tension between your earlier hypothesis or your earlier interpretation of this new information. Now, some people, if they're lazy, they don't want to do the reinterpretation work or the rethinking work. They'll just block out that new information. But if you're a good thinker, if you're a careful thinker, and if you are committed to the truth and you say, "Well, this is new information. It seems like it's true information. I have to find a way to make it work with my existing hypothesis, and that might mean changing my hypothesis and doing a certain amount of work."
- 1.79** Just to take a silly, not really a silly, if it's a cold winter night and your car is dead, but example of this, suppose it is a cold winter night, you go out to your car and you put your key in the ignition, you turn it and your car doesn't start. And then not only that, it doesn't start, it doesn't even crank. So it's not that it tries to start but fails to do so. It's just completely dead. Now you're a person of some experience with cars. So immediately you're going to say, all right, with all of this new data, cold night, car, probably the battery is dead. Chances are very good that's the first hypothesis that comes to mind. And you'll think, probably the battery is dead.
- 1.80** Then you might remember immediately, "Wait a minute. Just two weeks ago, I had a new battery put in." That's a new piece of information. You're going to then almost instantaneously then rethink, "Well, maybe it's not the battery, or it's less likely that

the battery was dead. New battery shouldn't be worn out after two weeks." But then you might remember, you watched the guy putting the battery in, and he seemed a little tipsy, maybe even a little bit drunk.

- 1.81 He didn't say anything at the time, but maybe he didn't fasten the new battery as tightly as he possibly could have and should have. So something has come loose in a couple of weeks. So now we're back again to thinking, "Oh, tipsy mechanic, cold night, battery, we're probable now." Then you remember maybe your dad, when you got home, checked everything, and your dad said, "No, no, everything looks really good, tight," and so forth. So then you're back saying, "Well, it's not likely the battery again."
- 1.82 Then you have new information comes in. Finally, you look at the light switch on your dashboard and you see that you left it on. And then you went away for some hours on a cold winter day, so again, the battery. So the point is, new information coming in, hypothesis, weak, strong, weak, strong, back and forth. And all of that is a certain character of mind. You're willing to go through that process, but then also being able to know what to do with the new information and how it affects your previous thinking.
- 1.83 Now, in more serious cases, if we think about what detectives do, where they go into an enormously complicated situation, they want to figure out who the murderer is, and immediately, it's just overwhelming the amount of information. They have a huge amount of sorting to go through. They're constantly coming up with new hypotheses; new information comes along. This is actually the way, if you read Sherlock Holmes or Agatha Christie or all of these very popular detective shows, they set this up intentionally, where all the initial evidence that they draw your attention to points in a certain direction. This guy is the suspect.
- 1.84 And then they let you dwell on that for a little while, and then they give you some new information. Then you realize, "I have to rethink everything, go back to the beginning, and now it's like, maybe it's her. She's really the suspect." And it's only through the commitment of that process and the willing to be flexible and to take into account all of the information and look at it at different ways that eventually the detectives figure out whodunnit. Now, we're interested in justice. Justice can only happen in complicated situations if we have people willing to go through that process of logic and are trained, as we want detectives to do so.
- 1.85 Now, other value questions with respect to logic. I'm going to say that logic is absolutely important to all of us as individuals. Just as we want our bodies to be as healthy and strong as they can be, we want our minds to be as healthy and strong as they can be, and logic is indispensable for doing so. But lots of life involves dealing with other people and the extent to which the parties involved, whether it's just two people or more, are committed to a logical process, those social relationships are going to go well.

- 1.86** We all know, from often bitter experience, that a lot of people are not committed to logic. That's in many cases why those relationships break down. There might be other factors at work, but they just make stuff up, or they make all kinds of logical leaps, or they don't listen to your reasoning. Of course, sometimes you don't listen to their reasoning because it's often hard to do. There's a complex situation. They've packaged it a certain way, and it seems reasonable here. You've taken the same information but packaged it in a different way.
- 1.87** But you need to be able to step outside of your framework, see how this person has packaged the information. That person has to also be able to do the same thing. And that's hard. And it requires a commitment. And so it's partly a character issue, but also partly a learning issue. That's in friendships, it's in romantic relationships, it's when you're dealing with kids, when we go out into the business in this world, and we're doing negotiations, when we're talking about very important things in our lives, like politics and what laws are going to govern all of us, what we think about our religious beliefs and how important those are to how we're going to conduct our lives.
- 1.88** To the extent one is able, on those really complicated things, to use logic effectively in a social context, those things are going to go well. A big part of it is self-defense. Let me give you a hopefully fun example of self-defense. Most of the examples we've been working through is, I gave you a certain amount of information, and there was all of the information available at that moment, and it was a question of, what can we do with this information? In many cases, though, in human reasoning, we don't have all of the information available, that an important part of what we to think about is left implicit or unstated.
- 1.89** Now, a lot of times this is consciously done because, by the time we are adults, we have a huge amount of knowledge. And in most social circumstances, we know other people also have huge amounts of knowledge and there's lots of shared knowledge, so we don't need to belabor the obvious. So I can say things like, you're heading out the door, and I'm your friend, and I say, "I saw that it might rain; you might want to take an umbrella." That makes perfectly good sense to you.
- 1.90** I'm not going to say, "You know what? It might rain, and when it rains, water falls out of the sky. If you are outside when water falls out of the sky, it will fall onto you, and then you will get wet. Chances are good, maybe you're going off to work, you don't want to get your work clothes wet. So, if you have an umbrella, what umbrellas do is, they open up and they block the rain from falling onto you, so you'll stay dry and then you'll be comfortable at work."
- 1.91** Now, all of that is true and all of that is relevant, but we've learned all of that. And so all of that remains explicit, or implicit rather, by the time we become to be adults. And we just say, "Hey, it's going to rain. Take an umbrella," and you fill in automatically all of the rest. We're very powerful minds in that capacity, for being able to assemble all of the relevant information and bring it to bear on a particular case.

- 1.92** But at the same time, in a social context, while that's useful, people will do these things in sneaky ways. People who know the logic and know the assumptions, they will put you in a position where they want you to answer a question or respond in some way, and they've given you some information, but the most important information has been left out. So standard examples in logic courses, lots of these ones are, people who are hostile, they will ask an attacking bundle question, as we call it.
- 1.93** So we're at our high school reunion, and I come up to you—we didn't like each other in high school—and I say, "Are you still that jerk you used to be, back in teen years?" Now, immediately you're going to be on the defensive, and that's the right thing to do so. But part of the issue here is that that is a bundled question because the guy is asking you for a yes or no answer. It's an interrogative. Yes or no? "Are you still the same jerk you used to be back in high school days?"
- 1.94** But the problem is, if you say yes, then what are you saying? You're saying, "Yes, I used to be a jerk, and I'm still a jerk." And he likes that answer. But if you say no, which is the other logical possibility, then what you're saying is, "No, I'm not a jerk anymore, but yes, I used to be." You're implicitly granting that assumption. In that case, what has happened is, in a social context, the person has bundled a certain amount of information, but assumes something as given. He doesn't even need to state it or defend it explicitly. "You used to be a jerk back in high school."
- 1.95** It's only to the extent that you recognize complex bundles of information, and you're able then quickly to unbundle those pieces of information, say, "Well, wait, there's two issues here, the past and the present. Which one do you want to take up first?" And so you immediately push it back. But also at the same time, going back to our analogical reasoning, chances are good you've dealt with other people who are obnoxious and verbally attacking and so forth in your life.
- 1.96** And so you've learned some lessons from them, and you say, "Oh wait, this guy is acting like all of these other people over in the past. Those were problem people. Those relationships didn't work out. I probably don't want to deal with this guy." So then you say, take all of that, apply it to this case, and you set the guy aside. Very useful logic as a tool of self-defense. One more motivational thing, that's in individual social context. If we elevate this to very important issues, where logic and a commitment to logic becomes absolutely crucial, not only to individual relationships, but legal principles, the conduct of law, and so on.
- 1.97** I like always to reflect on the example of John Adams before Independence, when John Adams was a lawyer, and the British and the colonial Americans, there was tension between them, borderline riots, and some British soldiers had shot into a crowd of Americans who were getting a little uppity. It's not quite clear who instigated it, who was in the right and who was in the wrong, but British soldiers shot, Americans ended up dead.

- 1.98** There was then immediately an outcry among the early Americans and beginnings of a lynch mob. "We're just going to storm, take all the British soldiers, and just basically kill all of them." That likely would have happened. It didn't matter which one did the shooting, it didn't matter who started or not. But the police were able to defuse the situation. The British retreated, things tempered down, and then there was going to be a trial. At this point, with tempers running high, the question of taking this to a trial is already a commitment to logic, and the logic commitment is bundled with a commitment to civil decency and justice.
- 1.99** The only way we are going to figure out in this situation, even though someone has died, what the right thing to do, who is guilty and who is innocent, is to put it before a court of law. And what we're going to do in the court of law is, we have two possible positions. The British acted wrongly or the British acted correctly. And we're going to gather as much evidence as we possibly can, and that's going to be a lot of evidence, including all of the eyewitnesses' testimonies.
- 1.100** We're going to cross-examine them. We're going to see if the arguments can be put together for guilty, for not guilty. The best way, put them in collision with each other. Then and only then, when we go through that process, will we have a best, just verdict. We'll probably get the right answer, but just by committing ourselves to that logical process of evidence and argumentation and counter-argumentation, are we showing what our commitments are?
- 1.101** One of the things that made John Adams' reputation was that he was willing, even though he was a colonial American, to be the defense attorney for the British and to make the best arguments possible for the British defendants in that case. It was a matter of principle for him. It doesn't matter my American loyalties or my British loyalties. Those are not relevant to the truth and the justice of the matter. The logic matters. We're going to go through this logical process.
- 1.102** That's logic and why it matters. What I want to do next is turn to more formal methods and start from the beginning and do logical analysis from the ground up. Some technical terminology and some foundational nuts and bolts with respect to the logic. One of the things that we do in reasoning, of course, is we take the existing facts on the ground or the ones that are available, and we've got them organized in a certain way, is try to break them down to as far as we possibly to their component parts.
- 1.103** But then also we take this chunk that we've integrated and we put it together with another chunk over here to build a larger structure. By analogy, I often think of a car and mechanics. You can look at the car engine, and it all works together. If you know what you're talking about, and this is not necessarily me, you can say, "Ah, that's the fuel injector, that's the break fluid, that's where the transmission is. Those are the pistons and spark plugs." There's all of those elements, and you can see how they are working together.

- 1.104** But one thing you do want to be able to do, if you're a good mechanic, is get really into it and say, "This is the transmission," and be able to take it apart and see all of its moving parts. A certain amount of what we will be doing is looking at relatively simple arguments, like the ones we've had on here. How do we take them apart, look at their component parts, and see how they are put together? Another part of it, this is typically more advanced, is then how do we deal with more complex arguments and networks of arguments all the way up? We'll work our way through this course as much as possible in there.
- 1.105** Now, the first thing is the word argument itself. In logic, we use argument precisely. If two people are just yelling at each other, that "you're an idiot," "no, you're an idiot," "no, you're a bigger idiot," "no, you're an even bigger idiot," that's not an argument as far as the logicians. That's just people yelling at each other. An argument is a structure of propositions where you take one piece of information and you make the claim that something else follows from that.
- 1.106** So if the one person had said, "No, you are an idiot because," and then gave a reason, then we've got two things. There's the reason, and now we have the conclusion, "you're an idiot." Let's try to come up with something a little less insulting if we possibly can. That then is to say, an argument is always going to have two components. One is the conclusion. What is being argued for? What are you trying to establish? What are you trying to prove, either for yourself or, if it's in a social context, what do you want the other person to come to agree to?
- 1.107** The argument has a conclusion. It also has a premise or premises. The premise or premises, those are the reasons that are given in support of the conclusion. Think of the conclusion as, what am I supposed to believe? And the premise is, why should I believe that? Now, that's the nuts and bolts, but that's how we'll be using these concepts throughout the course.
- 1.108** Here, let's read through this first argument. I just took it straight out of Aristotle's Politics. "In a democracy, the poor have more power than the rich, because there are more of them, and the will of the majority is supreme." Now, in this case, we have three claims. First, in a democracy, the poor have more power than the rich. We have another claim. We say there are more of them. And the "them", this pronoun refers to the poor, so we might substitute the word poor. There are more poor people in society than there are rich people.
- 1.109** And, that conjunction, the will of the majority is supreme. That's how democracies do what they do, by means of the will of the majority. To say that it's supreme, that's connected to the word power. The will of the majority, that's connected to the word democracy. We have a nice tight connection, but sometimes the verbal formulations are a little bit. Now, this is an argument, three pieces of information. Which one is the conclusion and which are the premises?

- 1.110** In this case, we would say, well, what he's trying to prove is that in a democracy, the poor have more power than the rich. I'd like to just indicate that with a little c. That's what I think the conclusion of the argument is. It makes sense to say that that is the conclusion. Well, why should I believe this, Aristotle? And he gives us some reasons. He says, "Well, there's more people who are poor in a democracy, and whoever is the more, the majority, they are the ones who have the most power."
- 1.111** So I reconstruct this in my mind. We say, there's more poor people than rich people, and in a democracy, the poor people have the power, then it follows that poor people have the most power in a democracy. Now, another thing. One thing that's useful is that, in English and various other languages, we have developed concepts or words that we use to indicate whether something is a premise or an indicator. Surprisingly, we call these indicator concepts, conclusion indicators and premise indicators.
- 1.112** You've got these largely automated as adult thinkers and conversers, but it's worthwhile to highlight explicitly the various kinds of concepts and which ones they are. In this case, I would, when I'm learning logic, circle that word. That word is a premise indicator. And anytime you see the word "because", it tells you that the person is now going to give you the reasons. They're going to indicate the premises. And it almost always means that the thing just before that is the conclusion.
- 1.113** So in a democracy, the poor have more power than the rich. Well, why should I believe that? Well, Aristotle says, because, or by the cause of, for these reasons. There are various other words that we might use that would be synonyms at this point. I could, in this case, instead of the word because, I could have put the word "since." The poor have more power than the rich, since there's more of them, and blah, blah, blah. Or I could say, for the reasons that, really just belabor the point, for the reasons that, or as, or for, and there's a whole number of them.
- 1.114** So what I recommend is, when you're practicing this method initially, and it's good to practice it this way, start to mark things up when you are looking at various arguments. Sometimes it's a little bit obvious in the easier cases, but it's a useful skill to have when you're starting to deal with more complicated pieces of reasoning. In many cases, you'll read some long passage and you know there's an argument in there, but the passage is kind of confusing, and it might not be your fault. It might be that the person is not a very clear writer, or it just might be that this is a complicated amount of information.
- 1.115** If you can identify that's a premise, and just put a little p next to it, that's a premise, that's got to be the conclusion, put a little c at it, and then go through that way, then reread it with that information in mind, your comprehension oftentimes will go up a lot more. In this case, we have the will of the majority. That's a premise, and there's another premise there.
- 1.116** All right, let's try this one. Doctors should never lie to terminally ill patients, for one ought never to lie, and patients have a right to know their condition. What would we

say in this case? Well, I've got this piece of information. Doctors should never lie to their terminally ill patients. You might be tempted to. You don't want to hurt the person's feelings. Then we have this claim here. One ought never lie. And we have this claim here. Patients have a right to know their condition. This one seems to be speaking about an obligation that one has, as a physician or as a doctor. This one is talking about patients and a right that they have.

- 1.117** We have this word in the middle, for. I would circle that. That is a premise indicator. If we read that, then to say, well, this is probably the conclusion. The person wants you to say, "You shouldn't lie to terminally ill patients." Well, why not? It makes sense to say, well, for these reasons. You should not lie, in the first place, and the patients have a right to know. If I reconstruct it, one should never lie, therefore, follows that physicians should not lie to their patients. If it's the case that, the other premise, patients have a right to know about their condition, then it follows that doctors should not lie to them about their condition. So it follows. This is a conclusion indicator.
- 1.118** Try this one. We can't go to the concert and clean up the apartment tonight. But as we are hosting that dinner tomorrow, we've got to clean the place up tonight. So the concert is out. We can't do both of these things. New information. We're hosting a dinner tomorrow. We've got to clean the place up tonight. Concert is out. We can't go to the concert. Now, in this case, I would circle this word, so. When we use the word so, we're almost always indicating a conclusion. This is what we're trying to get to.
- 1.119** In these cases here, I gave the conclusion first, then the supporting reasons. Conclusion first, supporting reasons. In this case, I'm laying out my reasoning, and then at the end, we get to the conclusion. How does this one work? Well, we have an option here. We can go to the concert or we can go to or stay home and clean up the place. Then, as I said, this one I have to give priority to, cleaning up the place.
- 1.120** Why? I give a reason for that. I've got these two options. I'm going to give this one a priority. Why? Well, because tomorrow we're having a dinner, so we have to clean up the place tonight, but we can't do that and this other thing here as well, go to the concert. So, big conclusion, concert is out. We're not going to the concert. End of that chain of reasoning.
- 1.121** Now, let's try this fellow. Rabbits are a type of fish. Since fish have fuzzy ears, it follows that rabbits have fuzzy ears. I'm glad to see smiles. Rabbits are a type of fish. Oh. Word since, what is since? It's not a conclusion indicator, it's a premise indicator. I could put the word because here. Then I've got this other claim, fish have fuzzy ears. I've got this little phrase here, it follows that. This is a conclusion indicator. It follows that. Well, something follows, logically follows. This is going to be a conclusion indicator. This is what I'm trying to argue for or prove. Rabbits have fuzzy ears.
- 1.122** Now, this is puzzling because when we see an argument like this, and we do see lots of arguments like this, not as blatant as this typically, we start to get confused because

there's obviously some bizarre information in this argument. But nonetheless, it is an argument, even if it is going to end up being a bad argument for obvious reasons. What we do need to do is develop our habit of being able to see what the argument is and not let the distracting falseness of this particular piece of information confuse us and lead us not to do whatever.

- 1.123** Good argument or not, nonetheless, the argument is, the person says rabbits have fuzzy ears. Now, interestingly, that's a true piece of information. Rabbits do have fuzzy ears. But that piece of truth is largely irrelevant to the truth of the premises here. We've got false information. Rabbits are not a type of fish. Fish don't have fuzzy ears, but nonetheless, somehow we still end up with a good conclusion. That's illuminating right off the bat then, to say, well, I've got false information, but somehow I ended up with the truth. Was it a coincidence, or is there something else at work here as well? But at least I know what the argument is, and I can put it in standard form.
- 1.124** Let me read through a few examples. As I want to say my hometown, Toronto, is going to win the championship because they have the best field goal kicker in the league. Train your ear to pick out the word "because." That's a premise indicator. He's claiming that the reason why Toronto is going to win is they have the best field goal kicker, therefore, they're going to win. The indicator concepts, what I would recommend is, for further self-study, is make yourself a list of premise indicators and conclusion indicators. Any good logic book will include those. You can also have lots of online sources that will do so.
- 1.125** Just make them part of your mental furniture. So you know premise, conclusion, argument, and you have a whole list of indicator concepts, more or less automated. This is very useful because the very first thing we always need to do is learn how to take arguments, however they come at us, and put them in what we call standard form. There's this premise, this premise, and then the conclusion, or however many premises they are. But that's not typically how arguments come at us. And knowing what the indicator concepts are will immediately do a lot of the sorting for you.
- 1.126** It's very common, particularly among adults, for people just to give information in any particular order. I can say, for example, Socrates is a human, and humans are mortal, so Socrates is mortal. Now, I give a little boost to the word "so" there. You know the conclusion is, so Socrates is mortal. But I can give you that same argument. It is the same argument, same information, but in a different order. I could say, since humans are mortal, Socrates must be a mortal, as he is a human being. That's exactly the same argument, different order in this case.
- 1.127** But I've got some indicator concepts that help you out, as he is a human being, since humans are mortal. So then you know what you're looking for. Or I could give you the conclusion up front. Socrates is mortal. After all, he's a human being, and human beings are mortal. In this case, "after all" is another kind of premise indicator. After all, here's

all of the other information, and after that comes the conclusion. So the more you practice this, the more automatic and the more fluid your thinking is going to be.

- 1.128** Next complication. We've indicated this before that, very often as adults, people do not give you all of the information. But what we want to do, if we're going to do logical analysis, is get the argument out fully and completely. We could do that very laborious thing about umbrellas and the rain and not getting wet. If we are, for example, going to be talking to a computer, we have to have all of that information out and feed it into the computer to put it together, at least in the beginning.
- 1.129** When we were little children, we had to go through all of that process and have a certain amount of experience to put all of that information together, so that we know about making umbrella-taking a habit under certain circumstances. This is another thing to start practicing. Almost always when you have one premise and one conclusion, there's almost always a missing piece. In that case, technically, we call the argument an enthymeme.
- 1.130** So if you're up on winning Trivial Pursuit-type games and knowing the terminology, the word is enthymeme, an enthymematic argument. That is to say, arguments that are incomplete are common, and they're not necessarily a bad thing. Sometimes we are saving people's energy. So I might say, Socrates is a man. Sorry, I might say Socrates is mortal because he's a man. It doesn't follow, he's a man, therefore he's mortal. I need to have a connection between the man concept and mortal. But then there's that obvious piece of information. Human beings are mortal, and we just fill that in.
- 1.131** When I was a young faculty member, I was at a party, and there was a woman who was making a joke from a feminist perspective. We were talking about another colleague who had said something a little bit off-color, and she said, "Well, of course he's a chauvinist. He's a man, isn't he?" Now, that's an argument, but it takes a little bit of massaging. So this is where I want to take a moment. Suppose I say, "Of course he's a chauvinist. He's a man, isn't he?" Now, in this case, we have two pieces of information. Claim that he's a chauvinist and that he's a man.
- 1.132** Now, we start to do the analysis. We start to say things like, well, what's she trying to get us to believe? What does she want us to believe? He is a chauvinist. I'll just abbreviate the chauvinist with C, that's the conclusion. There's no conclusion indicators here, no premise indicators, so we have to figure it out from context in this case. Well, why? The argument is, he's a man. This is actually put in the form of a question, but this is a rhetorical question.
- 1.133** One of the things we learn is, often rhetorical questions are to be translated into a declarative statement. But if you're trying to make a joke, or sometimes if you're trying to be sneaky, rhetorical questions are useful because you know that people have to do the additional work to figure out, what does this rhetorical question mean literally, and what other assumptions are involved? Now, in this case, it's not particularly

complicated, but what you look for is, we have he here, we have he here, he is connected to M, he is connected to C here. We need to connect M and C somehow, to make this logical tightness work.

- 1.134** Then the claims are going to be something like, men are chauvinists. That's the general claim. If men are chauvinists and he's a man, then it does follow, logically, that he is a man. That's the argument that we are dealing with. But then we would say in this case, probably this is the overstatement question. But you're not going to get to the overstatement, a claim, rather, that's being made, unless you do this initial work, or you can pull out, "Wait a minute. There is this assumption here. Your argument was an enthymeme." Now that we make this explicit, we can talk about this. Now, this is not what you want to do at the party, but this is the logical work that will have to be done.
- 1.135** All right, how about this joke? I hope it's a joke. "Every fourth child born in the world is Chinese. Mrs. O'Reilly is expecting her fourth child..." Now, good. That's a little bit of humor. But what makes it funny? Well, in order to get the joke, you have to do a certain number of things. You have to take "every fourth child born in the world is Chinese" and interpret it a certain way. One of the things that we learn as adults is that some formulations are somewhat ambiguous.
- 1.136** We could say every fourth child in the world who's born might mean, well, this one, then this one, then this one, then there's the fourth one. There's a sequence way of using that. But sometimes we say "every fourth child in the world" is to use percentages. All we mean is, not in sequence, which is 25% of all the children who are born in the world are born to Chinese people. We deliberately misinterpret that one to make it a sequence thing.
- 1.137** Then we make it more ridiculous by choosing an obviously Irish ethnic last name. Mrs. O'Reilly is expecting her fourth child. If she's Irish, she's not Chinese, then it makes it absurd, the inference, but also notice the inference is not made. The inference in this case is, this argument is an enthymeme, but we just use our logical sense and we say, "Oh, so the person is trying to say, Mrs. O'Reilly's child is going to be born Chinese. Ha, ha, ha."
- 1.138** All of that is going on, but what was interesting, if you monitor your own reaction, is that that happened in half a second. All of that happened. That's good. That's a sign that you are a mature reasoner, but we need to start at the beginning. Suppose we were explaining this to a computer or a four-year-old. A huge amount of information, go slowly and carefully. That's what we are in the habit of doing.
- 1.139** All right. Final thing, piece of terminology for breaking down arguments and putting us in a position to evaluate them. We know what the premises are, we know what the conclusions are, we have the ways of filling in any missing pieces that are there, so it's no longer an enthymeme, we have the complete argument. We have indicator concepts for conclusions, for premises, to help us sort out which one is which.

- 1.140** The next thing is going to be the terminology that takes us into evaluating arguments. Once we've got the argument in front of us, it's complete. We know what the premises are, we know what the conclusions are, any missing pieces, we have added them. It's complete. It's before us. Our next question, of course, is, is this a good argument or is this a bad argument? In many cases, this is important to go through these steps ruthlessly and train yourself always to go through these steps.
- 1.141** Because as we know, one of the things that happens in social conversations that don't go very well is that people are not careful in figuring out what the argument is before they launch into criticisms. They're very quick to start evaluating, but in many cases, they miss some subtlety that's in the argument or misinterpret it in some bad way. So at least at the very beginning, be careful, be careful, be careful all the way through.
- 1.142** So I've got three arguments out here. They are complete, but they're all basic arguments. So Aristotle is human. Aristotle, humans are mortal, so therefore, Aristotle is mortal. The concepts we want to introduce for evaluating are two concepts. One is the concept of truth, and the other is the concept of valid. These two concepts pick up two different aspects of evaluating an argument.
- 1.143** By analogy, suppose you were on a shopping trip, a little mathematics example. You went on a shopping trip, you bought a whole bunch of things, and then you get home and you say, "How much did I spend?" And then you start to write down all of the things you bought and how much they cost. Then you add all of that up and you get a certain number. Is that the correct number of how much you spent? It could be true. Sorry, it could be the right number, but that would depend on all of the things that you bought, you remembered them and you got the right amount.
- 1.144** And it would also depend on that you did addition correctly, with respect to all of those numbers. There's two possible sources of error. One is that you got a number wrong, the other is that you made an addition mistake. By exact parallel, truth and validity are capturing the same thing in logic circles. In this case, with respect to truth, we can apply, say, the truth filter. We would just say, well, here's a premise that the argument depends on. Is that true or not? Look at that premise in isolation and ask that one question. We'd say, yeah, that's true.
- 1.145** Humans are mortal. Look at that premise in isolation. Is that true or not? Say, yes, that is true. Now, the follow-up question, since we're doing an argument, and the point of an argument is to take some information and draw an inference from it or make an inference from it, if this is true and this is true, we put these two pieces of information, does it follow logically that Aristotle must be mortal? If our answer to that question is yes, if that's true, and if that's true, this has to be true, then the argument is valid.
- 1.146** What we are attending to is not the truth or falsity of the individual components, but the form of the logic or the structure of the reasoning that we engaged in. In this case, we would say, this follows logically. That's the same thing in logic as saying it's a valid

argument. Truth, do the pieces of information correspond to reality? Validity, have you structured that information and drawn an inference in a way that preserves truth? And both of those are important.

- 1.147** So in this case here, Aristotle is human. Humans are mortal. Therefore, water freezes at zero degrees celsius. Well, this is true, this is true, the same true information. But in this case, obviously, the conclusion bears no relationship whatsoever to the premises. It doesn't follow. If this is true, that water freezes at zero degrees, this is not the reasons why. So this is a bad argument. So in this case, we say this is an invalid argument. You can have perfectly good information, but not do good stuff with it. You end up with something that's not relevant.
- 1.148** In this case, rabbits are a species of fish. False. Fish have fuzzy ears. False. Rabbits have fuzzy ears. Well, we're supposed to see if the conclusion follows from the premises or not. If it were the case that rabbits are a species of fish and fish had fuzzy ears, would it follow that rabbits have fuzzy ears? The answer to that question is yes. This is the same pattern as that first pattern over there. This is a valid argument. The premises are false, but using that information and structuring it to draw this conclusion, this follows.
- 1.149** This is probably the first mental barrier that one has to get past. You have to be very comfortable with saying perfectly valid argument, even though the information is false. With that, we will pause on this first unit. We now have some skills under our belt, separating premises and conclusions, knowing what premise indicators are, knowing that we have to make sure we're filling in some missing pieces to have the entire argument before us, practicing, putting things in standard form, listing all of the premises, followed by the conclusion, then starting the evaluative work.
- 1.150** Go through all the premises and ask, given your knowledge, does that seem true or false? Whatever your state of knowledge is. Then the logic part that we especially are going to be, if that information is true or false, we set that aside for the moment. Does this conclusion follow, logically, from those premises? We'll go on to do more sophisticated stuff in the next lecture.

Lecture 2: Concepts & Our Cognitive Hierarchy

At a glance — Concepts and our cognitive hierarchy.

- 2.1 The 18th century thinker Voltaire, he of the razor-sharp logic and razor-sharp wit, once said, "When enlightened men go on arguing for a long time, there's a distinct possibility that the question is not clear." This is an occupational hazard for us as thinkers about abstract things and complicated things. It's very easy for our thinking to get fuzzy, in our own thinking, but also for people to be using words in overlapping but sometimes distinct ways.
- 2.2 So one of the important elements of logic is paying attention to the concepts that we are using to build up our propositions and make sure that we are, in our own thinking, clear how we are using the words that we are using and certainly in a social context. Now, one way of approaching this is to say, well, if you think of a standard argument, "All human beings are mortal, Aristotle is a human being, therefore, Aristotle is mortal, or mortal," well, already in there, I've got a number of abstract concepts, mortal, mortality, human being, a concept of all human beings.
- 2.3 And then relationships between those concepts start to develop. And we have a proper name. Aristotle is a particular being whom we've assigned a proper name. Do I know what all of those concepts mean? And in a social context, will we be using those concepts univocally? So your thinking is mapping onto my thinking as well. That's an important part of the logical analysis process, but it's also a part of seeing logic as an integral part of what is actually a broader human cognitive apparatus. I want to spend a little time putting logic in that slightly broader context.
- 2.4 Here, logic develops tandemly with developmental psychology. If we imagine for the moment a baby that has just been born, and we tell a story about likely human cognitive development, from just being born all the way up to mature adult thinker, it really is an impressive process. Partly, what makes this process impressive is that we typically think of humans coming from infancy to adulthood as being a process that takes a dozen years or more. You're not really an adult until you are significantly into your teens, and maybe not even then.
- 2.5 If you contrast that with the cognitive development of all most other animal species, even intelligent species like, say, dolphins and dogs and so forth, it's usually a matter of two years, three years, or four years until they have reached their cognitive peak, so to speak, and they are ready to function fully as adults. But then if we look at our three-year-olds and our four-year-olds, they're just getting warmed up. They're already impressive in a lot of ways, but they still have a lot to do.
- 2.6 I want to start by putting some developmental structures and some claims about them along the way. The first thing that we start off with is by having sensations. A child is born, the child is born fully healthy, you have five sense organs, and they immediately

start giving us information about the environment. So light energy, chemical energy, sound energy, tactile energy, more chemical structures with respect to tastes and so forth. So all of this information starts, say, impinging upon the baby fresh out of the womb, so to speak.

- 2.7** Fairly quickly and almost automatically, the child rises to the level of perception. What we mean by perception is integrations of sensations. So again, I'm appealing to some developmental psychology claims, but you take a fresh baby, born baby, looks, eyes, and there's just all this stuff impinging upon the eyes. But very quickly, babies are able to recognize their mothers, because obviously the mother has a vested interest in getting to know her child as soon as possible.
- 2.8** But at the beginning, there's just all these color patches and things moving around in the visual field. But at a certain point, that face with the two eyes and the mouth integrated as a unit, that's now a perception. It's not just a bunch of sensory stimuli happening. If the mother's face goes away but comes back, the child recognizes it as the same integrated unit. If the mother's face moves across the infant's visual field, it's not just changing visual fields all over, it's, there's background, visual face, and you can see the child tracking the mother's face.
- 2.9** Now, this happens also with the other sensory modalities. We start to integrate bundles of sensations into what we call percepts, and we are aware of objects now out there in the environment. Now, this also happens with most other animal species. What seems to be largely distinctive, this is another developmental psychology claim, it may not be that we are unique, but we are among a small group of beings that have the next cognitive power, which is what we call conception. That is to take a bunch of percepts and bundle them and integrate them into a new mental unit, but one that is now abstract, and we start to learn words.
- 2.10** So this happens immediately. This happens within minutes and then, with increasing sophistication, typically over the course of the child's life, even on into adulthood. This typically starts to happen when kids start having their first words. Because then what we know when they start using the first words is, they are using those words to designate classes of things.
- 2.11** So it might be the case that there's the mom and the dad, maybe they have a family dog, and maybe the neighbors have a dog. So at some point, the kid is going to learn the word dog, but then you might be out riding or the parents might be out riding with the kid, and there's another dog out on the street, and the kid will say dog. And that then is not a proper name for that particular percept. And dog is not the name for this particular percept. It's that there's this category that's more abstract, of a certain being, and the one at home and this one and the other one down the street are all members of that particular category.

- 2.12** Now, there is some evidence that some dolphins have some concepts, some chimpanzees have some concepts, and that's an open-ended question in animal psychology and human psychology. Human beings definitely have it, typically kicks in seven, eight, nine months, and then it's another open-ended process where we start to have all kinds of categories until, these are some estimates by linguists, the average high school graduate in North America has about 10,000 such words. Those words stand for concepts that he or she all understands. The average college graduate has about 25,000 such words kicking around.
- 2.13** Now, just a couple of examples about the difference between the conceptual level and the perceptual level, because this is very important. Standard examples we use is, you take a hen, a chicken, the female chicken. There's one species of chicken that is a speckled hen. And these ones, characteristically, are not all white or beige like a lot, but they have lots and lots of speckles on them. We can, as other animal species can, look at that hen and we can perceive the hen as a unit, as an integrate, and we can see distinctly all of the speckles that are on it.
- 2.14** But we have no idea how many speckles the hen has, just by looking at it. But we can, if we learn this, we can learn to count. Then we could take the chicken, probably against its will, and then say, I'm going to count every single speck on here. We end up, it's not going to enable me to perceive the chicken any better, but I might then reach the conclusion, because I have these mathematical concepts, there are 872 speckles on this chicken, and that's different from 871 and 874. The math is going to be different. I am thinking now conceptually about the nature of that chicken.
- 2.15** This is learned, and it's important to understand that it is learned, because this is where we have to start doing extra effort to figure out what we're doing with all of these bundled concept categories that we are using, what's included, what's not included, to make sure we're in a position to build more sophisticated structures. This is learned. There are some fascinating experiments that have been done.
- 2.16** So we use the example of counting. It might seem like counting is pretty basic and it would come more or less naturally to human beings. But there is some evidence that it is not something that comes naturally, that it has to be discovered and then transmitted by learning to each of the next generation. There were some anthropologists who were studying a tribe in one of the Indonesian islands, there's thousands of them, and this was a tribe that was still Stone Age in its technology and quite remote. So they just went and they started to study them. And the tribe, they were perfectly fine with this.
- 2.17** And they had some fairly sophisticated things, sophisticated buildings, so to protect them from the elements. And they were at the stage of agricultural because they had goats, and they had built enclosures to keep their goats at night. But they had a job assigned to the young boys of the tribe, which was, every morning, take the goats out of the enclosure and take them out into the meadows and forests where they could

feed, and at the end of the day, bring the goats back in. But it's important. Goats are valuable. These are an important animal, to keep track of all of your goats.

- 2.18** And so there would be a boy who would be assigned, when the goats were coming out of the enclosure. There's a big pile of pebbles there. Every time a goat came out of the enclosure, take one of the pebbles off this pile and start a new pile here. So that kid would just stand there doing this, and then all of the goats would be gone. And if you asked, how many goats does the tribe have? The answer would be this many. This pile of stones represents. But they did not have counting concepts. It wasn't for them, 1, 2, 3, 4, 5.
- 2.19** And this became very apparent and became important because, at the end of the day, the goats would all come back, the boys would bring them back, and one of the boys would have the job, take all the stones here. Goat comes in, goat comes in, goat comes in. It's a good day if you have no stones left over at the end of that process. But what happens if there are stones left over? In some situations, the anthropologist would be in, where they would say, oh, there are eight stones there. That means there are eight goats missing. We have to go out and find eight goats.
- 2.20** The problem was, there could be eight stones there, but the boys had no idea how many goats were still out there. It's like the speckled hen. There's just a bunch of stones there. If it was 15 or 13, so they wouldn't be able to go out, for example, and say, "We know we are looking for eight goats. Find the eight goats and bring them back." Or they go out there and they only find six goats. We've counted them, we bring them back, but now we know, even before the stones, that we still have to find two more goats.
- 2.21** So the point of all of this is, conception, abstract mathematical concepts, which is one important category, it's learned, and the precision of being able to use those kinds of concepts, logically, is an important learned skill. One other famous experiment that I'd like to talk about, it wasn't formally an experiment, it was actually a semi-formal experiment. It goes back to the 1880s, about the difference between perception and conception. There's still an ongoing question about how intelligent crows are, and they're pretty smart for birds.
- 2.22** But this guy was Sir John Lubbock. He was a hunter, and what they did, I'll draw a little picture here, there was some sort of a clearing here. There's a bunch of trees over here. So a little bit of woods. Then there was a path that came out of some other woods over here. Over here in the woods, right at the edge, there was a little cabin. And what typically would happen is that these crows, I suppose I should draw the trees in green and the crows in black, there's the crows that would come down and hang out here in the evening, to do whatever crows do in the evening, but there's a whole flock of crows.
- 2.23** So what Sir John noticed was that he would come through here. The crows would see him when he appeared, and then they would fly up into the trees for safety, and then

he would go into the cabin, and the crows would not come out until he left this cabin. Let me make it more clear that that's a cabin. That's the chimney. When he left the cabin and went away, then the crows would say, "OK, fine, it's safe for us to come back," and they would come down again.

- 2.24** So he was curious how smart crows are. So he got a buddy, and the two of them would come through here. Crows would fly up into the trees. They would go into the cabin, and they would try to trick the crows. So just one of them would leave, go back along this pathway here, and then what would the crows do? They would stay up in the trees. They would not come down. They know that there's two, only one left. There's some rudimentary—I don't know if I want to call it arithmetic yet, but they're able to perceive the difference between one and two, and remember the difference between one and two, and then that matters.
- 2.25** Well, let's try it with three. Three guys come in, three go into the cabin, and after the crows have flown up, two of them leave. What do the crows do? Big mystery, the crows stay up in the trees. They don't come down until the third person leaves. Then the crows know it's safe, and then they come down. Next iteration of this semi-formal experiment, four people come through, crows fly up, they go four of them into the cabin. Three of them leave. There's still one in there. What do the crows do? The crows come down and they start to feed again.
- 2.26** So, from their perspective, it's like their math is one, two, three, many, and it just becomes undifferentiated at that point. So when it was four, it was more like, many came, many left. End of the story. All right, now human beings, of course, we can operate at that level in our thinking as well, but we can be four, five, six, and we have these open-ended processes. The point is that that is all learned.
- 2.27** Now, the next thing that we start to do is, once we have words, if we go back to a developmental thing, children typically build up an arsenal of single words—blanket, food, drink, mommy, daddy, dog, etc. But then at some point, they start putting the words together. There's a further integration. So the sensations are integrated into percepts. Percepts are integrated into concepts, and we start to speak in sentences. And a sentence will be an integration of concepts.
- 2.28** So, Aristotle is mortal. So I know who Aristotle is. I know what mortality is. Probably not when I'm one year old, but Aristotle is tall. I know what tall is. I've got some height and some length concepts. Aristotle is tall. So I know tall, I know Aristotle, but I put those together as a new unit. Or the mother can say, "Where is your blanket?" And the child might then be able to say, "My blanket is in my room." Blanket, a possessiveness, room as a certain place, my room, relationship between the blanket and that particular place. That's pretty sophisticated. That's five or six mental elements all being integrated into one sentence.

- 2.29 And so then we know we are increasingly able to make more complicated sentences as we mature. So we can say things like, read in a story, "The quick brown fox jumped over the lazy dog and escaped from the henhouse with the prize clamped firmly between its jaws." So what was that? Sixteen, eighteen words, and it's easy even when you're four years old. That's how sophisticated we have become.
- 2.30 And then we start to do more sophisticated things, individual sentences. Parents, when they start reading to their children, we start to have storybooks where we'll take individual sentences and we start stringing them together to tell a little story about what happened to Peter Rabbit or whatever the story is. So we start to do what we call narratives, and we have complexes then of sentences that are connected in various ways, and we want them to be connected logically, so that the causal events of the story fit together and everything ultimately makes sense.
- 2.31 We also go on, since often narratives are fictional, then to develop more sophisticated theories, which are whole networks of propositions about the way the world works. Just as an example, I always think of at this point, because all of the younger kids that I know have gotten to this stage quite sophisticatedly, usually by grade 2. I remember growing up in Canada, even in grade 2, then when I moved to the United States, kids were doing the same thing. So there must be a universality about this.
- 2.32 But the weather cycle, and at some point, every kid in grade 2 comes home with a picture he or she has drawn of the weather cycle. And what it does is, just put it up here, is, I'll say, here's the sun, and here is a lake, and then here's a little stream, and then on it, there will be some water and there will be some squiggly lines that say this water from the heat of the sun evaporates and it goes up. So we put a little arrow here, and then it forms clouds. I should have paid more attention in art class instead of logic class.
- 2.33 But as it goes up, it also cools. So we build in this knowledge about cooling, and the clouds get heavier and heavier, and then it's going to fall down as rain, and then this is going to flow into the sea. Then there's going to be the ocean, and the same thing is going to be happening here. Sometimes there's the wind that blows the clouds over here, so sometimes it's raining. We have this sophisticated drawing. But we have lake, rivers, oceans, sky, heating, cooling, water in various states, and the kid gets it. This is our little theory about the weather cycle at age six or seven years old.
- 2.34 But that's a theory. That's a lot of abstractions, a lot of sentences based on certain concepts and observations and so on. And it's open-ended, so as we grow to adulthood, our theories get more complicated. We start off reading just short little books, three, four, five pages, mom and dad with the kids. We progress to chapter books, which might be as long as 30 pages or so. And then by the time we are young adults or into adulthood, we're hopefully reading novels or even multi-novel series, and we are keeping the story arc going and so on. So it's an open-ended, extraordinarily complicated process.

- 2.35 Now, logic is built in at all of these levels, keeping track of everything. But this is the level we are focusing on right now, at the concept level, making sure that we are clear in what these concepts are. So suppose we are now adults and we are comfortably working our way throughout, and we get into adult-level issues, and we want to have arguments about religion and politics. So does God exist, for example? We all bump into this question at some point, and we reason about it, and we want to reason about it; we want to make up our mind about it.
- 2.36 From the logic perspective, the first question would not be, to start plunging into the arguments for and against the the existence of God. The first question is going to be, when you say the word God, do you know what you mean? That should be a deeply personal question. I should be able to identify, in a clear statement, sometimes it might be in a formal definition, what I mean by God. It's going to be important for my own thinking, but then as we know, it's important for social reasoning as well, because that is one of those words that, by the time we enter into the discussion, there are dozens of major religions and thousands of historical religions, and how they use God is not univocal.
- 2.37 So the first question will be, before you and I, say, start having a discussion about, do we believe in the existence of God or not, what are we talking about? Can we get this concept clear? If we were going to argue about important things in politics, I want to say, that guy is a socialist, that guy's a conservative, then we immediately plunge into all kinds of arguments for and against socialism, arguments for and against conservatism, and so on. Do we know what those concepts mean?
- 2.38 We should, in our own thinking, know what we mean by the word socialist. But of course, we know, as soon as we enter into social context, different people mean different things by socialist. Again, there's a big history of socialism, as there is for conservatism. The logicians always want us to slow down, and they're always going to say, "Clarify the terminology first before you start the argument." If you're really serious, give your definitions of your key terms up front, so that everybody knows what you are talking about. You also know what you are talking about.
- 2.39 The models here would be things like in geometry, where we can say, what is a square? When we are young, so there's a square, there's a triangle, there's a circle. And your geometry teacher, in introducing these concepts, you have lots of observational experience of each of them. But before we can go on to do more sophisticated logical derivations of theorems when we're doing our geometry, what is a square? Then we look at the square, we measure the square, we figure out its important attributes.
- 2.40 We say, well, it's got four sides. The four sides are supposed to be enclosed. These angles are supposed to all be the same. If we measure the angles, they should add up to 360 degrees. We take all of that as raw material. Based on my observations, I've got all of the constituent concepts, and I'm going to try to define this more abstract thing. I might say something like, a square is a quadrilateral, four-sided thing. The interior

angle is adding up to 360 degrees. Or we might say with four right angles as interior angles. That would be my rough draft. Go to work on that, fine-tune it, and so on.

- 2.41** What about this one? Well, three sides enclosed. The interior angle should add up to 180 degrees because this also is a triangle. Three sides, the angles will add up to 180 degrees. Having done that preparatory work, a triangle is a three-sided figure, interior angles adding up to 180 degrees. That's precise. This one, a little bit tricky. How do you define a circle? It doesn't really have sides. It seems like it only has one, or it has one line, but this is also just one line.
- 2.42** We want to differentiate the circle from the one line, but it's hard to figure out this angle stuff that goes on in here. We might say it all adds up to 360 degrees again. So a line, blah, blah, blah, blah. So how do we capture this, the fact that it's not a straight line and so on? We might try a device like saying, well, imagine a central point. So it's a line, all of the points of which are equidistant from a given point and adds up to 360 degrees. Again, that's my rough draft. Continue to work. But the point of this is, before we can go on to do serious work in the geometry of mathematics, this definitional work has to be done.
- 2.43** In business, if you're going to talk about investment, almost all business textbooks will define their investment terminology, and that's very important. What's the difference between a stock and a bond? Well, they're both investment devices. Then you start to think, well, let's see, in stocks, what's important about stock? Well, it's an investment thing, but you own a chunk of the company. There's a certain amount of risk because the stock might make money, it might not make money, and so you might lose your initial investment or you might not.
- 2.44** By contrast, you can say, here's a bond. Well, it's another kind of investment, but in this case, it's a loan. You're loaning money to the company. You don't actually own it, but you have a guarantee from the company that they're going to pay you a certain amount of money each year. You're looking at these devices and what's going on in the relationship, and you have all the constituent concepts. This now puts you in a position to make something that would be a more formal definition. Stock is an investment vehicle or an ownership investment vehicle that carries risk. By contrast, a bond is an investment loan vehicle with a guaranteed rate of return. Again, rough draft, clarification, now we're more ready to go on.
- 2.45** One of the ones I like to think about a lot because this is one that's commonly confused in common distinction is the distinction between envy and jealousy. Suppose we are, I'm just going to use this as an abbreviation for psychology, the Greek symbol, psi. We're figuring out what goes on, and one important part of psychology is all of the emotions that we have, as well as all of the thoughts and conceptions that we have. There's a whole bunch of emotions that are out there.

- 2.46 But if we are trying to sort out our own mental furniture, what's going on in me, we're dealing with our children, and children are having all kinds of emotions, and we want the kids to describe how you are feeling, so I can figure out what's going on with you; or even if we're going to become professional therapists and psychologists and so on, all of this territory, we need to get sorted out before we can do sophisticated theorizing about it.
- 2.47 So one of the very basic distinctions is the distinction between these two, but often they are used interchangeably with each other, even though there is an important distinction here. So, in the case of jealousy, for example, what do we mean by jealousy? We start by saying, well, here's some examples of cases where we would use the word jealousy.
- 2.48 Suppose my wife, she's gone out for the evening, and I remembered I needed something at the mall, and so I went off to the mall and, walking through the mall, and then I see, oh, that's my wife, but she's with another man, and that man is hugging her, and she's hugging him back. This is a nightmare. This is totally made up. But what am I feeling at that moment? Am I feeling envious or am I feeling jealous? The right answer is to say, no, I'm feeling jealous in that case. So, I'll say, well, that would be an example of jealousy.
- 2.49 Then I'll try to unpack a little bit more what's going on there. But what would be an example of, say, envy? There's another made-up example. Suppose I drive a really old car and it's breaking down, and each day I get a new, I'm not sure if it's going to start up or not, but I don't have enough saved up to get myself a newer vehicle. And then I see one of my buddies, with whom I've always had a little bit of rivalry, drive by, and he's got some \$100,000 vehicle. It's exactly the kind of vehicle, I would just love to have that vehicle. At that point, what am I feeling? Well, I'm not jealous in that case. In that case, I am envious. He has this beautiful, say, new car. I'm feeling envious.
- 2.50 Now, those are some working examples. We might get some other working examples, but then we start to analyze and we would ask, well, what's the difference between these two? Well, in the case of jealousy, these are actually both involving possessions. I say, "Well, this guy has a car, and I have a different kind of car; and I have a wife, but it maybe seems like somebody else has her right now," so these are, we might say, both involving possessiveness, or possessions.
- 2.51 But what's happening in this case is, I think that I have this possession. She is mine, hands off. I think she belongs to me. I have no idea what's going on. I think I have here, but I fear, with this new evidence—there she is, hugging some other guy—I fear that I am losing her. I have her, but I am fearing the loss of her. That bad emotion is the fear of loss and the unexpectedness, given what you thought was the reality.
- 2.52 Over here, in this case, when I see the guy driving by in the beautiful car, well, I don't have a beautiful car. So this is an issue of, I don't have. I'm responding to something

that I don't have. I think of myself as not having that. I have this old beat-up, not-working-very-well car. And I would say that I fear that I won't get that. Because given my current budget circumstances, I can't afford a car like that. It's not very realistic that I can, we're going to get the money anytime soon. So this guy gets this great car, I don't have it.

2.53 So these are both about possession, where this is the fear of losing something that you have. This is the fear of not having or not getting something that you would like to have. And now we're in a position, I think, to come up with, say, a rough draft of a definition of the distinction between fear and jealousy, and then we can go on more explicitly and do serious work.

2.54 This would be important because, suppose you become a professional psychotherapist and you have a patient you've been working with, but you want to refer this patient to another psychotherapist. If you send it, well, the person has severe jealousy issues, but really you mean the person has severe envy issues, that's going to be destructive of the process when it picks up with the other psychotherapist as well.

2.55 All right. Carrying on, we should be able to do this in politics. If we are serious thinkers about politics, what are the important concepts? Well, starting in the American context, liberal, conservative, socialist, libertarian, fascist gets thrown around a lot; communist gets thrown around a lot, Democrat, Republican, etc. It should be part of your serious enterprise, before you enter into any decisions about what your views are or start to argue them, that you've got it all worked out, what the proper definitions of those concepts are.

2.56 I want to give an example of human beings. It's because this is an interesting one and a fundamental one. When we are working through these sorts of things, we are doing it somewhat case by case, working our way up from examples. As we start to deal with things that are increasingly complicated, we'll start to find all sorts of things that we are putting on our rough draft list, as I think of them here.

2.57 Then the important thing is to figure out from that rough draft list, this is a complicated situation. It's got all of these elements over here, and I'm trying to distinguish it from this situation, which has all of these elements over here, what's similar, what's different, and so on. But also there's an essentiality judgment. So one thing that is fun is to say, what if we try to say "human being"? And we'll often see this when you read more serious, shorthand, for definition, you're trying to try to define human being. And let's just go through this exercise a little bit.

2.58 So suppose we might say, well, what makes a human being a human being? Well, they have a body. They have a mind. Then we start talking about this part here. Should we call it a spirit? Should we call it a soul? Should we call it something more neutral, like the psyche? And what's the relationship between these sorts of things? We notice if we look at the body, the human beings, we can list all of the various body parts. But we

might then say, one of the things we want to be able to do, just as we're trying to distinguish envy from jealousy, distinguish conservative from liberal, is distinguish human beings from other animals.

2.59 We might notice that human beings are the only beings that have an opposable thumb. Now, I think that's true, but I'm not a human biologist or animal biologist, so let's just take this as a working example. But apparently, it's either very rare or it's exclusive to human beings that we have a thumb that we can oppose to the other fingers. That gives us more manual dexterity, and we can grip in various ways. Others, they can grab like this, but they can't quite make that opposing thing happen.

2.60 We might be listing all sorts of traits, right, out there. Human beings have an opposable thumb. Human beings, we might then say, are the animal with opposable thumbs. Say, well, it might be true, and it distinguishes this from other ones. Another candidate that was offered was to say, well, actually, the important thing about human beings—this goes back to Plato and Aristotle times—is that human beings are, more importantly, bipedal. If we look at other species, so gorillas are mostly on all fours, cows, horses, dogs, cats, squirrels, and so on, on all fours.

2.61 Some of the apes and monkeys can, for a short period of time, rise up, so there might be a judgment call. Then someone said, well, what about chickens? We might be willing to say chimpanzees or whatever are a borderline case, but chickens are also bipedal, and so this one doesn't seem to be able to work very well. Then suppose we then say, well, we are not only bipedal, but we are featherless. So we introduce this negative thing. So human beings are animals that are featherless bipeds.

2.62 And then the joke was, in history of philosophy, that someone took a chicken and plucked all the feathers off it, and then just threw it in the philosophers who are arguing about what it is to be a human being. So then you have a featherless bipedal thing, this featherless chicken in there. But obviously, we haven't done our work. So what else do we have to add here? We can work the body side of the equation. Sometimes we will get over to this side of the equation.

2.63 People will say, the thing we need to focus on is that the human being is not a body primarily, but more primarily, it's something on the psychological side of things; that we have a part of us, a spirit or a soul, that is different from physical stuff. And so what differentiates human beings from all other animal species is that they are animals who also have a soul or animals who also have a spirit. That, of course, takes us into arguments about whether souls and spirits really exist.

2.64 Others, they'll want to say, there's not really a differentiation between body and mind, they are two aspects of one integrated being. Nonetheless, what differentiates us in our minds or in our psyches is that human beings have the capacity for rationality. We might have the opposable thumb, and maybe there are beings that are pretty close to that. But in terms of our capacity for rationality, doing math, doing logic, thinking about

history, projecting the future, designing scientific experiments, complicated jokes, theatrical productions, and so on, all of these things that are expressions of our rationality, maybe that's the thing that differentiates us from all other beings.

2.65 Now, at this point, we are just working our way through four definitions, or four proposed definitions, and I don't want to terminate the process, where the point is that this is how it has to be done. It has to be done more, the more you're dealing with more complicated beings. The point then might be that we're trying to look for things that are meeting two creatures.

2.66 One is that they differentiate. That is to say, whatever it is that you're interested in focusing on, this is my concept, this differentiates it from all of the other ones. So does bipedal do it? Does opposable thumbs differentiate? Does the lack of feathers, does the rationality, does the possession of the soul? We have to think hard about whether, in fact, it does it.

2.67 And the other is the essentiality. Whatever it is that you are trying to explain or talk about, what are the most important things about it? The essential things that, if you didn't have that, you wouldn't have those anymore. On this kind of score, the way this next level of argument would be to go, would be to say, well, suppose you took a human being and instead of making it featherless, you took a human being who had suffered some accident and lost their thumbs.

2.68 You would say, well, if our definition of being a human being is that it's an animal with opposable thumbs, and then here you've got this guy who no longer has any thumbs, do we want to say, well, therefore, he's not a human being anymore? And that seems a bit harsh. It also seems wrong because human beings do have them, but these are not as essential to what it is to be a human being compared to other things. So we might not want to put that in the example.

2.69 By contrast, we can make a better case for rationality. Our rational capacity enables us to do language, get jokes, do math, think about our lives, be morally responsible agents, and so on. We might say, if we're trying to understand what it is that makes a human being a human being, rationality is a lot more important and a lot more explanatory than, say, having thumbs. We might then do the same thing and say, well, look, what happens if you lose your rationality? If someone has some severe brain damage or some severe Alzheimer?

2.70 We do start to say, well, it seems like the person isn't there anymore, or he's lost his identity or her identity, or something important is missing. Still human, but we are more likely to say this person is not able to do the things that are essential to human beings as before. That would then be a sign that maybe there's something more to that definition.

2.71 Now, other cases where these distinctions are important and being able to do the conceptions, all this definition work, is discovery. One of the things that we do is, then

we take our concepts and we have organized them definitionally and we see relationships between the various ones, is that they enable us to make a connection between things that we otherwise would not be able to observe.

- 2.72** An example I want to use here is from the history of medicine, involving the condition of scurvy. The historical story here is that the British Navy, early in its days when it was off exploring the world, trading with the world, conquering the world, and building up the British Empire into something fairly significant, was running into a problem that, when the sailors were at sea for a long time, they started to develop scurvy.
- 2.73** This would have a number of symptoms, one of which would be that your teeth would start to get loose and then start to fall off, and there would be other physiological symptoms as well. It was a serious problem, and they had no idea what caused the scurvy. But then almost accidentally, they discovered that limes cure scurvy. This is partly why the British then acquired the ethnic nickname of being Limeys, because the British Navy, once they discovered that limes cured scurvy, everywhere that they went, that was almost like their first question is, "Do you have any limes? Do you have any limes? Give us your limes. We want limes." So they became the Limeys.
- 2.74** This was an accidental connection. But if you add limes into the diet, then the scurvy symptoms go away, and so they start to do it more systematically. Now, the follow-up question, though, is, well, why? What's going on in the limes? And there you need to do some biological research, some chemical research. What nutritional needs do human beings have? What are the biochemical structures of limes? And so all of that project considers.
- 2.75** But also, sometimes it's pretty hard to find limes, because limes don't grow everywhere in the world. They don't even grow everywhere in the tropical world. But if we have, say, observed other kinds of fruits, for example, we notice that some of those fruits are somewhat similar to limes. Then when we are forming this concept of limes, and now we're adding to limes as part of that concept, limes cure scurvy, if we have perceived and categorized some other kinds of fruits that seem similar to limes, we might then ask the question, "Maybe those also can cure scurvy?"
- 2.76** They would start to say, limes are green and round and small. And lemons are typically a little bit bigger and yellow, but they seem to be in that same tart sweetness category when you taste them. So maybe it's the case that limes and lemons, while we have different concepts for them, and maybe we've organized them into yellow fruits and green fruits over here, that nonetheless, they are part of a broader category.
- 2.77** And that whatever it is up in this broader category, it's not the greenness of the limes that cures the scurvy, it's not the yellowness of the lemon, that it might be some common ingredient that they both contain. And so with this conceptual structure, what we then say is, we know the limes work to cure scurvy; what if we see other things in

this more abstract category, whatever we call this? We might call this category citrus fruits.

- 2.78** That will then lead us from our lime discovery to this abstract move to the case of lemons. Then we go get a bunch of lemons, we give them to our sailors, and we see if they cure the scurvy or not. Then it turns out, yes, they do, they work. Then we also then will say, well, what else is in this category? Then we might say, well, what about oranges? What about grapefruits? And so on. The point then just is that paying careful attention to our conceptual structure, and defining it and organizing it explicitly in this way leads us to more scientific discoveries that lead to better health and medicine results.
- 2.79** OK, as an example of generalizing on this issue of citrus fruits, and we know also we take fruits and we put them with vegetables, and then we will put those in as things that grow on plants, and so we have a higher abstract concept. But then we have all of the animals that we've learned, dog, cat, cow, horse, goat, human being, and so on. Then we have mammals. Then we've got other beings that we've learned, birds and so on, fishes and so on. We put those all together in hierarchical trees.
- 2.80** We might all say, then we take the plants and the animals. We have this very abstract category, living things. We have this entire hierarchical structure. The more sophisticated we are in our thinking is, if we know where things are in that hierarchical structure, we can get there very quickly. And then when it's important, formally, we can specify its location in that hierarchical structure and communicate that to other people.
- 2.81** So just as an example of an exercise that one would do, and this is something that I think it is fun to practice on a regular basis, anytime I am learning about something new, some new area, I will typically spend some time in this area. Most textbooks that are introductions to new areas, the important concepts that need to be learned by the incoming students are defined. In many cases, there are charts and hierarchical tree structures and places to get that person up to speed.
- 2.82** But just as an example, what would we do to make a hierarchical structure that enables us to then define each element's place in that structure? So these are all modes of transportation. And we might say they are all automobiles. We might notice that there's also going to be trains and boats and planes, and so then we would have an even more abstract category of modes of transportation that would be possible. But we're just focusing in here.
- 2.83** So what do we want to do? We start looking at the list, and some things seem to go together, some things seem to differentiate from each other. Sometimes it requires specific knowledge, so you might not know exactly what that means. You might have to go and look up, what does that word mean? So one word that strikes me as a sticking-out word here is this word, import. Those are cars or automobiles that are brought in

from foreign countries. And I know that the contrast to that is export, but export is not on the list here, so chances are good that I'm going to have to put export on the list as well.

- 2.84** Suppose I say that maybe that's what's governing this approach here. We have imports and, I'm going to put this in brackets, exports, or perhaps it is domestic, ones that are made inside this country. I don't know yet. I put the square brackets here because there's a good habit to get into, that whenever you're adding something to an argument or a conceptual structure that's not in the original, put that in there to indicate "I'm adding this." It's a standard device because of course, if you want to be fair to someone else's argument, you don't want to be putting words in their mouth that don't necessarily belong there.
- 2.85** If you find the person, you've reconstructed their argument, and it ends up making them seem like they're saying something ridiculous, it's always worthwhile to go back and make sure you're not adding something that has made it ridiculous. So, I'm adding this as a possibility. Then I would say, well, I know from personal experience, Expedition and Ford, that Expedition is a kind of Ford, but I know also that Ford is made in the United States, so I would put then Ford here, and then say the Expedition as an example here. And now I've got import covered, I've got Ford covered, I've got Expedition covered.
- 2.86** I know also Chrysler is made primarily here, and it's different from Ford, it's a different company. So I would put that there and cross that one out. I might say Toyota, and I know that Lexus is the luxury brand of Toyota, and that Toyota is not made here. So I would put, say, Toyota over here, and Lexus as a luxury sub brand of that one. That crosses out those ones, that crosses out those ones. Tesla, made mostly here, so I would then add that one in the domestic production. That looks after that one.
- 2.87** Volkswagen, BMW, and Accord. I might, from my knowledge, say, well, I know Toyota is made more over in Asia, in Japan. These ones are from Europe. If the geography matters, I might make a distinction—imports from Europe, imports from Asia, imports from wherever. Doesn't necessarily seem important here. So, at least in this first draft, I won't do that. So VW, I would put it here, BMW here, and now I've got Accord. I'm saving that one for last. What about Accord? BMW.
- 2.88** Here what I would say is, I know that Accord is a sub brand of Honda. It's the Honda Accord. Honda seems to be at the same level as Toyota, Volkswagen, and BMW. Those are different companies, but I don't have Honda on this initial list here, so I would consider doing this. Honda, equal level hierarchically with those ones, and then one of their variations is the Accord. OK, rough draft, but that's the exercise to go through, and one clarifies in one's own thinking, one can communicate, and one can leave things open for further additions of knowledge as they come along.

- 2.89 Now, this issue of definition is important work, but one of the things that's always very interesting is the connection between humor and logic and how, in many cases, the best humorists have not only a wicked sense of humor but a wicked sense of logic. They have very keen insights into how we use words, how we think, how we set up people's thinking; where you take them along this direction, you know they're going to carry along in that direction, and then you switch and you take them in this direction suddenly, and it's very funny. But it requires that they have a very sharp logical brain in order to be able to do so.
- 2.90 It also holds at the level of forming definitions. One of the strong versions of humor, of course, is word humor with respect to definitions. I want to give you some examples of this, and we can talk about these definitions. A committee. Well, what is a committee? "A cul-de-sac down which ideas are lured and then quietly strangled." Now, this is put in the form of a formal definition, but it's meant to be humorous. It's also, I think, meant to draw on our knowledge of having been on committees.
- 2.91 What's the idea of a committee? A committee is where we're supposed to get together as a group, share ideas, and then decide what we are going to get done. And so what makes this humorous is, he's aware of that, and he's aware that we're all aware of that, but then he sets up your expectation for something like that, but then subverts it. This is partly a dark truth because we do know that what often happens, against the purpose of forming a committee, is the exact opposite happens. Anybody with a creative idea gets stifled, or the good ideas don't get acted on, hence the bitter humor in this definition.
- 2.92 Cult. What is a cult? "A cult is a religion with no political power." OK. Now, this is an interesting one. It's a witty point but it's got an edge to it. There's some criticism of existing religions that seems to be built into this. Why so? Because typically, we know when we use the word cult, it's meant as a bad word. "Oh, he's joined a cult. We need to get this person out of the cult and remove the brainwashing from this person, get the person back to normal."
- 2.93 So the suggestion then here is, well, what's the difference between those crazy brainwashed people who've lost control of their lives and real religions that aren't cults? The suggestion here is, there's not really an important difference, except that ones have power and the other ones are marginalized and taken unseriously in our society. Then, of course, that means, wait a minute, not my religion, whatever that is. My religion is better than that. But it forces me to then say, well, in what way is my religion better than that?
- 2.94 Can I say, it's not just that my religion has captured a bunch of people and brainwashed them to believe certain things from youth, and we just happen to have a lot of power in society and nobody really questions us? If that's all I've got to go, then this definition has some traction. But if I don't want that definition to have some traction, I need to up

my game in terms of figuring out what makes my religion better than that religion. So it's humor, but with an edge, and it's a clever edge.

- 2.95** "Shaving is a male ritual that consists in spreading a potion across one's face and scraping it off with a sharp implement." See, my fellow guys are not with me right on this one because I shaved. They haven't engaged in this ritual, and maybe they have not engaged in this ritual because they agree with this definition, because the point of this definition is to reduce shaving to triviality and something that seems irrelevant. What's the point of putting some potion on your face and scraping it off?
- 2.96** Of course, the interesting thing is, he's making fun of people who shave, and he's trying to say, "This is why I have my beautiful beard," or whatever it is, because it seems kind of pointless. But for the joke to work, he has to, I've already erased it here, but eliminate the essential thing that really would go into a definition of shaving, which is to say, it serves a certain function of removing unwanted facial hair. That's not in there, but it's, of course, meant to be humorous with a joke.
- 2.97** One more. "Junk is something you throw away three weeks before you need it." Again, in the form of a definition, it's true; junk is stuff that we throw away, but then he's just reflecting on that wry experience that no doubt we've all experienced. I don't think I need this. My judgment was wrong. Ha, ha, ha. So good connections there. I always like to end with this quotation from Confucius.
- 2.98** While it is fun, if you are nerdy and geeky and you just like to have a clean and tidy mind, these exercises are important. They're always very important foundationally, in learning any new discipline. But especially if we want to protect the things that really matter to us, when we get up to talking about religion and politics and relationships and family structures and so on, there, when the stakes are the highest, it's equally important that we have our concepts at their highest level of clarity.
- 2.99** In this lecture, we have been talking about concepts and concepts as abstractions, and then as building blocks for later propositions and then logical structures in theories and narratives. There is an ongoing philosophical and psychological controversy over the nature of these concepts. And built into that controversy is whether our concepts are still keeping us in connection with reality, or if there is something subjective and arbitrary about them, that means we are engaging in a disconnect from reality, right at this level of concepts.
- 2.100** So for example, we would say all of us in this room are human beings, and we would say that we are all equally human beings. I've got this abstract category. I can see each of you as discrete, distinctive individuals, but nonetheless, we share something or we're all part of this broader abstract category, human being. Then the puzzle might be, well, if I then look at you more carefully, what is this abstract thing that we are talking about, this humanness or this humanity? Then I would notice that your hair is all equally, or

unequal, so to speak. You have different hair, different lengths, different colors, and so forth.

- 2.101** You are different shapes, different sizes. You have different weights. I won't do this, but we all have different smells, if we get close enough to each other. We make different sounds. In terms of every thing that we can possibly sense and/or perceive, we are all individual, unique, and different. Then suddenly we are going to this abstract level where we are saying, this one label "human" fits all of you equally.
- 2.102** There is a bit of a mystery, or at least a philosophical and psychological issue, about what we are doing when we go to that level of abstraction, and whether we are maintaining an objective connection to reality. In the history of logic, and the history of this part of logic, there have been three main positions that have developed about this issue, and logicians and psychologists continue to debate them.
- 2.103** One of the most famous individuals is a guy named William of Ockham. He was a brilliant late medieval logician and an epistemologist. But here was his position, and his position was to maintain that there is something subjective about concepts and that they are not objective. So the phrase is, "I maintain that a universal is not something real that exists in a subject, but that it has a being only as a thought-object in the mind."
- 2.104** So that then would be to say, the universal human, it doesn't exist in you. So if I say humanness or your being human is out there in you, that's wrong. He doesn't think that's true. What's out there is a unique particular individual with his or her own unique particular traits. This human, this abstract that we're all equally human, whatever goes into the definition of being human, that's only something in the mind, and we've made it up inside the mind.
- 2.105** That then is to say that we have a gap between the words we are using, as we go on to use these words, and reality. In some sense, we are being false to reality in using these concepts. That takes us in a certain more skeptical direction. This position is usually called nominalism, from the Latin, which is to say, these things are just names, they're not realities, just a bunch of names.
- 2.106** Now, an opposite position is often ascribed to Socrates and then most explicitly to Plato, which is to say, it can't be right that all we are is different names or different objects with different dimensions. We just have these subjective names that we've slapped on to things that are more or less similar to each other. You really are human. You really are human. You really are human. And you are just as human as she is. And you are human in exactly the same way that she is human. That's real. I didn't just make that up.
- 2.107** But Ockham and the others are right that when I perceive you, I don't perceive this abstract humanness floating around. I perceive particulars. So the answer then that Plato came up with is then to say, it must be the case that we have two levels of reality.

We have the level of particular physical reality, where you have your body and your features and so on. And when we are talking about humanness, there is an alternative reality in which there are these perfect abstract, what he called forms or templates, and that we have a mental ability to intuit these perfect abstract forms in the other dimension and see how they apply to their particular entities in the physical world.

- 2.108** We might, if we go back to examples of triangles, we might say, well, here's a triangle, here's another triangle. But if we actually took a really precise protractor and measured all of the angles, is it going to equal 180? Well, the Platonic answer is going to be probably not. This one might only measure 179, and this one's 180.5 degrees. You're never going to find a perfect triangle in this world. Everything has little bits of imperfection.
- 2.109** But there is a perfect triangle, perfection. The definition is perfect. So there must be triangularity in this higher world of forms, and that we have a cognitive capacity for seeing it or perceiving it or intuiting it that works with our perceptual capacities and so on. Now, this position is sometimes called realism, the idea being that abstract things are real and independently out there in reality, just not in the lower physical reality, in a higher physical reality.
- 2.110** Then the third position, more contemporary, Ayn Rand, wants to argue that neither that realist position nor that nominalist position is correct. And this is her description of it from her work in epistemology. "None of the traditional theories of concepts regard concepts as objective, i.e., as neither revealed nor invented." Let's just pause right there. So the Platonic position where we say abstract concepts are as an act of revelation to us, they're out there and they are revealed to us, the nominalist position is to say, no, we just make them up. They are inventions. Our position is to say those two traditional theories are wrong.
- 2.111** "But as produced by man's consciousness in accordance with the facts of reality, as mental integrations of factual data computed by man, as the products of a cognitive method of classification whose processes must be performed by man, but whose content is dictated by reality." So that's a lot of words, and we're just introducing what the issue is. But what she wants to argue is that it is objective, but when we are observing, and I see all of you in your particularity, and I'm sorting similarities and differences, all of that is objective, and you all have the features that you have.
- 2.112** But nonetheless, it's not just revealed to me. I have to engage in a certain process of abstraction. She wants to make a distinction between the process that the mind engages in and the nature of the reality that one is abstracting from, and that's the proper understanding. Now, that's just to describe three positions that smart philosophers continue to debate over the nature of concepts.
- 2.113** But I want to have that as a part of the record for the course in logic, because how do we think about these concepts? Did we just make them up? Should we strive to be

intuiting perfect forms more better? Do we need to attend to these conscious processes that we are engaging in, in a way that keeps us connected to reality or not? The further research and thinking about concepts stems from what position one takes on that controversy. Let's pause there.

- 2.114 As far as, we talked about the concepts of reality in this debate, does postmodernism fit into that concept, as far as logic? Because they don't really think of reality as—
- 2.115 OK, good. That's a great question. I used Plato from the ancient world and Ockham from the medieval world, and then Rand from contemporary times. But it is interesting that you find all three of those positions articulated in each of those eras. So the postmoderns whom you are mentioning would be in the shoes of William of Ockham. They are nominalists in that conceptual sense.
- 2.116 So Derrida, Foucault, and the others, they don't say, yet the words that we use, all of the language that we use is subjective. We're just arbitrarily carving up the world however it is that we want. Although typically they want to say that we ourselves, the way we're trying to carve up the world has just been conditioned into us by power structures who themselves have just arbitrarily decided to carve the world up in certain ways for their own value agenda. They are extreme nominalists, extreme subjectivists.

Lecture 3: Logical Fallacies

At a glance — Logical fallacies and good versus bad reasoning.

- 3.1 In logic, we like to emphasize the positive, good reasoning, valid reasoning, sound reasoning, keeping our minds in connection with reality, and figuring out the truth. But also, of course, we know we are beings that can make mistakes, all sorts of mistakes, and particularly on things that are complicated and things that push our emotional buttons. It's customary in logic courses, and I think for very good reason, to spend a certain amount of time looking at common patterns of invalid reasons.
- 3.2 Typically in logic textbooks or logic courses or wherever one gets one's material, these will go under the label of informal fallacies. They're called informal fallacies because they are fallacies, which is to say they are mistakes in reasoning, where the premises seem to provide some support for the conclusion, but they don't provide any support for the conclusions at all. So it's fallacious reasoning. But informal because there are different kinds of logical forms.
- 3.3 You can be doing deductive categorical syllogisms. We'll get to those in a little bit. You can be doing propositional logic, you can be doing inductive statistical, whatever form of reasoning you are doing. There are some common patterns of bad reasoning that pop up all over the place. So it's worthwhile to isolate them since they are so common, assign them names, and then just the act of going through that and doing some practice examples puts you in a position of intellectual self-defense.
- 3.4 Sometimes, with respect to yourself, it's very easy for us all to slip into certain patterns of reasoning, but also in social context, so that other people don't slip one past you. In many cases, there are people who are con men, the lawyers who have the bad reputation, who know certain tricks of the trade, so to speak, how to use illogic in a way that seems to be persuasive. So it's worth being on one's guard against those.
- 3.5 So the basic idea of logic is to maintain an objective connection. That is to say, our minds are trying to attend to the objective reality. So what goes on inside the subject, I'm not just making up the truth or just believing whatever it is that I want to be true. I'm not being subjective, in that sense of subjective; I'm trying to be objective. But subjectivism is the first one that we typically give a label to.
- 3.6 So subjectivism and, in its crudest form, is just to say, I want something to be true, and so I'm going to believe that it's true, or I'm just going to will myself into believing that something is true, even if I don't have good evidence for it. Even if I know there are some evidence that militates against it or leans against it, I'm just going to not pay attention to that evidence because, ultimately, I want this to be true.
- 3.7 Now, if we put that in standard form, it would just be looking like this. I want X to be true, and so therefore, I will believe X is true. Obviously, we call that subjectivism. The

reason for that is to say, well, what is the evidence for X as being true? It's just my wanting. That's just something that is with respect to me. Now, almost never does this happen so clearly. Even people will try to deceive themselves that this is the policy that they are operating on. But nonetheless, you can detect in yourself and in other people when this is happening.

- 3.8** Let me give you some examples to work on. Sometimes we have something with respect to our bodies, something acts up as one gets older or whatever, and something in the back of your mind says, "Maybe I should go to the doctor to get this checked out." That's a reasonable thing. Evidence something is wrong, doctor is expert at fixing things, I should get this checked out. I don't know what it is.
- 3.9** But if what's going on is, I don't want to go to the doctor; maybe I'm too busy or I don't feel like going to the doctor, or it might turn out that the doctor gives me bad news, and I don't want to hear bad news from the doctor. So what one will then do is just going to push out of one's mind the pain that one is suffering or whatever the symptom is, or whenever the idea of, "Go to a doctor," no, I'm not going to let myself think, "go to the doctor." And that's a policy of subjectivism.
- 3.10** In that case, it's like, I don't want there to be something wrong with me, so I'm going to act and believe as if there is not anything wrong with me, when the proper policy would be to pay attention to the evidence. If the evidence is that there's something wrong, get it looked after by someone who has the qualifications to do so. Now, very interestingly, there's an interesting connection between the logic and psychology here. Just put this out that women are much better at being objective on these issues than men are on these issues.
- 3.11** There are some studies that seem to indicate, keep the symptoms similar, women have those symptoms, they will get themselves to the physician much more regularly. Guys will be more likely to shrug it off. It's just a flesh wound. Maybe it's an issue about male ego. "I'm a tough guy," and you don't want to think of yourself as someone who gets damaged, for whatever the reasons are. There might be psychological sex differences that lead to subjectivism being easier or more difficult for us as individuals.
- 3.12** If I can be in true confessions mode, for example, I can look back and laugh on this, but when I was a college graduate student, I went on a road trip with a buddy, and we were driving across the United States, and we were in the middle of Nebraska, which I always think of as the state that never ends. You just keep driving and driving, and there's more and more Nebraska. And as poor college students, we had a car that probably should not have left the original city that we were in, so it was not clear it was suited for going that far.
- 3.13** But sure enough, we're out in the middle of nowhere in Nebraska, and there starts to be a sound happening from the engine. And you know that's not the right sound. What I did, I was driving at that point, was reached over and turned up the music. So this is a

policy of subjectivism. In this case, I don't want there to be anything wrong with the engine; and if I can't hear anything wrong because the music is so loud, then I don't have to think about it, I don't have to believe it, and so on. OK, so kind of a ridiculous example, but that's an example of it.

- 3.14** More serious cases of relationships, and we all likely have had failed relationships, and in many cases, after the relationship has failed, we say to ourselves, "I should have known better." There were lots of red flags or lots of indicators. But we do know at the same time, as part of human psychology, when we're in the early stages of a relationship, everything is easy, everything is wonderful, and you want the relationship to work. You're always looking for the good in the other person. It's very hard to get yourself to be critical and notice possible weaknesses, possible red flags.
- 3.15** If some of those do start to arise, it's very easy for us to not pay attention to them, shut them down. Even in the case of well-meaning friends or parents who might say, "I know some things about this person," trying to be gentle, you will shut that person down because you don't want to know certain things, given that you are in a certain psychological state. Now, again, some people are better or worse at it, but it's a set of skills that happen, and subjectivism is dangerous.
- 3.16** Relationship issues, religions. People often know there are issues in their religion they should think about, perhaps tidy up their religious beliefs, but they don't want to, or their political candidates or their political philosophies. "This matters to me. I really want my politics to be the right politics. I want this to be the right person. I want this to be the right religion because I'm already invested." And so, subjectivism is an occupational hazard constantly in all of those issues.
- 3.17** In business cases where you are trying to make a big decision, you're going to be putting a lot of money out there, but you came up with an idea, and it's your idea, you worked out the business plan, you really want it to work, it's easy for you not to take criticism well in that situation, and so you lose a lot of money. So the policy is, while emotions are wonderful, emotions are important. There's a huge place in our lives for emotions.
- 3.18** But deciding what's true and false, that's not the job of emotions. Since we are subject to strong emotions and we want to be subject to strong emotions, we have to manage our internal dialogue to make sure that we're not letting our subjective desires interfere with our objective assessment of the evidence. Ongoing challenge.
- 3.19** Now, second fallacy that commonly gets put on the list is the appeal to large numbers of people who believe something. So the label for it is the appeal to majority. And in its crude form, it is, I'm going to believe that something is true because lots and lots of people believe that it is true. Sometimes it's a little narrower, and that's a lot of people in my peer group believe that it's true. So I'm going to believe that it is true as well.

- 3.20** And this is a fallacy for the most important reason that, if what you have only is that other people and lots of them believe something to be true, that doesn't tell you in your own mind whether it is true or not. You don't have access to whatever thinking they did or did not do to make up your own mind whether the thing is true or not. So many people believe X, therefore I believe X. So it is a failure of independent thinking.
- 3.21** Other people believing that something is true can be a clue that maybe there is something to it. You might ask them, why do you believe that? What's your evidence for that? What's the reasoning for that? Why did you come to believe whatever it is? Then you go through the same process and see if you reach the same conclusion of it. Or, of course, if you disagree, you have to, have to disagree.
- 3.22** Now, if you don't do that, then another reason why it's obvious this is not a good policy of mind is we just look at all of the historical examples where lots and lots of people throughout history have believed lots and lots of things that are wrong, ridiculous, crazy, destructive, and so on. So one of the lessons of history should be, it might be, if it is the case that there's just a huge mob of people suddenly latching onto some idea or it's very popular, it might very well be wrong.
- 3.23** So the famous examples are ones that we laugh about now, or at least we roll our eyes about, involve things like people believing that the Earth is flat. And for many people, it would be just something they picked up by osmosis. Everybody else seems to be suggesting it, but they've not actually spent any time thinking about it, asking what would count as evidence for or against the Earth being flat or not. They're just carrying on someone else's belief.
- 3.24** How do we find out if a woman is a witch, if we up the stakes a little bit? Well, what you do, everybody believes this, and there are dozens and hundreds and thousands of trials conducted along this way, is you take a woman to a body of water, you tie her to a long pole that has a fulcrum, and you dunk her so that she goes underwater for a long period of time. And if she is not a witch, this is partly in a certain kind of religious context, God will intervene to show her innocence, and he will protect her while she's under the water.
- 3.25** On the other hand, if she's a witch, she will also engage in various witchery type of things to protect herself. So if she can survive this repeated dunking, it probably shows that she's a witch, in which case you have to burn her at the stake because you've proved that she's a witch. And unfortunately, if she doesn't survive, maybe God didn't just choose to intervene. She was innocent and just a normal woman and not a witch. Well, too bad. But that's a destructive practice based on just following a practice and a belief that seems to be widespread in that cultural atmosphere.
- 3.26** But my favorite historical example involves the early days of electricity and magnetism, back in the 1700s, when you can take this thing that just looks like a piece of rock and then take some iron, and then somehow they attract each other. We don't know what

the causal mechanism is. It seems a little bit mysterious and magical. But many humans, being what they are, they then start to ascribe all sorts of things to these magical powers. It's like the crystal bracelets and this, that, and the other thing of our own time. But one of the very common beliefs that started to sweep through early America at the same time was that the magnets had the power to determine and tell you if your wife was cheating on you or not.

- 3.27** So the test was, you would take one of these magical magnet things, and of course, you have to pay me a lot of money for this, but you take this home, if you were worried about your wife, and put it under her pillow at night. And if she is an adulterous, the magnet will know, and it will drive her from the marital bed, and then you can take the appropriate consequences. Everybody else is doing it. Where's my money? I need to get one of these magnets. The fact that lots of people believe it doesn't provide any support at all. Let me put a box around this one, so I can put another box around this. This is the appeal to majority.
- 3.28** Now, as with the subjectivism, when you start putting it out clearly and logically like this, it seems ridiculous, and it is ridiculous. But there is the question why so many people seem to commit the appeal to majority so frequently. Peer pressure, going along with the crowd, and just saying things like, "Well, you don't want to believe that. Nobody believes that anymore. Everybody else is doing this." There does seem to be a psychological element that makes us prone to believing whatever the majority of people want to do.
- 3.29** Now, it might be just laziness. I don't want to have to think about these things. These other people seem to have thought about it, so I will just rely on their judgment. But it might be something else. There's some very interesting psychological experiments that have been done on this issue of committing the appeal to a majority for conformist reasons. The most famous name associated with this is a psychology professor named Solomon Asch. I want to spend a little time just showing this experiment.
- 3.30** There's lots of variations on Asch-types experiments over the subsequent decades. I believe these were first done in the 1960s and different types of constituents and variations on the theme. But the bottom basic variation was this. Professor Asch was teaching at a university, has lots of students, and he uses his students as subjects in these experiments. He says, "I'll pay you a couple of bucks." He would tell them a story.
- 3.31** He says, what we want to do is bring you into the lab and do tests of visual acuity. We're going to show you some lines and shapes of different shapes and sizes. What we want you to do is tell us whether these, say, two lines or two shapes are the same, or if you can see that there's a difference between them. Sometimes the differences will be really blatant, but sometimes they will be quite subtle. We're testing people's ability to discriminate subtle perceptual differences.

- 3.32** Here's a big whiteboard, say something like this one here. It's on a stand. We're going to bring the—that's a chair. We're going to bring the student in. Years of art lessons paying off right there. Just imagine that the guy's head is facing that way. Then we got some projector back here, and we're going to put projected images. This guy just has to say, "Same, different, same, same, different, different," right, and so on.
- 3.33** But we have two other students. We say, well, we're going to do this so that we gather more data at the same time. Also here, so these guys will be sitting here as well. This person is the test subject. We're not really interested in visual acuity. These two people look like students, but they are working with Professor Asch, and they are told to make mistakes in their answers at certain key times.
- 3.34** What we want to do is see when this person can see whether they actually are the same or actually are different, but these two say something else, what does this guy say? Does he go with his own independent judgment? I can see. Or since there's two of them and only one of him, and they're saying something different, will he say what they are saying? So, I'm going to say, I don't know, we have a secret Professor Asch commitment there.
- 3.35** So on the board, we start, say, putting two lines. I'm just going to keep it as lines to keep it the same. So we have two lines. Are those lines the same or different? These people can see, everybody can see, they are basically the same. These guys say the same. This person says, "Oh, they're the same." And then we do the same lines, different orientations, some shapes, things are the same.
- 3.36** Then we start to mess with the guy a little bit. We make this one just a little shorter, but we know that people from this distance can see that difference. These two people are quick to say at this point, "Oh, they're the same." The question is, what does this person say? Majority of cases, this person will say, "They are the same." He will agree with these two test subjects, but you know that he can see that there's a difference. Majority of people, and sometimes it's girls, sometimes it's boys.
- 3.37** There are later experiments where you have people of different age groups who count as being part of your peer, different racial groups or ethnic groups, and so on. The common data is that in the early stages of this experiment, this person will say something he or she knows is false. The only reason seems to be that to say something true would put him in a position of disagreement with the majority of people. Most people are not willing to express disagreement with the majority, at least in this experimental setting.
- 3.38** These are standard American kids, most of these initial experiments, but a lot of them have been duplicated around the world. It makes it even more pointed if you start to bring in cultural issues, because typically Americans have a culture of saying, "No, just believe whatever you want, think for yourself, be your own person," and so on. We don't have a culture that, on the face of it, compared to many other cultures,

encourages conformity in going along. We seem to have a culture that encourages being independent and so forth.

- 3.39** But even in such a culture as that, a majority will still do this. All of this is being filmed by some hidden cameras. You'll mess around even more. You'll make that line even shorter. At some point, the line gets short enough. These people will start to defect and say, "No, it's different." Even those two continue to say that it is the same. The interesting thing is, when you look at some of the films, you see the people's faces, looking at the screen, the images. "I can see that clearly they are different. I'm ready to say different. Then I hear this guy says same, this guy says the same."
- 3.40** There's that whiplash head. "What am I hearing?" Then you can see the face change, this anguish. "What do I say?" Then at the point where they are willing to defect, they apologize. They'll say things like, "I'm sorry, but they're different." So something psychologically is going on. Again, this is not just isolated individuals. You're at the point where you have to apologize for stating something publicly that is really obviously true.
- 3.41** If it's difficult for many people in these cases, then when we get to the more complicated cases, the more controversial cases, that might be then part of a psychological explanation for why the appeal to a majority is such a popular fallacy, and we see it all over the place. Which then means, by implication, if our goal as logicians or people who are committed to logic is, monitor this in myself, of course.
- 3.42** But if I'm a parent, if I'm working with other people, how can I encourage them not to be lemmings or conformists, such that they are willing to speak the truth when they see the truth for what it is and not to start lying publicly, which, of course, sabotages your own psychological states as well?
- 3.43** Next fallacy I want to turn to is a variation on the appeal to majority, and it is called the appeal to authority. As the name suggests, the crude version of this one is to have not just your peer group or lots and lots of people, but one or a few people who are authorities or have some reputation. They have a PhD in something, or they are the CEO, or they have some fancy title or claim to fame. And this argument then says, well, this person believes X or says that X is true, and then, therefore, I believe that X is true.
- 3.44** So often it will be the case, an English literature professor will say, "Here are the five books that are the greatest books ever written, and you ought to read those books." And so many people will then say, "Well, I want to be considered an educated person just like this professor, and so I will make a point of reading those books, or at least saying that I've read those books, or know what to say about those books because I want to be in the more elite crowd."
- 3.45** Or the president believes a certain piece of legislation will pass, or in some versions of parenting, parents will encourage this: "Children, don't think for yourself so much. I'm the parent. You should believe what I tell you as the parent." Now, in that case, it's not

a willing obedience to the authority person. It's an enforced obedience from the higher authority.

- 3.46** Let's put it formally on the board, the appeal to authority. All of these fallacies also have Latin names. This one is *ad verecundiam*, but we'll just stick with the English, appeal to authority. The blatant form of it is, some expert says X is true, and then I believe, therefore, X is true. Obviously, it's a fallacy in one respect because it's a failure of intellectual independence. The fact that the expert might properly know that something is true does not put you in the position of knowing why it is true. You're just taking it on faith that this other person says so.
- 3.47** Other obvious reason why this is a fallacy is that almost all important controversial issues, again, we can appeal to history. People who are experts have been wrong, believe all kinds of ridiculous, crazy, sometimes not ridiculous or crazy, but nonetheless, wrong things about various important matters. If everyone's policy is just "whatever the experts of our generation say, that's what we're going to believe," then we're setting ourselves up for stagnation.
- 3.48** It's also the case, as we know, even in our own generation, on almost every important controversial issue, there are experts who disagree with each other. And so, appealing to experts is going to be a nonstarter strategy for you actually figuring out what's going to be true. Nonetheless, this one is a favorite of the advertising agency. There's a whole genre of advertisements where they'll have experts in one field, but they will bring them in nonetheless to promote something in some other field.
- 3.49** So I don't know, you get some, what would it be? I don't know. You get a famous actor to say, "This is the best beer in the world," holding a glass of beer and looking satisfied. Well, this person might have expertise in acting and be a famous actor, but that does not translate into expertise about beer-making and the quality of beer. Certainly from the perspective of the audience, say, "Well, I don't know why this person thinks it's the best beer, but I'm just going to buy that beer." In that case, it's an appeal to authority, but it works on lots and lots of people.
- 3.50** We'll come back to, in a moment, what might be going on psychologically, but we'll flag that just for a moment. Now, that then is to say, sometimes the appeal to authority happens in celebrities, movie stars, or famous people endorsing things or believing things. So whatever the global issue is, you bring in a famous musician who's on some side of a political issue on the other side of the world—the idea being, this famous musician has lots of people and is an authority for those lots of people who like that particular music. And so their opinion on this foreign policy, politics issue, is supposed to have influence with those people.
- 3.51** Now, the more complicated cases, where *ad verecundiam* or the appeal to authority happens frequently, is though when it's an expert who has authority in his or her field, making a pronouncement in his or her field. So it's not the celebrity endorsing

deodorant. It is maybe a celebrity telling you about acting techniques or a musician telling you about their creative process and what you can do to become more creative in your music.

- 3.52** Or you go to people who are experts because you know they have knowledge that you do not have. You get sued on some matter of a business issue, and there's lots of complicated law at stake. You don't have time to get a law degree. You go to a lawyer, you pay that person the money, and you ask that person for expert advice. So, that lawyer might then say, "X is true. This is the best strategy. This is what we should do. These are the arguments that we should use." It is, of course, tempting and to some extent understandable why so many people will say, "OK, fine, you're the expert. I will just do whatever it is that you say."
- 3.53** Other analogy, or other example rather, is, again, there's something wrong with you. You don't have time to get a medical degree. You can poke around online to get some information, but you go to the physician, and the physician says, after examining you, "This is what's wrong with you. This is what I think we should do. Here, take these pills or have this surgery or whatever." You know that person has expertise. It's understandable then to say, "OK, I will take those pills, I will get that surgery," and so on.
- 3.54** But even here, I want to say it's still a fallacy if you act as described so far, that what logic will say, as a matter of intellectual self-defense, as a matter of self-responsibility, is even when you are talking with your lawyer, talking with your physician, you should ask them why. It can't simply be, "My doctor said that this is the case." The doctor says, "This is what you have. This is your condition." You should ask, "What makes you think, or what symptoms are you saying?"
- 3.55** The doctor should be able to, in ordinary person's language, say, "Well, you have this symptom and you have this symptom and you have that symptom. 94% of the time, people who have that symptom, this is the condition. But it might also be this other thing, and we might want them to check it out. Since you have this problem, we are going to do this surgery." You say, "Well, why that surgery? What are the other alternatives? Or what are the other options at this particular case?"
- 3.56** The physician should be able to say, "Well, there really only are three options that we can do here. I settled on this one because it has an 85% success rate. This one only has a 40% success rate. This one has a 30% success rate. Here's the literature you can go and have a look at and so on." Now, at that point, you are still going to be trusting your physician up to a certain point, but he or she has explained a huge amount and is putting you in a position to make a much better independent judgment about whether the policies are appropriate or not.
- 3.57** But even then, in a lawyer, if it's a really big case, or in a physician, if it's a really important issue, a part of responsible self-thinking and self-decision-making might be

to say, "Well, I am going to get a second opinion. That makes sense. I can see where you're coming from, but I still have some questions. I know sometimes things are cutting-edge. Sometimes people have narrower understandings of the literature, even if they are experts in that field. I'm going to get someone else."

3.58 Maybe your doctor is 60 years old. I'm going to get a 30-year-old doctor. Maybe the 60-year-old doctor isn't keeping up with the newest literature, but the 30-year-old doctor is more likely to have read that. On the other hand, the 30-year-old doctor doesn't have as much experience as the 60-year-old doctor. So maybe the best on this very serious medical issue is to do a compare and contrast of both of them. That's to avoid ad verecundiam.

3.59 Now, obviously, that's a lot of work, and it requires that you absorb a lot of information, a lot of information processing, making some important decisions. Maybe one of the reasons why people appeal to authority a lot is just a laziness. They don't want to do that work. Maybe it's a matter of social issues. You don't want to seem critical of this person in authority, like you're being a little bit rude by asking them hard questions, asking them to justify themselves. So it might be some social pressure, but override that. You can always find a diplomatic way to ask the important questions that needed to be asked.

3.60 As with the appeal to majority, there do seem to be other psychological issues that might be involved, aside from the certain kinds of emotional issues or the laziness issues that we've studied or mentioned so far. This also has been subject to psychological study in the form of formal psychological experiments, to try to test this appeal to authority. The most famous name in this area is another psychology professor whose name was Stanley Milgram. Again, experiments done back in the late '50s or the 1960s, subject to a certain amount of replication, lots of variations.

3.61 These ones are trickier because they start to run afoul of human ethics experimentation boards. They have to be more subtle now about how we're going to be testing for appeals to authority. But nonetheless, let me give you a picture of the kinds of experiments that Milgram did to illustrate that, in addition to this being just a strictly logical independence of judgment issue, there might be other psychological factors at work.

3.62 Now, in this case, think of a two-chambered room. Perhaps a close analogy would be a police interrogation chamber. We've seen them in the TV shows, in the movies, where you put the prisoner or the suspect in this room. And then here you've got, let me see if I can draw it very lightly. Imagine that's a two-way glass, where people in this room can see into this room, or it's actually one-way glass. You can see into that room, but people in this room cannot see back into this room. They just have a mirrored image. So I'll put a little arrow just to say that that's only that way.

- 3.63 In this room, what we have is a chair, and we've got a machine. On this machine, we've got a big electrical dial and, in big letters, say the word volts. This part of it is all in red, and there's danger if the voltage gets up to that point. There's an indicator of what your level of voltage is.
- 3.64 What Professor Milgram does in this case is, again, he has a large student body, he says, "I'll pay you a few bucks if you come and you work with me in the lab. I want you to be my laboratory assistant." And what he tells them is, "We're going to bring people in and we'll sit this guy here and we're going to attach him to the machine." So there's wires that come out, and they hook up to this guy with the electrodes at various sensitive points of the guy's body. There's also, one more color, there is a switch here for on, and then if you put over there, it will be off.
- 3.65 What we're going to do is, "I'll explain that we have good reasons why we're going to do this," I'm Professor Milgram right now. But we're going to turn this on and give him electrical shock sometimes, and other times we'll turn it off. But what we're trying to do is, we're trying to see if people's learning abilities, their memory, their attention improves through exposure to mild amounts of pain. So we give people tests, we ask them all kinds of questions, and we see how well they do on those tests.
- 3.66 And then we give them a very similar test, but every time they get the wrong answer, we give them a little electrical shock, and we see if that causes them to pay more attention and to do better on the test. So I want you to work with me to do this test. So we have some observers over here. OK. And that's going to be, say, Professor Milgram. And he's got to, I'll give him a hat because he's an important guy. And there's his clipboard, right, with all of the questions. And he's going to be speaking through some loudspeakers and so on.
- 3.67 But what I want you to do as my student assistant on this is to be in the room with this guy, and I want you to be in charge of the switch. So your job is to control the switch. And we'll have a microphone hooked up, so I'll tell you when to switch or when not to switch. We're going to ask this guy a bunch of questions. When he gets the answers correct, no problem. But once in a while, when he gets the answer wrong, I'm going to ask you to turn the switch from off to on and deliver a little shock to this guy.
- 3.68 Now, initially, it's on very low setting. It's going to be mild shocks and so on. Now, full disclosure, behind the scenes, I would not tell you this if you were this person here, but this person is an actor who has been hired by Professor Milgram to give wrong answers at prescribed times, get the right answers at various times. And in fact, there is no voltage at all going along here. But this person, you don't know that. And so our question is going to be, when I give you the instruction to turn that switch on, what will you do? So we've controlled all of the variables, and that's what we are interested in seeing.

- 3.69** So let's start the experiment. Professor Milgram asks, and the voice comes over, what is, let's just do some warm-up questions, what's 2 plus 2? This guy says 4. What's 13 plus 8? This guy says 21. What's 16 times 47? 184. Professor Milgram says, "Sorry, that's wrong. Please shock him. Or just please turn the switch on." What does that person do? Vast majority of cases turns the switch from off to on. And this person is the actor who knows it's coming. You can see he gives a little twitch.
- 3.70** Let's turn it back off, carry on the test. Who was the author of Moby Dick? Herman Melville. Who wrote House of Seven Gables? James Joyce. Wrong. Shock, please. Shocks him again. Now, we play around with this. At one point, Professor Milgram then says, "Would you turn the dial up? We want to have more." People will continue to shock the person who gets wrong answers.
- 3.71** Now, as you might expect, as this goes up and more and more, these people start to defect. "I don't want to do this anymore." But there are a significant minority of people, again, mostly young Americans, but adults, this will get into the red zone. This person is crying. This person is saying, "Ouch, that really hurts." This person is saying things like, "I don't want to do this anymore. I want to stop." And Professor Milgram will say, "I just have a few more questions," and asks those questions, and a significant minority will continue to shock.
- 3.72** Now, why is this? That's the question. And it's not because they are sadists. They've just been waiting their whole life for a chance to shock people for getting wrong answers to trivial questions. We've isolated as many things as we possibly can, and it seems to be, "This guy's the professor. He's running the lab. He's my superior on this order, and he said I should do this." So I, in a way, stop thinking. Stop thinking I'm a responsible agent, the one who's making the decision here. I am a, maybe overstating, I'm a robot, just doing what the real power and the real decider makes.
- 3.73** So very disturbing set of experimental results from Stanley Milgram. And this is on the psychological side of things, because anybody, if you just stop them on the street, all of these people, if you said, "If someone doesn't know who the author of the House of Seven Gables is, do you think it's appropriate to run electrical current through their body?" All of them will say, "No, absolutely not. That's sick. What are you, a weirdo? I would never do that." But put them in this situation, and significant numbers of them would. These are not bad people, but under this set of circumstances, authority is seductive in some way.
- 3.74** These all have English names. They do have Latin names behind them, but there is one of the very common fallacies that the Latin name has stuck, and we don't use English variations of it very often. When you hear people talking about ad hominem, it's this particular fallacy. The Latin simply means that, instead of attacking the person's argument, you attack the person. Or instead of attacking the ideas, you attack the man, or you attack the woman, you attack the person. Ad hominem is the argument.

- 3.75** Again, this is something that happens very frequently. Someone will say, "I believe that X is true, and here are my reasons for it." And then the person who is responding will not attack or pay attention to the reasons. Instead, they will attack the person personally. There's lots of versions of doing so, like old-fashioned sexism, for example. If it's a woman who's making an argument, you don't want to have to deal with it, then the old sexist response is, "Well, she's just a woman." And so you just dismiss it on the basis of that, when obviously as a woman, she can have really good reasons for her argument, and you have an obligation to attend to those arguments.
- 3.76** One of the very frustrating things we all know on social media these days is how quickly people will say, "Well, you're a member of this race, or you're a member of that religion, or you believe in that politics." And even though the person is making a point that's a point worth paying attention to, has data, has arguments, and so on, the person is not willing to address those. They want to just dismiss them by dismissing the person.
- 3.77** So, the way this argument goes in its crude form, ad hominem, is, just to say, X says that Y is true, but X is full in the blank with whatever thing you don't like about that person. Therefore, Y is false. It might be that older people want the kids to shut up. So they'll say, "Well, you're just a kid. Wait until you are older." Even if, as kids, as we know, can have good ideas, good arguments, be thinking through various things. So just dismissing their say-so on the grounds of their age is another form of ad hominem.
- 3.78** Now, the way this often happens is in charged areas, as we might expect, politics discussions, religion discussions, family debates that get out of hand and various so on. But let me give you a couple of examples from business. Here's one. We're arguing over taxes. Should they be raised, kept the same, or lowered? Here's an argument that says, "Business professionals say the economy would improve if taxes were lowered." Presumably, they've given some arguments for that. "But of course, people in business just want to make money, so they would say that." And so we have that sarcastic tone in there.
- 3.79** Now, presumably, there are arguments about the taxes. If taxes were lowered, this is what would happen to the economy. Maybe it's good data, maybe it's good argument, maybe not. But if all we're doing is saying, we're not paying attention to your argument, we're saying you have a motive, you're a business person and you want to make money, and that's supposed to discredit your argument, well, it doesn't. It's irrelevant to the argument.
- 3.80** Nonetheless, we do know that, and this might be part of the reason why this argument is so common, if you have a constituency of people who listen to you, who don't like business people, who are already on the other side of the taxation argument, if you can find a way as to attack this person, get them not to pay attention to that person, it might work, but it's not working logically. It's working through cheap and dirty tactics.

- 3.81** Another example I hear a lot in philosophy, one of the great giants in the history of philosophy is Friedrich Nietzsche. Everybody reads Nietzsche and gets excited by reading him. Just so that everybody finds some things in Nietzsche that are incredibly outrageous, but other things that are incredibly penetrating. I would agree with most philosophers that he's worth spending some time with.
- 3.82** But he was famous for being harshly critical of religion in general, but particularly the forms of religion that were extant in the 1800s during his time. There's a lot of really interesting arguments there and a lot of really important arguments for religious people and non-religious people, too, to pay attention to. But it will be a mark of a person who's not interested in the issues seriously to find various ways to say, "I don't want to have to read Nietzsche. I don't want to have to be dealing with counterarguments of Nietzsche. It makes me too uncomfortable."
- 3.83** Maybe it's subjectivism. I want to just believe what I believe, without having to deal with this philosopher as well. So, the mark of that kind of person pops up very regularly in social media, and even in semi-professional philosophy discussions, is the biographical fact that toward the end of Nietzsche's life, he did go crazy. He became intellectually dysfunctional. One of the leading theories was that he contracted syphilis at an earlier age, and syphilis can do nasty things to you because there wasn't a cure for it at that time. That's not the only hypothesis.
- 3.84** But the faulty argument, or the ad hominem, then would be to say things like, "Well, yeah, Nietzsche has all of these arguments about religion, this, that, and the other thing, but he went crazy. So we don't have to pay attention to that guy." And so his later craziness, which is not relevant to his earlier arguments, and the arguments stand on their own, and even independent in the Nietzsche case, we know that sometimes people who are a little bit eccentric or a lot out there can nonetheless still come across things that are interesting, important, and you have to attend to them on their own grounds.
- 3.85** Another common version of this one is, and this might explain part of the psychological attraction of ad hominem, is pointing out hypocrisies in other people. So someone might be making an argument for something, but then you notice that in their behavior, currently or perhaps in their past, they've not been saying things or acting in a way that's consistent with what they are arguing for right now. And you will then notice that and then say, "Oh, you're just a big old hypocrite. Therefore, this argument is no good."
- 3.86** For example, it might be, say, that I'm running for political office, and we're having big debates about public education. Should we reform public education? Should we have more private schooling? But suppose for whatever reasons, I am an advocate of public education, but I'm approaching it seriously as a politician. I'm saying I'm doing these studies, I've got these arguments for public education. I'm not necessarily opposed to

private education, but I'm running on a platform of strengthening the public education sector.

3.87 Now, this happens, but you might then say, "Oh, well, you, 20 years ago when your kids were going through school, you didn't send your kids to public schools. You sent your kids to private schools. So you're saying you're in favor of public schools, but you sent your kids to private schools." So there's an inconsistency there that is worth raising. Why did you send your kids to private schools?

3.88 It might be that it was 20 years ago, different context. Maybe I had different beliefs, back 20 years ago. But if the cheap shot is just going to be, "Well, there's an inconsistency in what you say now versus what you said then, therefore, I'm going to ignore your arguments now on the benefits of public education," that would be to commit ad hominem.

3.89 But we're going to see in political campaigns when people are emotional and they're looking for the cheap shot, why an argument like that would be attractive. There also, I think, is something about the human condition when we are monitoring ourselves and certainly monitoring others, that we seem to be very attuned to hypocrisies or inconsistencies in what people say versus what they do, what they say now versus what they say then.

3.90 Even in little children, if you have any parenting experience, they're very quick to pick up on the slightest inconsistency in policy, as announced or practiced by their parents. It might be that this is something that we are wired, psychologically, to be especially attentive to and to be especially irritated by it, which makes this argument very attractive.

3.91 Another example might be, I don't know, suppose you go to your physician and your physician says, "Really, we weighed you and this, that, and you seem like you're carrying a few extra pounds. You should go on a diet." But then your physician is sitting there, belly hanging out right over the belt. And you might then say, well, you know what the temptation is going to be. But don't give into that temptation because it may very well be the physician is exactly right to say you are carrying a few extra pounds, and there's good reasons why you should pay attention to this medical advice.

3.92 Whether I am doing a good job of maintaining my weight is an important issue for me, but it's not an important issue for you. The fact that I might be overweight does not undercut the value and the legitimacy of my arguments for you being there. I want to do one more in this session, and then I want to turn to a controversy and a historically relevant controversy.

3.93 This other important fallacy is called the appeal to force. In Latin, *ad baculum*, an appeal to a stick. So, metaphorically, the idea is, you're engaged in a discussion, you're engaged in an argument with someone. I'm supposed to be presenting my reasons for my belief. You're presenting your reasons for your belief. That's the right way to go. But

I introduce a threat into the discussion, or I introduce some force element into the discussion. The idea then is, since I am threatening you, you will start to say different things or you will start to behave differently.

- 3.94** This is a "might makes right" approach to argumentation, or saying that rather argumentation doesn't matter, rather who has the power matters. That's how we are going to decide socially. The appeal to force, or *ad baculum*, it's just X is true because I say so, backed up with threat. Sometimes the threat is veiled, sometimes the threat is explicit.
- 3.95** Organized crime, one of the standard things, we see this in the movies. A shop person opens up a new shop, and then some tough guys come around and they engage in conversation with the new shop owner, saying, "This is sometimes a bit of a dangerous neighborhood. It's always good to have some extra security to make sure nothing bad happens to your shop, particularly when you're not here, or even if you're here or your kids are helping out in the shop. Wouldn't it be awful if something bad happened to the people in your shop? If you slip us some money on a regular basis, we will make sure that none of those bad things happen to you and your shop."
- 3.96** Now, in that case, that's an appeal to a threat. If they were to say, "You should give us money on a regular basis," they don't really have a good reason for that independently. They said, they are manufacturing. They are the threat. "We will mess you up and mess up your shop if you don't buy us off." So that's an appeal to force, and it's obviously inappropriate.
- 3.97** Parents will say, the kid says, "Why should I eat my vegetables?" "Because you will get a spanking if you don't." Now, typically, we don't make that argument so much anymore. It's a generational shift, but parents are tempted sometimes to appeal to force, because at that point, saying the kids said, "Why do I need to eat my vegetables?", to get into a discussion about nutrition and so forth, you don't want to go there at that point. You just want to tidy up and go watch TV or relax. So this is a short-circuit of the reasoning process, but it still is a fallacy.
- 3.98** Much of terrible education. This is a scandal in some higher education faculty, where we're trying to train young people to think for themselves and to be aware. Lots of controversial issues and arguments on both sides and to present all of that, but many professors we know are less interested in education and more interested in indoctrination.
- 3.99** "I'm the professor. I have my views, say, on these political issues or these religious issues or whatever it is that I am teaching, and I want you all to believe what I believe. I will make it clear in various ways that if you say the wrong thing in class or you say the wrong thing on the test, you will get a worse grade." Now, I'm obviously going to be sophisticated enough not to come out and say this explicitly, but savvy students pick up on this.

- 3.100** So part of, I think, professional education responsibility, given the stakes that are involved here, young people, we don't want to be teaching them less than that. What they should say is just whatever the authorities want them to say, and to say it out of fear, which is what the ad baculum or appeal to force argument is.
- 3.101** If I say, this would be even too crude, "I know you're all writing your essays on the French Revolution. But in the course of this lecture, I've made it clear what my views on the French Revolution are and what the right political lessons are. It's almost the end of the semester. Don't forget who gives the grades in this class." That's pretty naked, but people know that that's going on, and so that's going to short-circuit the thinking of the students over the course of whatever period of time that they are writing the lesson.
- 3.102** We know some religions, when children start to ask pesky questions, challenging questions about the religion, will threaten various sorts of things. The big one is, "You will go to hell if you ask questions like that." So ad baculum is built into some forms of religion that are not trying to encourage independent thought. Same thing occurs in political discussions. "I'm the father, and around in this house, we vote this way." Election season is coming and to make it very uncomfortable for anybody to articulate a possibly critical political opinion about my views because I am the dad and I've got power over you.
- 3.103** So, obviously it's a temptation. If you want to get people to believe what you want, the hard way to go is, you have to present your evidence. You have to present your data. You have to present your reason. You have to be patient with the person as they process through all of that. You have to handle their uncomfortable critical questions, their offbeat questions, their critical questions, and have good answers to all. That's very hard.
- 3.104** In many cases, if I'm not very committed to objectivity and independent reasoning of the people I'm dealing with, just using my power to force in the short term the result that I want, it can be tempting. And then on the other hand, a lot of times, unfortunately, people, when they are on the receiving end of that, they also learn the lesson, particularly when they are young, "I shouldn't think for myself. I shouldn't raise hard questions. I should just do what they tell me in order to get through this process." But that ends up being hard on their character, submerging yourself in that way, and has the damage of turning into a habit.
- 3.105** All right, in the case of all of these fallacies, what we have are extreme ways and a lot of different ways of not paying attention to the connection between evidence and/or the premises and the conclusion, short-circuiting that process by means of various kinds of devices. All of them are illogical, but some of them have some psychological power because human psychology is very powerful.
- 3.106** The lesson of all of them is to resist all of those temptations and all of those cheap tactics, and make sure that, in your belief formation, whatever you are believing, your

belief is tied as tightly as possible to the available evidence and the available data, making that your policy of action. Now, that policy, historically, is controversial because many philosophies, many politics, many religions have found logic uncomfortable when they sort through their positions. They have inconsistencies and/or things that they don't have good data for.

- 3.107** So the policy of having to be reasonable in that logical sense, they will try to subvert it in various ways. Some famous historical examples of this controversy. Thomas Henry Huxley in the late 19th century stated it this way. "In matters of the intellect, follow your reason as far as it will take you, without regard to any other consideration." At some point, there will be other considerations. You have to take them in, your emotions, how they affect on other people, what other people believe and how they feel about things. But first obligation is, what does your reasoning intellect teach you?
- 3.108** "And negatively, in matters of the intellect, do not pretend that conclusions are certain which are not demonstrated or demonstrable." So we all would like to be certain in our beliefs, but sometimes the way it is, is we only have a certain amount of data, and something is suggested as a hypothesis. I'm considering it as a hypothesis. It might be that I'm in a situation where I know I've got inconsistent data and a couple of possible hypotheses. Well, that's where I am in my thinking, and I face that squarely.
- 3.109** I might have a lot of data, but I know it's not certain yet. But I really want the closure on this. Well, resist that. Say, "This is very compelling. I'm not yet certain." And only let yourself feel that certainty and make that commitment until you've got the certainty. Always be ruthlessly honest about how much support you have for whatever beliefs that you have. No contradictions, or if you have contradictions in your thinking, face them up and then back away from believing either side of the contradiction until you get it better sorted out in your thinking.
- 3.110** Now, the most famous historical antagonist to that viewpoint is represented by Tertullian, one of the early church fathers. Back in the times when the Roman Empire was still strong but starting to go into decline, Christianity was an ascendant force; and there was a lot of tension between the Greek and Roman schools of philosophy that thought Christianity was not true—some of them thought it was ridiculous—and some of the early defenders of Christianity who recognized that it had some serious possible inconsistent and outright contradiction problems, but nonetheless wanted to be committed to it.
- 3.111** So, Tertullian is one who went on the offensive in response to that, against the idea that logic matters, assessing the evidence matters, or worrying about contradictions matters. So on issues, for example, that are parts of faith, here faith meaning things that you don't necessarily understand, might not even seem reasonable to you, but nonetheless are part of the belief system that you want to commit to. If you're going to maintain your belief, it seems like you can't be logical and rational at that same time.

- 3.112** So was Mary really a virgin? Did an infinite God become a finite being in time and space? Could Jesus really have died and then really have come back to life? Can we really say that God is one, that we're really serious about monotheism, but then start talking about God and Jesus in different language, and also have the Holy Spirit somewhere in the mix as another element as well?
- 3.113** The point of all of this is not to say that, if we are smart theologians, that we can't address these issues and maybe find ways to overcome what seems to be tensions and inconsistencies and perhaps contradictions. But should we have to do that in the first place, particularly if we are the average person who wants to believe in this doctrine? Are we going to sell people, "No. Be super logical about everything in Christianity and don't believe it until you've got all of those things worked out for you. You've got all of those possible contradictions solved in your own mind"?
- 3.114** So Tertullian's strategy is to say, "No, logic is not for religion. Logic and faith are opposite. I'm on the side of faith, and that means I'm going to critique logic fundamentally." So the famous lines are these ones: "The Son of God died." How can God die? He's supposed to be immortal. "It is immediately credible," that is to say, believable, "because it is silly." It is silly, but I believe it. "He was buried and rose again. It is certain because it is impossible." It's impossible. But it's something that I have a psychological attitude of certainty to.
- 3.115** Now, this is not logic, this is faith. We have a huge then controversy over human cognition, whether the Huxley position of faith—no, you don't believe stuff, particularly if it seems inconsistent in contradiction to you—and Tertullian taking the position, no, you have to have faith, and if that means you have to set aside logic, you set aside the logic.
- 3.116** Another variant on the religion some thousand years, most famously perhaps represented by Galileo, who—Huxley was an agnostic, probably didn't believe, but thought we should be rational. Galileo was a believer in a god, but thought one should be rational. Tertullian wanted to believe, but on non-rational grounds. The essence of Galileo's argument is to say, look, if you're going to be a good believer in God, then you have to believe that God made the world. He made it a certain way, so that scientists can study it using logic and scientific method.
- 3.117** God made us, and he gave us certain faculties. He gave us our senses. He gave us our concept faculty. He gave us our reason. He gave us the ability to do logic and mathematics. So he wants us to use those things. He wants us to use our God-given faculties to study nature, logically, rationally, scientifically, to come to know him and his creation. So believing in the doctrine should ultimately be consistent with being logical.
- 3.118** And so, now what we then have is a three-way debate—Huxley thinking, if you are logical, you're probably not going to be a believer; Tertullian arguing or stating rather, if

you want to be a believer, you have to be illogical. And Galileo arguing that we should be able to reconcile being logical with belief. We'll pause there.

Lecture 4: Fallacies & Cognitive Biases

At a glance — More fallacies and cognitive biases.

- 4.1 All right, more fallacies. We're in the midst of looking at a pattern of argumentation that is fallacious. It has the appearance of premise leading to a conclusion, but there is no logical connection at all between the premise and the conclusion. Instead, what seems to be going on is people are pushing each other's psychological buttons and trying to manipulate a certain conclusion or get it for free in a way.
- 4.2 So, appeal to majority. Lots and lots of people believe this doesn't mean anything for the truth of what it is that they believe. Authoritative individuals believe or are urging something. That fact alone does not provide support for the truth of the premise. The fact that you want something to be true, that you would really like something to be true, provides no evidence for it actually being true and so on.
- 4.3 The fact that you don't like a certain person making an argument—your dislike of that person or disapproval of that person does not mean there's anything wrong with the facts or the argument that that person is pressing. What we need to do is just train ourselves to recognize when that pattern of linguistics or arguments is happening and red flags go up and one is less likely to do so.
- 4.4 There's another common family of argumentation called begging the question. There's a whole family of them. There's a lot of variations on this fallacious mode of reasoning. It is really like empty calories, but it has the appearance of taking you where you want to go. Now, the name begging the question goes back to some old Latin, "petitio principii," where the idea of begging the question is—the question is the thing that you're supposed to be proving, what's your conclusion—but you're trying to get it for free in a way, and that's the begging part.
- 4.5 Or you haven't really earned the right to that conclusion; you've begged it in some way. Almost always what it means in a simple or more complex form is whatever it is that you are trying to conclude, you assume that in one of your premises. Having assumed it in your premises, you've already got it in your conclusion, so it makes the work of argumentation a whole lot easier.
- 4.6 In very crude form, it would be—when you reduce all of the manipulations out of the way—X is true, and at some point in your premises, you've got X is true. Now, almost always, though, it's not that crude. Instead, what happens is people do various machinations or linguistic manipulations.
- 4.7 Now, one form of that is just restatement. We know in English and in most languages, there's lots of different ways of saying the same thing. We have lots of synonyms, we have phrases that stand in for particular words. Sometimes we use metaphors and clichés and so on, and they stand in for a particular concept. A restatement would

simply be to say whatever your conclusion is, you just said the same thing, but in different words in the premises. When you cancel out the differences of the words, it ends up being the same content.

- 4.8 Got some examples here on the board to get us started. Capital punishment. Take this first one. Capital punishment is justified because there are good reasons for putting people to death. Now, this looks like an argument. It is an argument. We have the because here. That's a premise indicator. We know this is the premise. There are good reasons for putting people to death. Before the because, almost always is the conclusion: capital punishment is justified.
- 4.9 So the argument is: there are good reasons for putting people to death. Therefore, if we're putting it in standard form, capital punishment is justified. But when we look a little more closely, we say, well, what is capital punishment? This phrase, capital punishment just is the legal system putting people to death. It's the same thing. It's a restatement of what capital punishment is. Then this one here, it's justified. Then we have this phrase here, good reasons.
- 4.10 Well, what is something to be—well, you have good reasons for it. If you have good reasons, then you have justified. This is two sets of phrases that are saying the same thing. This is one word, but then a short phrase saying the exact same thing. In this argument, I've just said the same thing repeatedly, and that then is begging the question.
- 4.11 Another example here, switching to democracy. Let's read it. Democracy is the proper social system as society should adopt majority rule. I've got a "should, proper," so normative concepts here. But then, let's look at it a little more closely. Actually, we have an "as" indicating. This is a premise indicator. This is my premise, "as" giving my reasons, and then this must be my conclusion.
- 4.12 We say, "Well, what are we talking about here in my conclusion?" I'm talking about democracy. Then we say, "What is democracy? Normally, we say, "Well, it's majority rule." Majority rule and democracy are the same thing. It's a synonym. What is it to say that something is a proper social system? It's a social system with this normative evaluation concept, proper.
- 4.13 Well, is that not the same as saying society should adopt? Should adopt the proper thing. Society, social system is proper, what you, society, should adopt, that's just a restatement. It seems like I've got an argument here. These are all nice words, but these are empty. These are begging the question.
- 4.14 So restatement. One thing, then, when you're dealing particularly with matters of some complexity is to make sure that it's not simply restatement, that the premises and the conclusions, the premises have more information. The conclusion is drawing an inference, not simply saying the same things in different words.

- 4.15 Now, another form of this is a kind of redefinition that we can do. What we can do in an arbitrary way is take a concept that we know people ordinarily use in a certain way, and then in the course of argument, substitute a different definition that takes you to where you want the argument to end up.
- 4.16 Now, that's an abstract formulation. Let me give you an example here. Suppose we're arguing about religions and morality and whether some religions are more moral than other religions and so on. So this is a made up example, but I'll be Q. I will say, "All Buddhist are virtuous. That's a very strong claim. And then you can be Z. You would say, "Wait a minute." You know a Buddhist, this guy named Singh. You say, "Well, he does such and such," various unvirtuous things, whatever those are. "Look at Singh's behavior. He's a Buddhist, but he isn't virtuous."
- 4.17 And then I say, "Well, obviously Singh isn't a Buddhist." You say, "Well, why not?" I say, "Well, because he isn't virtuous." Then you would say, "Well, he goes to temple. He reads the scriptures regularly. He calls himself a Buddhist, comes from a Buddhist family. So why isn't he a Buddhist?" You say, "Because he isn't virtuous." That's to beg the question.
- 4.18 What you have done is take what's the ordinary understanding of what is a Buddhist. Someone who thinks of himself as a Buddhist, goes to the Buddhist temples, read Buddhist scriptures, etc. But you've substituted your definition, which is to say if it doesn't meet your criterion of virtue, any deviation from that, then suddenly that person is not a Buddhist. That's a contentious, slippery way of avoiding having to argue for the issue at stake.
- 4.19 Now, a more complicated one, aside from these arbitrary definitions being substituted and restated—and this is where one really has to be paying attention and do one's work—is what is more formerly called circular reasoning or arguing in a circle. Now, the crudest form of this is going to be: one says A proves B. You might have some good evidence that A leads to B, and then B proves C. C proves D, but then you find that the argument from D proves A.
- 4.20 What you've done is you've gone in a circle here. You don't have anything outside of this circle, any facts independently supporting this whole circle of reasoning, and that's illegitimate. Now, what makes this challenging is that in complicated matters, particularly cutting edge matters, when one's developing an argument, say, over a whole chapter, it might be that the A proves B part occupies the first page or two.
- 4.21 You say, "OK, finally, we've got A proves B." Then it takes two or three pages for the person connecting B to C. Four or five pages later, we finally are getting to D. Then, in the wrap up, finally, the person is, in effect, pulling the rabbit out of the hat, but it's there that they make the argument for D proving A. You have to have digested, say, this whole 25-page chapter and be keeping track of the argument as it goes along the major stages at each point.

- 4.22 Then step back and say, "Oh, wait a minute, we've just gone in a circle over 25 pages." That's hard work. In some cases, of course, this happens over the course of a whole book, which means one's note-taking analysis has to be very diligent in order to catch the begging of the question.
- 4.23 Let me give you a silly but fun example of this begging the question. Suppose you want to get a loan, so you go to the bank and you're talking with the bank officer and you say, "I would like to take this loan out." And the bank officer says, "Well, OK, fine, but do you have any collateral? So if you don't pay back the loan, we can take something of equivalent financial value to cover the loan." And you say, "Well, no, no, unfortunately I don't have any collateral, but I'm an honest person. I always pay my loans back on time. You can trust me. Give me the loan."
- 4.24 And the bank officer would then, quite rightly, say, "Well, I don't really know you. I don't know your track record. I don't have any way to know whether you are really a person of character and that I can trust you or not." You say, "Well, that's fine. That's why I brought my friend with me." So you did bring a friend with you. "My friend will vouch for me. He'll tell you what a great guy I am, that I'm trustworthy, that I always do what I say that I'm going to do." This is better. A friend vouches for me as evidence, witness evidence, so to speak, for the bank officer.
- 4.25 Now, then the bank officer, of course, is going to say, "Well, I don't really know your friend either. So I don't know whether I can trust him when he says that you're a good guy who's going to pay the loan back." And then you say at that point, "Well, that's fine. I can vote for my friend. He's a really good guy. He wouldn't lie about these sorts of things."
- 4.26 And there I've gone in a circle because my friend is vouching for me and I'm vouching for my friend who's vouching for me. And the bank officer is caught then in a circle. He doesn't have any independent verification of anyone's trustworthiness or integrity on these particular matters. All right, now, again, that's over the course of two minutes.
- 4.27 Chances are good; you can catch that one, but it's the shorter ones like that that flag the thing to your attention so that when—you start to suspect, when you're dealing with longer things that you've come in a circle, you can slow down and check those things out a little more.
- 4.28 Now, another example of begging the question is typically called the question-begging epithet, where in the course of your argument, you will slip a word or a phrase that supports, illegitimately, the conclusion that you want to support. The most standard examples involve lawyers trying to manipulate jurors in various ways. I say I'm the lawyer for the state and we are prosecuting someone who is suspected of murder, but we've not yet proved that this person has committed the murder. My job as the prosecutor is to lay out all of the evidence systematically to show that this person is the murderer.

- 4.29 But suppose in my opening speech to the jury, I say something like this: "Ladies and gentlemen of the jury, I will now proceed to prove to you that this vicious killer is indeed guilty of the crime." Now, that's a question-begging epithet, which is a word or phrase that we've used because I have called this person a vicious killer in my opening statements. But I have not yet provided any evidence for this person being a vicious killer. That's my job. But I'm not allowed to call him a killer until I have proved that he's a killer—much less call him a vicious killer until I have proved that there is viciousness.
- 4.30 Or a politician might say, in the heat of a debate, "This is a monstrous policy, and I will go on to show why this policy that's proposed is monstrous." Can't call it monstrous until you've proved or given the evidence for its monstrosity, whatever it is. The idea here is that you're trying to cheat because you are smuggling in your conclusion in your opening statements, so to speak, and that's illegitimate.
- 4.31 Another version, in some ways, this is the most fun version of this, but it's very common, is begging the question by means of what are called complex questions. I will put a question out there, and it's a serious question. Sometimes it has a somewhat rhetorical force to it. Part of the challenge is you have to take the question and turn it into whatever the declarative sentences the person is really meaning, the literal sentences and so on. But it's called a complex question because in the formulation of the question, you again have built in the conclusion that you want just into the formulation of the question.
- 4.32 Up on the board here, if we approach at some family reunion or high school reunion, I haven't seen you for a while, but I ask you: "Do you still gamble away your paycheck every Friday night?" Now, it might be true and it might not be true, but that formulation, particularly if there's a crowd of people around you and I'm doing something attacking, immediately puts you on the defensive—partly because it is formulated, again, as a yes or no question where you can't really say "yes" without admitting something you don't want to admit, and you can't say no without admitting something that you don't want to admit.
- 4.33 There's two questions that are built into it. Did you, at one point in the past, gamble away your paycheck every Friday night? That's one question. And now, do you still gamble away your paycheck on Friday night? So there's a question about the past. There's a question about the present. But this person, me in this case, has bundled those two together in a complex question.
- 4.34 The sign of a complex question will almost be that it's phrased as a yes or no. You have to affirm or deny the question, but you can't, or you find yourself resisting wanting to do so. Because in this case, you want to say, "Yes." Admission? Well, that means I did and I still do. Or you want to say, "No, I don't do it anymore," but you've already then seemingly admitted that you did in the past. Have you stopped beating your wife? Would be another one. Drop that one into a social event. The poor guy doesn't know what's hit him.

- 4.35 Why should we continue—in a political—why should we continue to be exploited by the greedy capitalists? You see that in social media discussions. It's assuming we are, and why should we continue? Well, we shouldn't. There's a whole set of assumptions built at the end of that one as well. Moms and dads to their kids: "Do you want to be a good boy and go to bed?" "Do you want to be a good girl? Eat your vegetables, etc." Of course, the poor kid has not taken logic yet. He's not in, "Wait a minute, Mom, that's a complex question." But maybe they can pick this one up. This is a part of the training here.
- 4.36 There's two things here. Do you want to be a good boy? Well, probably the answer to that question is, "Yes." Do you want to go to bed? "No." So one answer to this complex question is inappropriate. The kid doesn't want to say "yes" because then they're saying they want to go to when really the little boy doesn't want to go to bed, and he doesn't want to say "no" because he doesn't want to say he's not a good boy. So it's sneaky parenting. It might be a useful tactic, of course, when you're desperate, but, logically, it is fallacious.
- 4.37 In politics, you see this fairly regularly. People will assume that their approach to politics embodies all virtue and all goodness, so they will bundle a statement about their politics with morally charged language. "Do you want to be a progressive socialist person?" There the assumption is, socialism has a monopoly on being progressive, so we just lump progressivism in with socialism.
- 4.38 Do you want to be—religions do this, too. "My religion has a monopoly on proper approaches to morality. Do you want to be an upstanding religious person?" Or "Don't you admire these upstanding religious—" whatever denomination you want to praise? Again, making the association implicitly that being a member of this religion entitles you to use that moral labeling.
- 4.39 But you need to argue for that. You need to argue, to say, "Here's my religion, here's what it stands for. Here's my morality, here's what it stands for. There is an intimate connection between these two things, and that connection is better than what you find in the other religions." That's a lot of work. It's a question-begging cheap shot to go there.
- 4.40 So let's say, sometimes this gets piled up together. "Do you want to be a good boy and go to bed?" I mean, suppose your kid had been secretly reading logic books, and said, "Wait a minute, Dad. That's a complex question. That's begging the question. That's an illegitimate fallacy." And you say, "Shut up or I'll smack you. Get to bed." And then the kid would say, "Wait a minute, Dad. That's an appeal to force. That was in the earlier chapter, also illegitimate." So logic can be dangerous for parenting.
- 4.41 One possibly apocryphal historical example that I like to think about involves one of the King Charleses back in the day, and he suspected that his court intellectuals—this was in the early days of science—were just making up a lot of stuff. They really didn't know

what was going on. So he put this question to them as a test, but he put it, not that they knew it was going to be a test. So he put this scientific question to them. He made a little competition associated with it.

- 4.42 He says, "What's interesting is, if you have observed a fish in a bowl of water, a live fish, but you fill the bowl of water right up to the brim—so if you had a little more water, it would spill over, but just enough so that it doesn't spill over, that the fish can swim around in that water and it won't spill any of the water. But if you have the exact same amount of water and you put a dead fish in it and fill it up to the brim right again, it will spill over. Isn't that weird? Why is this?"
- 4.43 And so his court intellectuals then get busy and they start to come up with all kinds of explanations and so forth. None of them work. The reason why none of them work is that King Charles has given them a complex question. It is not true that the dead fish in the full body of water will cause it to spill over. But nonetheless, they just took the complex question and then devoted all of this energy to useless intellectualizing, when built into that complex question was something they should have detected and checked.
- 4.44 In that case, the first thing to do would be not immediately go to the speculation, but go back to the beginning and say, "We have some claims here, some empirical claims. Live fish in that amount of water won't spill. Dead fish in the same amount of water will spill. I need to check that, check the data, and unbundle the question before going on to do the explanations."
- 4.45 Now, another pair of fallacies that go together, they go by the names equivocation and amphiboly. It's big fancy labels for things that are unclear or or ambiguous in formulation. The only difference is that, in equivocation, we have words that could mean two or more different things. The fallacies typically occur, this is an abstract formulation of it, when you start off in the argument using the word this way, but at some point you shift in the argument to using the word in the other way.
- 4.46 Or in a social context sometimes the argument will break down because one person has started using a concept in a particular way, but the other person at some point gets away from that and starts using the concept in a different way. Then obviously the reasoning is going to go awry in the course of doing so. Amphiboly is the case of particular words, not them alone, but rather phrases that can be interpreted in different ways.
- 4.47 Here's a silly one to begin. It's a way of insulting someone. You say your argument is sound, nothing but sound. Now, in that case, sound is one of those words that's a congratulations word in logic. Your argument is very strong, good reasoning. It's sound. But we also use the word sound to talk about ephemeral sound waves. They come and go, and sometimes they're empty and they don't mean anything. So this starts off

sounding like it's praise, but quickly switches to an insult. That's an equivocal phrase in action.

4.48 We use the words trunk. This would never happen in an argument, but it's an equivocal phrase. Are you talking about the long protuberance on the front of an elephant? Are you talking about the tail end of a car where you can store stuff? Are you talking about a big wooden chest up in your grandmother's attic where there's stuff stored in it? So trunk is potentially equivocal.

4.49 A fence. Is it a boundary between your property and the next property? Is a fence someone who deals in stolen goods? It's the same words. Now, those are crude and quite distant, so we're not likely to confuse them very often. But as, again, things get complicated, they get more serious, and particularly in cases where we're a little invested in what the issues are, the temptation to equivocate and use words in this slippery way becomes more tempting.

4.50 Let's try this one as an argument here. Voting is a right. The government should enforce rights. Thus, the state should enforce voting. Let me pause for a moment. We have this word thus. It's a formal conclusion indicator. Thus, therefore, hence. So we know the conclusion is, the state should enforce voting. Well, why? What are the premises for this? Well, one premise, voting is a right. The government should enforce rights is another premise. You put those together, the state, therefore, should enforce voting.

4.51 Now, there's an equivocation in here because this can be interpreted in more than one way. When I say that the state, for example, in the conclusion, should enforce voting, what does that mean? It depends on what you also take the right to be. So does voting as a right mean this is your prerogative as a citizen, your choice as a citizen? You have the access to the political power to make your beliefs on political matters heard and counted for something in the decision-making process. You have the right to that.

4.52 But also as a right, it's your choice or not, you can say, "Well, actually, I don't want to vote, so I don't choose to vote." But we have this enforcement issue that comes up in the second sentence and in the conclusion. When we're talking about the government enforcing things, sometimes we mean the government would make you do things. If we take the enforcement in this case to mean that, when probably the other person means enforce the other way, you have this free choice right, and so your right to vote or not vote, that's what the government should be protecting, protecting your choice.

4.53 In the second interpretation, the choice gets taken away. No, the government is supposed to enforce rights. Voting is a right, so therefore it's going to enforce voting. So it goes around and rounds people up on election day and makes sure they go to the voting poll booth and cast their vote. Now, that would then be an equivocation on the nature of a right and the nature of enforcement that's appropriate to the right in this case.

- 4.54 Again, this is more subtle, stakes are higher. Of course, we do know there are some democratic systems that do enforce voting in the one way. It's just an option, a power that you have. You can exercise it or not. Others that say voting is a right, but in a democratic context, everybody has to participate, and so they make people engage in the voting process.
- 4.55 Here's another example. I give you this sentence here. People who sell drugs to children are the scum of the earth. Chances are very good you'll say, "Well, yeah, of course. They're awful." But then if I were to give you the rest of an argument that said something like, pharmacists sometimes sell drugs to children, the logical conclusion, it seems, would be that pharmacists are the scum of the earth, and something has gone wrong in that case.
- 4.56 So when we talk about people who sell drugs to children, that phrase, immediately at the beginning, you were thinking of the grubby people, the illegal people, the people who are callous about kids and so forth, whereas, really, I was wanting you to think about pharmacists. But because this phrase is ambiguous, it's not clear which one is which. The argument works if you assume we're talking about the nasty people, but it doesn't work if we are talking about pharmacists. We're talking about legally or illegally, and there's a whole background context that we need to be attending to.
- 4.57 Some connections between logic and humor. Again, humorists are very often very good with logical expectations and wordplay and so forth. Equivocation is one of their favorite places to spend some time. The famous comedian George Burns died a few years ago. He actually lived to be well over 100, smoking huge cigars all of the time, constantly. But one of his jokes was, "If you live to the age of 100, you have it made because very few people die past the age of 100."
- 4.58 Now, where's the equivocation in that? Well, it could be, one interpretation is, very few people die past the age of 100. What that means is, by the time you get to 100, most people are gone. So there's very few who get there in the first place. That's the true interpretation. But the fun interpretation is to take it the other way, to say, well, once you get to 100, very few people die past the age of 100. That means you've got it made in the sense of, you're not going to die or you're not very likely to die, because you've made it to 100. So two different ways of playing with that.
- 4.59 Psychiatrist joke. One out of four people has a mental illness. So check three of your friends. If they are OK, it's you. That's right. Again, one in three does not necessarily mean in every group of four, it's going to be one of them. Doesn't mean if you've sequentially checked one, two, three, the next one is going to be the person. It's the statistical percentage of a large group when we don't know the constituent numbers.
- 4.60 It used to be popular in some farmers' fields where there was a bull, the farmer would put up a notice saying, "The farmer allows walkers to cross the field for free, but the bull charges." So it's a fun way of saying there's a bull in this field, so beware. But we

have free and charge used in two different ways—charge as a monetary cost. Well, the farmer is not charging you in that sense. He's not saying you have to pay in order to cross the field. What bulls do when they come at you, we also call that charging. It's non-monetary. It's something else to worry about altogether.

- 4.61 Then in the ongoing jokes about men and women and marriages and so forth, a linguist noted, the shortest sentence in the English language, "I am." It's two words, three letters, but then waggishly adds, "Could it be that 'I do' is the longest sentence?" Making a joke about marriage, in this case, I do, can be just taken as a certain number of syllables, certain number of words, short words in both cases, there's only two them, but we also know that I do is a performative in the context of marriage, and marriage is a kind of sentence, but sentence not in the linguistic sense, but sentence in the punishment for your misdeeds, whatever those are.
- 4.62 Take life with a grain of salt, as we all know that cliché, plus a slice of lemon and a shot of tequila. The equivocation in that case is, life taken with a grain of salt. That little cliché. You put salt on something, the idea is that you moderate the claim to a certain kind of chemical reaction. You neutralize whatever these fiery things are going on. Take it with a grain of salt, calm down, not such a big deal. In this case here, literally take life with alcohol. That will help you get through. And that's the joke formulation of it.
- 4.63 OK, let's move on to amphiboly. Amphiboly is grammatical constructions, not particular words or phrases. So all those rules about grammar and using commas appropriately, dashes, parentheses, word order, in complex sentences—all of that needs to be attended to by clear writers and clear thinkers, so that you don't set your readers up for an amphibolous formulation. Here's an example here. I'm going to flag this one a couple of times.
- 4.64 So all that glitters is not gold. Now, it's an old cliché. The cliché is meant to say, just because you find something glittery, glittery things are attractive to you, don't make the mistake of thinking that it's valuable. There's a surface and depth issues, but then also a lot of glittery things, or things that seem valuable, are not valuable over the longer time. Gold is, of course, glittery and valuable, but not everything is like that. So it's meant metaphorically to be a warning against appearances and so forth.
- 4.65 But if you take this literally, another way of saying it is that all that glitters, we also then take that sometimes to mean everything that is in that category, all of the things that are glittery are not gold. And that's not true because gold is a glittery thing. So is it really saying some things that are glittery are not gold? So be on your guard. Or is it really saying literally all that glitters is not gold? The point is that, for your alert reader, they're going to have to pause and think about it to make sure what is the most likely interpretation in that case.
- 4.66 Another example of the exact same amphibolous formulation—all married couples do not sleep together. Now, if you take that literally, this is all married couples, they don't

sleep together. That sounds a bit harsh, and it also is not true. What this person probably means is, it's not the case that all married couples sleep together. But the positioning of this "not" in this overall formulation has made it amphibolous. Does this mean all married couples don't sleep together? Or does it mean some married couples don't sleep together? And you want to, for the interests of clarity and economy, not have amphibolous formulations like that.

- 4.67 This one here, a little bit amusing. Woman without her man is lost. This then becomes a lesson in punctuation because that can be interpreted in two different ways, depending on where you put the emphasis in spoken language or where you put the commas in written language. Do I say that the right way to interpret this is to take this as the noun phrase, woman without her man? A woman without her man, comma, is lost. That would then be to say, women who don't have a man, they are lost.
- 4.68 This is a little bit metaphorical, right? To say she doesn't know what she's doing with her life, et cetera, she's not fulfilled. That's a possibility. But if I put two commas here—Woman, without her, man is lost—and that then is the opposite. That then is to say, man is lost if he's without a woman. Woman, without her, the man is lost. So amphibolous formulations, we want to avoid them. So whichever of those you want to argue for and be popular at the party, make sure you put your commas in the right position.
- 4.69 Some funny ones, sometimes unintentionally funny. Sometimes when I'm writing letters of recommendations for my students, and of course, you want to praise up your students, but there are some students who are a little bit marginal and in a dark humor mode, you want to play around with the idea of sending a recommendation that's not really a recommendation. If I said, for example, "I cannot recommend this person too highly," well, it's like, it could be praise. I could be saying, "Here's all these recommendations, all this great, high, wonderful stuff, and I just can't say enough high stuff about this person in this recommendation."
- 4.70 That's one, but the opposite is to say, "I can't recommend this person too highly. So there's high recommendation, and I can't do that." But the formulation all by itself doesn't say which way it would go. Or, "Nobody would be better for the job." Ambiguous on the nobody, in that particular context. Are we saying people who are good for the job, there's this guy and there's nobody else? So nobody would be better than this guy for the job.
- 4.71 Or, this is rather pathetic, would you say, nobody, nonexistent, not having anybody do that job would be better than this guy. And that's the worst possible interpretation. But chances are good, in the context of a letter of recommendation, the person might give you the benefit of the doubt, but you don't want to leave that up for grabs.
- 4.72 From an old cookbook, instructions: "Serve the meat when thoroughly stewed." This was an old-fashioned cookbook. Stewed is an old way of talking about having had too

much to drink. How do you know when the meat is done? Well, after you've had enough wine or whatever it is during your — you are the one who is stewed. Now, finally, the meat is likely to be ready or whatever. In fact, the cookbook is trying to say, no, it's the meat that has to be thoroughly stewed. It has to be in a certain condition, not you, the cook, in a certain condition.

- 4.73 Book review, the author, these are real ones now. "The author warns against a number of logical fallacies in his book," the logic book, of all things. Now, is he saying that the author of the logic book says, "Here's a whole bunch of logical fallacies," and is warning you, the reader, about the logical fallacies?
- 4.74 Or is it, the author has written the logic book and reread it, but now it's been published and says, "Man, there's a lot of logical fallacies in my writing. I made a lot of mistakes here. I'm going to warn you up front that there are these logical fallacies in the book." Now, literally speaking, that's an amphibolous formulation. We probably know which one the person means, but it should be embarrassing that a person reviewing a logic book would use that kind of formulation.
- 4.75 All right, this is an old newspaper account, a real one. The Graf Zeppelin, I think it was almost exactly a century ago, was it not the big hydrogen-inflated balloons that were sometimes capable of transatlantic transactions? And so back in the day, that was a big deal in the early days of transcontinental flight. And this was a real newspaper clipping from the time.
- 4.76 "The Graf Zeppelin was leaving the Lakehurst airport." It had come across from Germany, I think, and now it was going to have its return flight to go back to Germany. "Among the last to enter was Mrs. Smith, the lone woman passenger. Slowly, her huge nose was turned into the wind, and then, like some huge beast, she crawled along the grass." I can't even say that with a straight face.
- 4.77 Now, this is an amphibolous formulation in a relatively short paragraph because, at the beginning, we introduced the Graf Zeppelin, which is a big cumbersome-shaped being. Then we introduced Mrs. Smith, and we don't have any descriptors of what she is like physically. But Mrs. Smith is the most recent thing we were talking about, and then we start using pronouns. Her huge nose. Whose huge nose are we talking about? Who is the beast lumbering across the grass? Hopefully, it's the Graf Zeppelin, but this is embarrassing newspaper writing because of the amphiboly.
- 4.78 Want ads have always been a source of really fun ones because, in old traditional want ads, you had to pay by the word, so you would try to cut out as many words as you possibly could, and often it would be the clarifying words that get cut out. So, "Selling an antique desk suitable for lady with curved legs and large drawers. Also, mahogany chest." OK, well, very well. We'll skip over the next one and go immediately to appeal to emotion.

- 4.79 Appeal to emotion. This is a complicated one on both the logic side and the emotion side, because arguing passionately about what one believes in, what one thinks is true, can be perfectly fine, as long as that is built upon a logical foundation. So you've done your homework, you've got your facts, you've got your data, you've checked your logic, and it's an important issue. Having some passion in the presentation of your argument is perfectly legitimate.
- 4.80 Appeal to emotion is a fallacy when the emotion substitutes for the logic. That is to say, one doesn't really have much by way of facts, data, solid argument. Instead, one appeals to all the language that we know pushes people's buttons easily. There are concepts and issues where we know people are coming in with an emotionally charged state in a particular way, and so we can use language to play on that.
- 4.81 So a lot of mob psychology and mob rallies in a political context are very good examples of this. People come out for a mass rally, they're already excited, they're energized, whatever the issues of the day are, they're charged up about that. And then you bring in a speaker, often a politician, who knows how to play the crowd, get them whipped up even into more of a frenzy. In many cases, the speaker does not have to have good arguments or make particularly clear points.
- 4.82 But if you use language well, then people will go away, hopefully, they will go away without destroying stuff. They will go away feeling like they've learned a lot, they've heard a lot, and they are more affirmed in their commitments. So demagoguery, certain kinds of populist politicians, particularly now in the last 100 years or so, since we are much more Democratic and Republican, mass participatory politics, and we have the technology to record a lot of the speeches on sound and video, they're useful to study for people interested in political rhetoric.
- 4.83 The cases of politicians who use logic integrated with rhetoric very well. Martin Luther King is an outstanding example of that, according to logicians who've done the studies here. Adolf Hitler is typically used as a documented example of the others. Very long speeches. Often there's not very much content to it, but he's very dramatic and very charismatic, and he uses certain kinds of language. Equally, as inspiring as people listening to Dr. King felt, people listening to Adolf Hitler felt. So the outcomes tended to be the same, even though one was more empty.
- 4.84 Now, menus is an example here. Often we go to fancy restaurants. They want to do two things. They want to let you know what you are ordering off the menu, so they have descriptors. But as we know from reading the menus, they almost always use the most elevated language possible to describe whatever the menu is. So, filet mignon sounds foreign, exotic, and so on. But I could describe this literally as chunk of dead cow. It's what it is.
- 4.85 This tells you as much or more, but you'll never see this on a restaurant menu. The emotional connotations are wrong. You don't want to think about cow. You don't want

to think of things as chunk. You want it to be properly prepared. You don't want to think about dead things. You want to think, "I'm in an exotic, fancy, eating a special meal," and so on.

- 4.86** Egghead is a pejorative meant for someone who is an intellectual. Talking about the same group of people, people who think a lot about ideas, so forth, calling someone an intellectual here, praise, calling them an egghead, is a mild insult. Same thing with respect to more recent phrases like nerd and geek. You call someone a geek, it typically means someone who has unusual habits and unusual ways of dealing with things that are a little bit outside of the mainstream. Nerds, more interested in intellectual matters or really drilling down into computers or math or whatever it is.
- 4.87** What is interesting, since language evolves, is that in many cases, what starts off as an insult then comes to be adopted by the group and they will start to wear it as a badge of pride. "Yes, I am a nerd. I'll get a T-shirt, nerds rule. Be nice to nerds. You're going to be working for one, one of these days." All of that sort of stuff. There's an ongoing verbal battle over phrases and the emotional connotations that are associated with them.
- 4.88** When we are debating complicated issues, to what extent or not should a person's race or sex or gender or religious background and so forth be a factor in making decisions about hiring and promotions and who gets into college and who does not, people who are on the opposite side of that will try to say, "Well, it's a kind of discrimination, but it's just discrimination in the other way." So the most literal true statement should be, it's a reverse discrimination. If we're trying to make up for discriminations that went the other way, we're just reversing it, but it's the same thing.
- 4.89** And the implication of this phrase is that there's something wrong with it. By contrast, people who are proponents of it are more likely to say, what we are doing is something positive or something affirmative. So discrimination has a kind of a bad taste to us, and just reversing it doesn't sound like progress. But if I can say we're taking action in an affirmative direction, trying to make up for past wrongs, then we use that label. So both of them carry emotive content with it.
- 4.90** In political issues, when we're dealing with people who are engaged in non-standard ways of protest, violent ways of protest, sometimes organized, sometimes not, that are affecting lots and lots of people negatively, lots and lots of victims, is this person a terrorist or is the person a freedom fighter? Very often people who approve of those tactics in that context will say, "Whatever the person's cause was, I approve of that cause, and there's a freedom component of some sort in that cause, and so this person is fighting for freedom."
- 4.91** People who disapprove are more likely and more quickly to say this person is a terrorist, because it's hard to find any redeeming qualities in just terrorism for the sake of terrorism itself. Or "Marxists versus commies." This is a more neutral person, in a

description of what the person stands for; commies is slang and a little bit derogatory formulation.

- 4.92 So one thing to attend to is, when you are reading a passage or you hear someone, or of course, also yourself, if you are worked up in the heat of the moment, is the language that you are using, emotional language, which is fine, but is it carrying content, or is the emotiveness of the language doing the work and not actually any logic to it?
- 4.93 Here's a slightly made-up example, taken from social media. Some of the debates over climate change, and some people believe that oil is the main culprit, and behind the oil, it's oil companies are the main culprit, so they want to attack those guys. Of course, there are issues and arguments to be addressed on this whole complex of issues as well. But this my massaged version of the social media post.
- 4.94 "The only possible cure for the present climate crisis is the elimination of all oil and coal companies. Finally, my friends, we must put an end to the diabolical conspiracy of the oil profit-mad vultures of Big Oil. Don't let the billionaire tycoons wrap their slimy tentacles around our children's future and squeeze it dead." OK. Now, that's some strong language. Slimy tentacles, profit-mad. In some quarters, Big Oil is taken to be a bad thing, squeezing things dead and so forth.
- 4.95 But if we take a moment and say, "Is there any content to this?", the conclusion seems to be, we need to completely get rid of oil companies. Why? Well, diabolical conspiracy. What diabolical conspiracy? Why is it diabolical, which has religious overtones? Are they really inspired by the devil in doing so? Oil profit-mad. Well, OK, of course they want to make profits, but does that make you crazy or mad? Is that the only thing that is going on in their minds? Et cetera, et cetera.
- 4.96 If you unpack each of them, it turns out that there is a neutral way of stating that. But even when you translate it into the neutral language, it's not advancing an argument; it ends up being rather emotional. Now, emotional language, I can't overstate this, has a perfectly good place. It doesn't mean you have to turn yourself into a robot or an unfeeling person making various arguments. You just have to make sure your emotions are tied to the logic.
- 4.97 We have more legitimate uses for poetry as well. In many cases, poets are very good. The good poets will have something to say, something important and significant to say. They could, of course, say it quite neutrally, but one of the things we do like about poetry is a careful attention to the kind of language that emotionally accentuates whatever the message is that the poet wants to communicate.
- 4.98 In this case here, this is a title of a famous Romantic poem from about 200 years ago. But suppose, instead of calling it or titling it, the poet had said, instead of "Ode to a Grecian Urn," I said "Poem for a Greek Pot." OK, well, an urn is a type of pot. Grecian, poem, a type of pot, and so on. This is more standard. Grecian sounds, as an adjective,

a little more elevated. Poem, a little bit neutral; ode is more precise and a specific type of poem and so on. So this works better in poetry, but it carries this information, but adds the emotive language, and that's perfectly fine.

- 4.99** Another fun one and a popular one we call a false alternative. As the name suggests, the person presents an alternative to you. It's an either-or, choose this or choose this. But it's a false alternative because there are, in fact, more alternatives. Sometimes those other alternatives are a little more subtle, or they're not immediately present in the context that you are operating in. But the person wants to force a decision in a certain direction, so they will limit the range of alternatives available to your mind and try to get you to adopt one of those.
- 4.100** I want to start with a joke on this one just because it's a classic joke from the comedian Steve Martin, now older, but one of the comedians of my generation, anyway. So he tells this joke, when he was younger. He comes in at the start of the set and he says, "I'm feeling a little bit sad because my girlfriend's not with me anymore, and I'm a little bit broken up about that." Now, then immediately people are there, "Well, Steve broke up with his girlfriend. Oh, poor guy. That's too bad."
- 4.101** And then he says, "Actually, she's dead." So you get the elevation from breaking up to dead. Oh, then there's nervous laughter at this point. Then he says, "I'm partly broken up because I feel a little bit responsible for her death." And now we all want to know, what's the story here? He says, "Well, we went to a party together, and I had a couple of drinks, but not too much. But she was really having a good time. And she was drinking a little bit too much and then a lot too much. And then she came up to me and she said, 'Steve, I think I've had too much to drink. I want to go home now.'"
- 4.102** And then Steve said, "Well, I was having a really good time and I didn't really want to leave at that point. So I said a few minutes. So she went off and had another drink, and I continued to have some fun. Then she came back, 'Steve, I really need to go home now.'" And Steve said, "I really didn't want to leave the party because I was having such a good time, so I shot her."
- 4.103** And that's the joke, but very dark humor. So what is setting this up, though, is a false alternative. So at that situation in the party, you're thinking, OK, the alternative Steve faces is, take her home, drive her home, be the responsible boyfriend, or maybe call her a cab to send her home. Those are the alternatives that are legitimate to be facing her.
- 4.104** So out of the blue, apparently in his mind, shooting her is another alternative for the circumstance, and that's the one that he chose. And that is kind of a false alternative, Steve, because you do have all these other options that are a lot less extreme than shooting the person just because you're having a good time at the party. So that's a false alternative, a rather blatant one.

- 4.105** The example we looked at earlier as a complex question, it also is a false alternative. "Do you want to be a good boy and go to bed?" So the mother or the father is presenting it. Either there is this package, you go to bed, or you're a bad boy. That's the implication. Obviously, there are other options. If you don't go to bed, it doesn't mean you're a bad boy necessarily. It could be that you stayed up for another 30 minutes and you're still a good boy and so forth.
- 4.106** Often in political contexts or religious contexts, people want to turn things into an either-or, and they don't want to deal with people socially in an inappropriate way. So you're either with us or you're against us. That's the bottom line that is communicated. But often it's not the case that you're either with or [against]. Sometimes, of course, you have to make a choice. What you stand for, you don't have any alternatives. But lots of times, it's an issue and you don't care about that issue. So you're not for or against on this issue. Or you're undecided. Maybe I'll be on your side after I think about this more, but at this point, it's a false alternative to say either-or.
- 4.107** Suppose this is a class involving testing. We've talked about certain material. You're in the midst of taking a test, you're writing away. At the end of the test, I'm your professor, and I come up to your desk where you're writing, and I take your examination, and I say to you, "You cheated," and I proceed to tear your examination up and deposit it in the trash. You say, "What?" You're surprised. "I didn't cheat." And I say to you, if that went well, "Prove it. Prove to me that you did not cheat."
- 4.108** In that context, if that's all I've got, I have committed what logicians call the fallacy of appeal to ignorance. Now, you probably at this point would say, "How would I try to prove that I didn't cheat?" You've been sitting there and you're doing your writing, you didn't look right or left, you didn't have secret cheat sheets. None of those things exist. So there's nothing that you can proffer as evidence of your non-cheating. If you didn't cheat, there won't be any evidence of non-cheating, because not cheating is a non-existence.
- 4.109** By contrast, I am the professor. Maybe I have a responsibility to make sure that student work, or part of my responsibility to make sure the student work is reflective of what the student knows. If I want to make the claim that you cheated, then I can point to evidence, if there is such things as evidence, because cheating requires certain actions taking place.
- 4.110** I can say, "Here we were videoing. I can see that you were looking at her thing for a certain number of minutes and copying down what she wrote." I can look in your ear and see evidence that you have a remote speaker in there and someone is feeding the answers to you, or you have a cheat sheet. Aha! I pull it out from under your desk, and so on. In that case, if there were certain actions that actually occurred in reality that are cheating actions, there will be evidence for it, and it's my obligation to provide them.

- 4.111 In that situation, we call it an appeal to ignorance because you are ignorant in this context of not being able to come up with anything to support your side of this disagreement. I am appealing to your ignorance. You can't prove that you didn't cheat. Then I'm going to take that as evidence for you did cheat, and that justifies me throwing your examination away. Appeal to ignorance is an illegitimate mode of reasoning. I want to make a claim something exists, it's up to me to prove it.
- 4.112 Now, the reason for this is that we all start from a default position of not believing billions and trillions and so forth potential things. Things are not in our heads. We don't have to start off with some sort of obligation to explain why a certain idea is not in my head right now, why I don't believe a certain thing. It goes the other way around.
- 4.113 For me to justifiably or logically have some idea in my mind, I have to show what led me, what facts of reality or inferences from facts of reality, led me to have that idea in my head. If I just randomly come up with some idea and say, "I want to believe that idea," and you haven't proved why that's a bad idea, you've never considered that idea before, therefore, I'm justified in believing whatever random thing in my head, that's appeal to ignorance.
- 4.114 A slightly dated example, going back to my mother's generation of college, when colleges were much more ruthlessly segregated between boys' dorms and girls' dorms; and the girls' dorms had very strict policies about times when young men were allowed to be in the dorms, where they could be in the dorms, and when the curfew kicked in and they absolutely had to be out, and these were very strictly, strictly monitored.
- 4.115 And my mother tells the case of a young woman she knows who was accused of having a boy in her room at the wrong time. And it turns out that this was just a revenge accusation from one of the girl's enemies, but she reported it to the house mom or whoever. "She has boys in her room, or a boy in her room," and the house mother goes up and says, "You had a boy in your room." And she says, "No, I didn't." She said, "Well, prove it. Prove that you didn't have a boy in your room," right, at that.
- 4.116 Well, how would you do that? This is before video cameras and various sorts of things like that. But if she's correct, and there was no boy in her room at that, there won't be any evidence of that, because non-existing boys don't leave traces of where they have gone in the world. In this situation, the obligation is with probably the first girl who allegedly reported it; but certainly, the house mother, if you're going to prosecute or, that's too strong, if you're going to pursue this, rather, have some actual evidence for why you are putting this person in the hot seat.
- 4.117 Now, this takes us to an important distinction that we have to insist upon. We call existence claims versus nonexistence claims. There's a difference between saying or thinking or considering that something exists, versus the position of someone who does not consider that something exists, doesn't believe that something exists or things not

actually existing. The idea of logic is that we're trying to keep our minds connected to reality, and the only thing that connects our minds to reality is evidence from reality.

- 4.118** We always start from the position of asking, if you want to say something exists or something is real or something is true, you start from having the obligation to provide that evidence for it being there. If something doesn't exist, then it doesn't leave any traces. It doesn't leave any evidence. Non-existing things don't interact with anything else in the world, so they can't be detected. They can't be detected because they can't exist. So it's just an empty void.
- 4.119** And so asking for evidence of nonexistence is a very different kind of claim. It's almost impossible to achieve that. That's why we have a closely related principle that we call the burden of proof principle, or sometimes it's called the onus of proof principle. So sometimes this is a rhetorical issue. Whoever starts the discussion, starts the argument, puts something out there—let's talk about this—that person does bear an immediate onus or burden of justifying why we should be talking about that. That can be partly a motivational issue.
- 4.120** But if you are going to make a claim, "I think this is true, I think this is reasonable, I think this is a probability," and so forth, you can't immediately say, "This is my claim. Now, you disprove this claim." That would be to shift the burden of evidence to the other person. If you are the one who's making the claim, you start with the burden of putting some evidence behind that claim. Shifting the burden is almost always illegitimate. Particularly, it's illegitimate if you are allowed to just put these things out there, shift the burden to the other person, and then put something else out there that shifts the burden to that person, because you're never going to get anywhere.
- 4.121** Let me give you an example of that. I will pick on this gentleman here for a moment. Suppose there was some terrible crime that happened last night, and I'm a police detective, and I don't have any evidence, right, one way or the other, and you just happen to be the first person I come across, and I say, "Prove to me that you didn't commit that terrible crime last night." Yeah, exactly. And you're saying, right, your immediate reaction. "What? What terrible? Oh, OK, there was a terrible crime. Well, it wasn't me."
- 4.122** OK. So personal testimony. Says, "Well, I don't know you. I can't trust you. Prove to me that you are a reliable person and you're not lying." How would you prove that you're not telling a lie? So then you've got another burden, and you have to start to think, right, and so forth. "How could I tell him?" He said, "Oh, well, wait a minute. I just flew back from Switzerland last night when the crime was occurring."
- 4.123** So you whip out your passport, and there you've got the Zurich stamp on your passport from last night and your Swissair flight ticket and so forth. And so you say, "Aha, see, I can prove that I didn't commit that crime because I've got these other kinds of evidence. I've got a passport with a stamp. The dates are all right," and so forth. I say,

"OK, well, that's fine." Now I'm a clever detective. I say, "Well, prove to me that your passport is not a forged passport, that you didn't pay someone a lot of money to make this fake passport so that you could be secretly committing this other crime as well."

4.124 What's going on here is, and I can play this game all day, and maybe you're pretty smart and clever, you can play the game for a pretty long period of time, but you're never going to be able to prove that you did not commit the crime, if that's the game that we are playing. I'm consistently asking you to prove a negative. Prove that you didn't commit the crime, prove that you didn't lie, prove that you didn't have this passport for it. Prove that you didn't run. So that constant negative language is always a clue that something is going on.

4.125 Instead, we say the proper thing is, if I'm going to make an accusation of him in particular, I need to have some glimmer of evidence that ties you, and I can start with that, and then on the basis of that, build a further piece of evidence. That is the burden of proof principle and the appeal to ignorance. Immediately following the burden of proof and the distinction between positive claims and nonexistence claims, and what we then more formally call the appeal to ignorance, is a long, long history. I want to indicate some philosophical issues behind that, ones that have important implications for law and for religion. It's one that has gone deep in human history.

4.126 In Roman jurisprudence for the first time, there was this burden of proof principle, and the claim there was, the necessity of proof always lies with the person who lays charges. You want to sue someone in a civil court, if you're the plaintiff, you have an obligation to present the evidence behind your suit. It's not that the person has an obligation to disprove the charges that are just thrown out against them. In a criminal case, the police and the entire court system have an initial obligation to come up with the evidence.

4.127 So we say, if you are functioning as a police officer, as I was in the earlier example, I have to have some probable cause. That is to say, I have to have some evidence that the cause involved this particular person, and it's probable. It doesn't have to be certain at this point or even particularly compelling, but I have to have something before I can start interacting with the citizenry and subjecting them to further investigations.

4.128 If I, as a police officer, gather a significant amount of evidence, and I take it to the prosecuting attorney, and the prosecuting attorney agrees this is worth taking to court, the person is arrested, we bring the person into a court, we are going to have a defense. We are going to have a prosecution. They both will have a chance of making their case. But the making of their case, the obligation of making the case, is asymmetrical. The burden of proof lies with the prosecution. If the prosecution fails to prove it, the defense does not necessarily have to do anything. If, just on the face of it, the prosecution fails to make the proof, then we are going to consider the person not guilty and let the person go.

- 4.129** Another way of formulating this, tightly tied into the principle of the burden of proof, is the presumption of innocence. When you enter into civil society, the presumption is you are an innocent person; you are free to go about your business, and police officers can only interfere with you if they have some grounds to think that you are no longer an innocent person. In a courtroom, even though the person has been arrested, the prosecution thinks they've got a case. In the eyes of the court, that person is still innocent until he or she is proven by the prosecution to be guilty.
- 4.130** We don't want the prosecution to be playing the same kinds of games of shifting the burden of proof over to the defense. I'm the prosecutor. I don't have any evidence. I don't have any logical arguments. You, defense attorney, prove that your client is not guilty of the particular client. That's an illegitimate move. Always, it will be objected to and so forth.
- 4.131** Now, there is a purely logical reason for this; that is that distinction between existence claims and nonexistence claims. If the person is guilty, that is to say the person did something, something exists in reality, there will be causal interactions and we can get the evidence for that. That's why, logically, we put the burden of proof on the prosecution. If it didn't happen, it's a nonexistent thing. There's no non-evidence that the defense can come up with to bring in. You can't bring nothing into the court and say, "Here, look at the nothing." Because the nothing doesn't do anything for you, it's a completely empty nest.
- 4.132** But there is another philosophical issue at work. It's one that's often contested, even to this day in some legal jurisdictions, because not all legal jurisdictions around the world now, or certainly historically, have had a presumption of innocence or had a burden of proof principle. In some cases, it has been the opposite, and that was for philosophical reasons. There would be the presumption of guilt. We assume that you are a bad person, and you have to show in some way that you're on the up and up, and then we will perhaps let you go about your business. Then even more formally, when we bring you into a court of law, we will require you to show that you are innocent.
- 4.133** Now, sometimes this stems from a philosophically very different view of human nature. If, for example, you think human beings are born bad, awful, terrible, we're all little monsters or whatever, that's the default understanding of human condition. Sometimes we might put this in strong religious language and say that everybody is born in sin, in original sin, and we expect that the badness and the wickedness of human beings is going to dominate their thinking and dominate their actions.
- 4.134** If that's your understanding of what it is to be a human being, then human beings are guilty from the get-go, and the innocent or good behavior is something that's going to be exceptional and rather rare. Then you're more likely, in a judicial system, to say, "Well, why should we presume that this person is innocent? Chances are very good from the get-go, what we know about human nature, that this person is guilty."

- 4.135 So you're going to much significantly lower the burdens of proof on the prosecution, make it a lot easier for the police to arrest and bring people and charge them. Your standards of proof from the prosecution will be a lot lower, and you will correspondingly elevate the amount of work that the defense has to do to try to show that they were not guilty in this particular case. So this philosophical issue about whether we think people are born basically good, whether we think they are born basically bad, or whether we think they are born neutral, tabula rasa, morally, and might become good or bad, has important implications for legal procedure.
- 4.136 It's also interesting, if you look at some of the methods that were considered legitimate for hundreds and hundreds of years in Western jurisprudential circles until, in the early modern world, they were gotten rid of, you take, for example, trials by ordeal. Suppose, I don't know, there's something that has happened in the village, and I'll pick on that guy this time. I think that he is the bad guy, so I will bring him up to the court of law. "Did you do this bad thing?" Of course, he's going to say, "No, I did not do that bad thing." But we all know that everybody's a liar because they're all wicked, so he's probably lying.
- 4.137 But what's my evidence that he did that thing? I don't necessarily have it, but we can figure it out by asking him to undergo a certain ordeal, and you will be willing to undergo this ordeal to try to prove your innocence. So here's the ordeal. We'll pretend that this is an iron bar. We'll take this iron bar and heat it up until it is red-hot. Red-hot means that it will ordinarily burn and damage terribly any human skin that it comes in contact with.
- 4.138 We will ask him to put his hand out. This will be laid in his hand, and he will voluntarily—so we're asking him to initiate something, to engage in some action here—clasp it and hold it for a certain amount of time, and then unclasp it. Then we will have the evidence of whether he is innocent or guilty. Now, how is this supposed to work?
- 4.139 Well, the evidence here is going to be based on a certain religious presupposition. If he is an innocent person, if he's not a guilty person, that's kind of unusual, for someone to be innocent, in fact. But God protects the innocent. Does he not? It might be an unusual circumstance that this guy is in fact innocent, but if God really is protecting the innocent, then God will protect him in this particular case. If we, after this certain amount of time, have him release it, and his hand is fine, then he has proved his innocence.
- 4.140 But of course, if not, he has failed to prove his innocence, and the judicial system takes him away, considers him guilty, and he is gone. So that is the exact opposite, partly driven by different commitments to logic and burden of proof principles, partly driven by understandings about metaphysics and how the world works and our understanding of human nature.

- 4.141 One more to take with you. If we take that jurisprudential example of this, you can think about this one more on your own time. Burden of proof principles in discussions of religion. Logicians will argue that, obviously, if we're going to believe in the existence of God, we have evidence, we have arguments and so forth, so we pay attention carefully to evaluating the arguments for the existence of God. And the open question is, do I think the arguments work? In which case I believe. If I don't think the arguments work, then I disbelieve until better arguments or better evidence comes along.
- 4.142 All of that is perfectly legitimate from the perspective of logic. But the logicians will insist that certain tactics that are sometimes used involve shifting the burden of proof inappropriately. So if we want to talk about the existence of God, well, we are making an existence claim. We are making the claim that, in the universe or in reality, there is such a being or beings, and we're going to call him God for the sake of this argument. This person really exists.
- 4.143 If I am making that claim, then I bear the initial burden of putting evidence to it, putting logic to it, and so forth. It would be an appeal to ignorance if I am wanting to say, "God exists, and I would like you to believe in the existence of God," and then shift to you to say, "Well, prove that God does not exist." If you cannot prove that God does not exist, then I am reaffirmed in my belief in the existence of God.
- 4.144 That is an illegitimate shifting of the burden of proof, so work that one through. Obviously, in jurisprudential matters, political matters, religious matters, the stakes are high. This can be a temptation, but it's one that sometimes it also can be quite subtle. So, resist and practice.

Lecture 5: Deduction: Categorical Syllogisms

At a glance — Deduction: categorical syllogisms.

- 5.1 We're now going to turn more to emphasizing the positive. Good arguments, valid arguments, as we like to call them in logic, and then even upgrading to sound arguments. Last couple of sessions, we've been focusing unrelievedly on the very common bad arguments that we see fairly regularly, the so-called informal fallacies.
- 5.2 We're going to be looking at a class of arguments that we call syllogisms; I'll say why they're called syllogisms in a moment or so. There's a lot of them, and a significant number of them are valid. We will be able to figure out which ones are the valid ones. We'll practice some, so we start to train our logic muscles, logic memory muscles, to recognize those patterns. We'll be able to more easily differentiate the ones that are invalid. Again, developing some memory muscle, so to speak, in our logic brains, so that when we see their sense, those patterns, we'll be less likely to fall for them or to attend to them a little bit more.
- 5.3 Now, a syllogism is any argument that has two premises and a conclusion. There's a division of those into three different types—categorical, hypothetical, and disjunctive. Over this session and the next section, we're going to develop techniques for identifying those and evaluating them. Before we plunge into that, I want to touch base again with the terminology from our very first session, where we introduced the concept of truth and validity.
- 5.4 Once we've done the work of figuring out what the argument is, we've got the premises identified, we've identified the conclusion, often we start putting them in standard form, premise, premise, however many there are, conclusion. Then we turn to the evaluative work. Is this a good argument, logically speaking? There are the two virtues that we are looking for. Of course, we want to have the premises be true.
- 5.5 When we're making an argument, we start off with an initial logical obligation of, if I'm going to believe something or reason on the basis of something that I believe, I should make some effort to make sure that it's true, that it's a fact, or that it's justified by some previous logical reason, and I can take it as reasonably true. We will look at all of the premises individually in the argument. Is that true? Is that true? Is that true? Or however much. Then if we have questions about it, we do our side research to reaffirm ourselves whether the argument is true or not.
- 5.6 Typically, we don't spend a lot of time on truth in logic classes, largely because we're arguing about everything in the universe and we're developing tools to enable us to argue about anything in the universe, but we don't have time in the logic class to independently figure out everything in the universe or even go research all of the various areas.

- 5.7** In that sense, logic recognizes that arguments are going to be good if they have true premises, but that's the job of the physicists, the mathematicians, or you in your ordinary life, to gather the data, the facts, to make sure your premises are true. We are more likely to specialize in what you do with those premises, how you combine them and recombine them, sometimes slice and dice them and repackage them, in order to try to make an inference, to make a mental move from the premises to the conclusion, to say that this conclusion follows from these premises.
- 5.8** We will take it more or less for granted that the premises are true or arguable about. What we're more interested in is what's going on structurally or formally in the move from the premises to the conclusion. That is where we use the concept of validity. What's the structure of the argument? What's the pattern of the argument? What's the form of the argument? Then is it a good one or not?
- 5.9** If it's a good pattern, and we've tested it, we know it's a good pattern, we will say it's valid. If it's a bad pattern, we will say it is invalid. The clarification is, and technical terminology, just automate it, truth is a matter of the individual premises. Do they fit the facts or not? Validity or not has to do with the logical moves or the illogical moves that one is making.
- 5.10** In this case here, let's use these three as test examples. First one, all humans are mammals and all mammals are mortal, so all humans are mortal. Now, this is a categorical syllogism. It has three categories— mammals, humans, and mortal. Then those three categories are connected in various ways. What we're interested in for the logic is the nature of that connection.
- 5.11** But our basic work is, we would read that through and we would say, there's a "so" there. So I would circle the so or underline it. Sometimes it's good to have different colors. This is a conclusion indicator. This is my conclusion. All humans are mortal. These must be my premises because all humans are mammals and all mammals are mortal.
- 5.12** Now, first evaluative questions. Is it true that all humans are mammals? As far as I know, true. All mammals are mortal. Is that true? As far as I know, yes. So check, check, check. But we're more likely in a logic class to say, OK, well, this really is an issue for biologists. This really also is an issue for biologists. We're not biology experts here, so let's just take that for granted. It seems like it's true, but what we want to know is, if that's true, and now I've shifted into a hypothetical voice, if that is true, and if that is true, does it follow logically that this has to be true?
- 5.13** There's no question in your mind that it has to be true. And in this case, we would say yes. In my evaluation, we say, yeah, fine. We have true premises. Just abbreviating truth with the T. And if this is good logical structure, we would say it is valid. And by valid, we mean the conclusion follows necessarily from those premises. If it doesn't,

there's a big question about it, or we can show that in fact that question is justified and the conclusion does not follow, it is invalid.

- 5.14** Now, here's the first contrast example. Let's read this one through. All rabbits are birds, and all birds eat whales, so all rabbits eat whales. And you go, "Oh my god, what's wrong with this guy?" Well, lots is wrong with this. So we start our first evaluative questions. We have the so. All rabbits are birds. Is that true? Well, no. All birds eat whales? No. So we have false premises.
- 5.15** The question of validity, if, hypothetical mode. Suppose we took for granted that this were true. And suppose we took for granted that this were true. Would this follow necessarily, logically? The answer is yes. This conclusion follows logically from those two premises. So this is a valid argument. Often a big psychological hurdle moving from ordinary language, where we sometimes use valid a little loosely, just generally affirmative, to precise logical language, is to say, in one way, this is a completely ridiculous argument, but it's perfectly valid, and I love it, logically speaking.
- 5.16** When you say you love it logically, you're not attending to the truth of the premises. That's an important question, but that's not our question right now. We're interested in the structure of the argument. And in fact, the structure of this argument is exactly the same as the structure of this argument. They're just changing the content. But we're going to abstract from the content and focus on that pattern.
- 5.17** That's a very good pattern. It's a very useful pattern. There's lots of them. This pattern is actually, is structurally called Barbara. All of the medieval monks who first systematically identified all of the possible syllogisms, they gave them each a personal name, and this one is Barbara. So meet Barbara. She is lovely. She's always valuable, always valid, always reliable.
- 5.18** Now, this one here, all humans are mammals, and all mammals are mortals, so all humans like jokes. This is our other contrast one. Well, two premises and a conclusion. Here's our so, so we know this is our conclusion here. All humans like jokes. We do our evaluative work. Well, is this premise true? Yeah. All humans are mammals, well, yeah. All mammals are mortals, yeah. So again, we have true premises.
- 5.19** Then we ask, well, supposing those premises are true, does the conclusion follow from them? In this case, we say, no, obviously not. This conclusion has almost no connection at all to any of the premises. Yeah, it's true. Here we're talking about humans, and here we are talking about humans. But here humans are connected to mammals, and here humans are connected to jokes. There's no connection in this argument between being a mammal and being a joke-liker. We would need more information to be added, but that information is not here.
- 5.20** This conclusion does not follow from this information. This is an invalid argument. That pattern also, unfortunately, is used rather frequently, not in really obvious cases, but it comes up often enough. It's one of the irrelevant ones. What we do start to learn is,

what we want is, in the premises, for this to be linked to this, humans are linked to mammals, mammals then to be linked to something else, and then you're looking for a kind of overlap that enables you to link the humans to the other thing. But it becomes very complicated what kind of link and how much information one is getting in all of those cases.

- 5.21** So that's our rehearsal of this language of proof and validity. Introduce one more concept. If an argument has both virtues, it has true premises and valid reasoning, where the logic is good and the information that is incoming that is good, let us just say, it checks all of the boxes— that is the gold standard of argument. We call those arguments sound. It's a sound argument, and that's the highest praise you can give to an argument.
- 5.22** You can say three good things about an argument. It has true premises. That's not enough. It can be a valid pattern of reasoning. That all by itself is also not enough. But truth in all of the premises plus valid reasoning is a sound argument. In this case, we only have one argument that meets that set of criteria. So this argument, we would say, is sound.
- 5.23** This argument is unsound because it has false premises in it. So we would reject it on that grounds. Maybe we wouldn't reject it. Maybe we would say, this is a rough draft; maybe you need to go back and rework it in some way. But right now it is unsound. This one also is unsound. Good information, but you botched the reasoning with respect to it. It is also an unsound argument.
- 5.24** All right, now let's turn with that rehearsal and background to our next major module, which is to deal with syllogisms. And syllogisms are arguments that have three components, the two premises and a conclusion. They're a basic form of reasoning. I've got examples of the three kinds of them. The ones over here that we looked at, these are all syllogisms, two premises and a conclusion. We call these ones categorical syllogisms.
- 5.25** The reason for that is that they are talking about the connections between categories. So one category of things. I'm going to start doing circle movements a lot here. Imagine this is our category of humans. Imagine all of the humans are in that category. Or if you think of what's going on in your mind as analogous to a file folder, like a filing system, and all of your files are interconnected to each other. You can open each of the the folders in your system and put more information as you want. But then you want to start making connections between the different file folders.
- 5.26** So these are the categories. So we have the category of humans, the category of mammals, and the category of mortal things. Then we're making some assertions about the connections between things in those categories. Same thing here. We have some different categories. We have rabbits, we have birds, and we have things that eat whales, whatever. That's probably a pretty small category, but there's a category there.

- 5.27** Then we have these same categories here, but now we have another category, things that have a sense of humor or like jokes. We can work out all of that. That will occupy most of our time for this session. But there are other types of syllogisms where it's not simply categories that are in question, but rather, as in the example here, we are thinking hypothetically and talking about causal connections or at least logical implications from whatever the content of our reasoning is.
- 5.28** If he is guilty, then he was in the victim's apartment, but he was not in the apartment, so he is not guilty. Now, this is a different argument because what's very important to this argument is this if-then structure, that formulation. I'm not saying there is a relationship between this and this, as I am saying here. I'm saying all of the humans are also mammals. There is a positive relationship.
- 5.29** I'm explicitly jumping to hypothetical mode here. If he is guilty, then he was in the apartment, taking that in this case—there's my conclusion indicator— as a premise to reason further. We will say these are hypothetical syllogisms. We look for an if-then or semantically equivalent statement to an if-then statement.
- 5.30** This second one: Either she's a genius or she made a lucky guess. She's no genius, so it must have been a lucky guess. Now, here we have genius, people who make lucky guesses, and so on, and her. We're talking about the relationship between those three things. But what's driving this argument is not a simple categorical overlap, but rather this formulation, the either-or structure.
- 5.31** Then we have a category of syllogisms, two premises and a conclusion, but it's driven by an either-or statement, an alternative. Then, so we will call these, sometimes these are just called alternative arguments, but more formally, we call them disjunctive arguments. A disjunct is basically a fork in the road. You're going this path, you can take this disjunct, or you can take that disjunct.
- 5.32** Then these ones are the ones that we are going to call categorical. Those ones are all categorical as well. There's no if-then statements involved in it. There's no either-or. It's just simple categories and their relationships. In this course, we always have to maximize the Canadian content when we can. Some Canadians are mathematicians, and no Canadians are bad at hockey, so some mathematicians are not bad at hockey. Second premise, of course, is true—not really.
- 5.33** We're now going to be focusing exclusively on categorical arguments or categorical syllogisms, and they are the most complicated of them for reasons that I will explain. In our next session, we'll turn to hypothetical and disjunctive arguments. We'll be able to deal with them in one session because they are less complicated to deal with. Being able to develop the tactics or the methods for sorting out the valid ones from the invalid ones is relatively simpler, but we need more apparatus for the categoricals.
- 5.34** A categorical syllogism has premises and a conclusion, but the first thing we need to focus on and get comfortable with are the categories and what we're doing with the

categories in the propositions that make up the premises. As we have seen in the case of this one here, we will have, all humans are rational and capable of self-agency. We look at that and we say this has three categories.

- 5.35** There's the human category. We imagine everything in the human category. If we were to draw a circle, everything in that circle is the humans. Anything outside of that circle, non-humans, anything that isn't a human. Rational, another category, all of the things that are rational, things that are capable of self-agency. We'll start to do this with increasing frequency because the easiest method for determining the validity or not, visually, involves what we call Venn diagrams, and they involve using circles.
- 5.36** So what we would just start doing is saying, here's a circle. I'll just label that with an H. There's all of the humans. And ahead of time, we know that we're trying to say there's a relationship between humans and other categories. So in this case, we have rational things. And abstractly, so far in our diagram, what we're just doing is saying, well, these would be things that are human, but if they are outside of the R circle, they're not rational.
- 5.37** And these here would be things that are inside the human circle, but also inside the rational circle. Those would be things that are both human and rational. Anything here, this would be something that's rational, but it's outside of the H circle that would be non-human, non-human rational things, whatever those might be. Again, we're in hypothetical mode here. And then here, outside of both circles, would be things that are neither human nor rational, whatever that would be. Maybe this marker is not a human, it's not rational. The marker lives here, so to speak, on this diagram.
- 5.38** Then we would do the same thing with capable of self-agency. Then we would draw a circle that looks like this, make it a little bit prettier there, and I would just abbreviate that, capable of self-agency. Those would be all of the things that are capable of self-agency. These are things that are capable of self-agency, but they are not in the H circle, so they're not human. They're not in the R circle, so they would be not rational. These would be things that are capable of self-agency and rational, but not human.
- 5.39** This would be humans that are capable of self-agency, but not rational. These might be empty places. These are just logical possibilities and so on. Then again, outside, things that are neither capable of self-agency, not rational, not human. Then in the middle, things that are human, rational, and capable of self-agency. What we're trying to do then is capture all of the possible ways these categories could or could not be related to each other.
- 5.40** We would do the same thing with this here. We have three categories—dirty, stinking, and liar. We have three circles like this. Here's all the stinking things. Here's all the dirty things. Here's all the liars. Then we want to know what exists where. Then John, well, there's John. He's a real thing. Typically with individuals, we just consider them their own category. This is the universe of John, the John category. Then things that are

funny, things that are sophisticated. Samantha, she's her own category. Things that are lean, things that are elegant. And so we'll work toward diagramming them.

- 5.41** Now, when we look at all of the possible category statements, we start from a proposition, and we see that all of the categorical syllogisms will have two propositions that serve as premises, and then one that serves as the conclusion. If we look at each of the individual premises, they all have two categories that are linked to each other. That sentence or that proposition asserts a relationship between those categories.
- 5.42** All humans are mortal. That is to say, I've got the category of humans, category of mortals, two categories, and all of the humans are also in this other category, mortals. What we have then is, all, in that case, S are P. I'm just changing my lettering because, in the logic universe, for the initial descriptions, we use S and P as our variables for the sentences.
- 5.43** One thing we can do is, say this is the category of S and the category of P, and all of these things are also in that category. That's a very strong claim. Sometimes we want to say that not necessarily all of them are in that category. We want to say, some of them are in that category. We pay attention to the quantity that we are asserting. I might, as in the one we looked at a moment ago, say some Canadians are mathematicians. I've got the category of Canadians, I've got the category of mathematicians, and I'm asserting a category relationship between them.
- 5.44** But I'm saying, some of these ones are also in this category over here. I'm going to put that here. Some of the S's are P's. This is our subject, this is our predicate, this is our copula, and we call this the quantifier. It tells you what quantity you are dealing with. This is talking about the category universally. This is a universal quantifier. They were saying just some particular amount. It's not universal, it's a particular quantifier.
- 5.45** Now, what else could I say? I could say, well, actually, take these two categories; there's no overlap between them at all. I could say, none of the S's are P's. I might say, no fish are angels. Fish are fish, angels are angels. There's no overlap between those things. I can say, no S are P. Now in this case, I'm also speaking my quantity. It's a negative quantity, but it's also universal. The whole category, all of them, none of them are in this other category.
- 5.46** Sometimes I want to say, my information is that there is an overlap between these two categories. Sometimes I know only that, for some of the members of the category, there's not an overlap with the other category. I got these two categories, and all I know is that some of these ones are not also in this category over here.
- 5.47** I might then say, some Canadians are not bilingual. So I've got all the Canadians, all of the bilingual people, and I've met some Canadians who speak one language, so they are not bilingual, but that's the only information I've got. We would say, some S are not P. Now, the first big claim that logicians will make about categorical language, categorical

propositions, is that every categorical proposition is one of these four. They all boil down to one of those four.

- 5.48** And this is, everything likes its terminology. We also, for shorthand, give these labels. We will call this one the A proposition, and so forth. We carry on. I proposition. The universal positive is the A prop. The particular positive is the I prop. This one, the universal negative proposition, we call it the E proposition. And then the particular, but still negative proposition, we call that one the O proposition.
- 5.49** So clearly, whoever came up with the names said, "Let's just take the vowels and run with—A, E, I, and O." There is strong claim, there's no U proposition. All of them reduced to that. Now, let me say some propositions, because we know in English, there are lots of different ways of saying the same sort of thing, and how logicians will massage the language to show that, really, it reduces to one of these four propositions.
- 5.50** So suppose we're talking about fishing and I want to say, many fish are caught each year. That's the English formulation. Many fish are caught each year. Well, we have fish and things that are caught each year. That's our category. What do we do with the many? And the logic answer is to say, well, it's not all. Anything less than all for us is just some. So that is the equivalent of saying, some fish are caught each year. That's an I proposition.
- 5.51** Not all dogs are fast runners. So we have dogs, fast runners. We're saying, not all dogs are in the fast runner category. What do we do with this "not all"? Well, if it's not all, then it must be only some, but we're also saying it negatively. What we really are saying is, some dogs are not fast runners. Not all is the same as, actually, some are not. So that is an O proposition.
- 5.52** Most Americans aren't communists. Americans, communists. According to the information given in this proposition, most, well, it's not all, so it's some. It's just anything less is some, and are not. So some Americans are not Communists. Another O proposition. Every Catholic is a Christian. Catholics, Christians. In this case, it's like a positive relationship, but it is every. For us, that's universal; that's all of them. That is an A proposition. All Catholics are Christians.
- 5.53** Being a liar and being virtuous are incompatible. A cumbersome way of saying, we have liars, virtuous, saying it's incompatible. That's a strong statement. What does that sound like? Sounds like saying, no liars are virtuous. There's a breach there. That is an E proposition. No liars are virtuous. All but a few of the delegates are present. Delegates, people who are present. All but a few. Well, it's not all. It's just some. Some of the delegates are present.
- 5.54** To be a tortoise is to be slow. Tortoises, slow things. Then this somewhat archaic formulation, to be a tortoise, that's to say, there's something built into tortoiseness. That's to say, all of the tortoises have this proposition. So all tortoises are slow. That is an A proposition. Not every man can be bought. Men, people who can be bought. Not

every is not all. Some men are not buyable, or not in the category of things that can be bought. That's an O proposition.

5.55 Without exception, each new generation of college students scorns the standards established by earlier generations. Now we're increasing the verbiage. New generations of college students, those who scorn the standards set by earlier generations. We have complicated phrases to pick out the categories. We're saying, without exception, that's an all statement, all of them. It's repeated each new generation. That's another all statement. All new generations are things that scorn the standards established by previous generations.

5.56 Then elegant and sometimes slightly archaic language, faint heart ne'er won a fair lady. I think that's from Shakespeare. I could double-check. Faint heart ne'er won a fair lady. OK, so we've got the faint-hearted. We've got those who win fair ladies. What is faint heart? Well, it's somewhat metaphorical for timid, scared, not enough courage in this particular context.

5.57 Won a fair lady. Well, it's not like in the lottery. This means to be successful in one's romantic pursuits. The fair lady is whoever you're attracted to. We have, presumably, people in this case who are timid and fair lady-winners. We're saying no, the ne'er, archaic for never, no faint, no timid, are winners at love. We're going to translate it that way.

5.58 All right, that said, this is all categorical syllogism. I recommend you spend time practicing this. English language is very complicated and, in many cases, poetic, some cases, archaic, but it's very rich. So get into the practice of being able to take complicated language, when you know you're dealing with a categorical syllogism more likely, and stripping it down to its simplest, most essential, in finding out which kind of proposition that you are dealing with.

5.59 I would just say in my personal experience, when I'm reading philosophers, the language is often quite complicated. In many cases, the most work one has to do is just figuring out what the person is saying in the simplest possible language. Once you do all of that, figuring out whether the argument is valid or not, still some work, but it's typically less work, so practice wrestling with the language.

5.60 What we know is, we have four proposition types, the A, E, I, and O, if we're comfortable with those abbreviations. Then we start to put those together in various combinations. This is where we have our first level of complexity. I could say, for example, all of the S's are P. I could then say, all of the P's are M. I could then conclude, well, therefore, all of the S's are M. That's one kind of argument.

5.61 In this case, I've got three A propositions, or three universal propositions of positive. But I could also make an argument that keeps that broad structure, all S are M. That's fine. All P are M, therefore, all S are P. Now, in this case, I also have three universal positive propositions, three A propositions, and the difference is the internal structure

of the propositions. In this case, the common term for both of the premises is the letter P, whatever that stands for.

5.62 In one case, it's the predicate. In one case, it's in the subject position. In this case here, the linking term or the common term is M, but in both cases, it appears in the predicate position. Those are different arguments. The structural position makes a difference. I could also, though, say, I don't want to talk necessarily positively all the time. I might want to say, all of the S are P. Draw a line here, separate these, and none of the P's are M. Therefore, I might want to conclude, no S are M.

5.63 All of the S's are P's, none of the P's are M's, and I'd infer that none of the S's are therefore M's. In this case, I have one A proposition and two E propositions, or one positive, two negative propositions. I might say, no S's are P, no P are M, therefore, no S are M. In that case, three negative propositions. Now, there are other possible combinations of A propositions and E propositions and so on, but we also know, I might want to say, some of the S's are P's, and some of the P's are M's, therefore, some of the S's are M's.

5.64 In this case, three particular positive propositions, three I propositions. You're probably also thinking, does that make sense or not? Chances are good you have a question about this one. This one will be one of the invalid ones, but we will show why. I might say, all of the S's are P's. Some of the P's are not M. Therefore, some of the S's are also not M, and so on and so forth.

5.65 Now, we are doing this somewhat unsystematically. We wanted to have on the board some A propositions, some E propositions, some I propositions, and some O propositions. If we were to start doing this systematically, how many possible combinations are there of A propositions, or A propositions with E propositions, or combinations of A, E, and I propositions, and we work our way through, it turns out that there are 256 possible categorical syllogisms. That would be all of them, and you could work them out systematically. On the board, we have only six out of the 256.

5.66 Now, what we can do, of course, and if you have a lot of time on your hands or you are ambitious in a certain direction, you can go through, figure out all 256, make sure you've got no duplications in your list of 256, and test each one individually. Is this a valid pattern or an invalid pattern? And then memorize them all. That's not what we are going to do here. What we are going to do is suggest that there are some tricks or devices that we can engage in, or some very simple graphical methods that will enable us to take any argument that we have a question about, put it to a quick test, and determine whether it is valid or not.

5.67 Now, some of these will be more obviously. This one here we met earlier. This is Barbara. She is valid. All S are P, all P are M, all S are M. Just memorize that one. Anytime you see that pattern, just say, "Oh, that's Barbara. I've known her for a long time. She is legitimate." All of the S's are M's, all of the P's are M, all of the S's are P's.

We looked at this one earlier. This one is illegitimate. This is an invalid argument. This is the predicate position, both being the same.

5.68 I could argue, all dogs are mortal, all cats are mortal, therefore all dogs are cats. Oops, doesn't work. Or I could say, all squirrels are mortal, all penguins are mortal, therefore all squirrels are penguins. Same problem, same mistake. Perfectly good information and so on. This is another one that is useful to recognize the pattern, be suspicious of it, and just memorize that. There are going to be a lot of other ones, though, when you see the pattern and it doesn't jump out at you. Is it logical or is it not logical? For that, we want to have the tests.

5.69 Now, of the 256, there's a small wrinkle, theoretical issue we can talk about called material implication. You can follow up on debates and discussions over material implication. It's a very small issue in logic, but depending on what your position is on that or how you are using the logic, out of the 256 possibilities, only 24 are valid. And that means 232 are invalid. There's a lot more ways to mess up in one's reasoning than there are to stay on the true and narrow.

5.70 And the other position with respect to material implication is even harder if you take that route. Then only 16 out of the 256 are valid, 240 are invalid. What this then means is, when you start to see the words flying fast and furiously and the phrases are complicated and things are moving along fairly quickly, if you can take some time, slow the process down, and do some quick checks, then that will save you some time.

5.71 Now, I'm going to turn to the Venn diagram method, the graphical method. I like that one the best for the initial stages of learning logic and as you practice more and more becoming very good, because as we know, logic is very abstract. In many cases, you're dealing with abstract issues; or if you are learning them for the first time and reasoning about them, they seem complex and abstract to you, the Venn diagrams are great because they are graphics and trying to capture information in pictorial form helps us learn. It brings things down to Earth, so to speak, in a way that's very useful.

5.72 There are some other methods that have been useful. It is possible to just learn four rules for all good arguments, or at least for all good categorical syllogisms, and you can memorize those rules and apply those. That's fun. Sometimes the rule method is useful as well, but I prefer the Venn diagrams for the visual.

5.73 One thing is, this here violates one of the rules for categorical syllogism, that is to say, the middle term. The middle term is the term that appears both, in both premises. So P is the middle term here. The middle term here is also P. The claim is that, in all valid arguments, the middle term has to appear once as subject and once as predicate, so anytime you see it both right appearing. So one of the rules is, don't have the middle term be in the predicate position both times.

5.74 Another rule is, anytime you have a negative premise, whether it's a none are, or a some are not, then your conclusion has to be negative. If you learn that rule and you

look at an argument and it's got a negative premise, one of them is negative and the conclusion is positive, that's going to be an invalid argument, and so on and so forth.

- 5.75** But let's turn to Venn diagrams. How do we translate the abstract propositional information in the syllogism to a graphical form? We started to play around with that a little bit earlier to set the stage. The first thing that we always want to do is, we look at the argument, or if we're just starting with propositions, and see how many categories are we dealing with, and we will draw a circle to represent each category.
- 5.76** Initially, abstractly, we don't know if the things in that category overlap or they don't, so we'll leave open both of those possibilities. We'll draw this circle and this circle with some overlap, but also some areas where there's not overlap, and then we'll carry on. There is a way to do one more circle that I know of, in more advanced categorical syllogisms. If you have three premises and then a conclusion, there still is a way graphically to represent it. It gets more difficult after that, just because we have a two-dimensional space and we're trying to capture multiple dimensions. We will stick with the three-dimensional ones right now or the three-category arguments as well.
- 5.77** In this case, what we would do is, say, take number one. Samantha is here. She's her own category. Here are all of the things that are lean, and here are all of the things that are elegant. You know that, according to this proposition, that we're trying to say that Samantha is both of these things. We want to indicate somehow that, on the diagram, this is where Samantha lives, or this is where Samantha hangs out, so to speak.
- 5.78** What we could do, and one of the symbols is, is if we're talking about one thing, it's just put an X. The language is going to be that X means that there is something there. To indicate that there is something right there, we could put an X. But this something is a little bit generic. What we typically do then is say is, any time we're dealing with the whole of a category or something that represents the whole category, we're going to use shading. I'll just call that shading. That means nothing is there.
- 5.79** These will be our two graphical conventions. There's either something there or there isn't something there. Those are the only two options. We have something that indicates the one, something that indicates the other. In this case, what we know about Samantha is that she is both lean and elegant. It's not possible for Samantha to here. There's nothing there, we would say, because that would be Samantha, not lean and not elegant.
- 5.80** We could consider this. We'd say it was not there either, because that is, would be Samantha, elegant but not lean. We know there is nothing there. Nothing there. Then what about this category here? Well, that would be Samantha being lean but not elegant, but we are told that she is elegant, so we know there is nothing there either. That's been ruled out as a possibility. If we know that Samantha is a real person and not just a made-up one, she would say that's where she is. Samantha is there.

- 5.81** Now, try the one with John. Things that are funny, things that are sophisticated. What is the relationship asserted by this particular proposition? Well, we know, rough and readily, we want to have John somewhere also in the same as the F circle, but not in the S circle, because the assertion is that he's not sophisticated. All right, so we can't say that John is here. This would be John in the sophisticated circle. This would also be John in the sophisticated circle. We want to say, I want to rule that out. There's nothing there.
- 5.82** We've got sophisticated things. We got John over here. No overlap between the two. But we do want to say that John is also in the funny things circle. John is not over here. He's an unfunny John. We said that's not the case. There's nothing over here. That means, if John is a real guy, this is the material implication point. He's a material being who actually exists. We'll say this is where that something is, the John stance.
- 5.83** So Samantha is lean and elegant. John is funny but not sophisticated. The diagrams are similar but importantly different. This carries graphical information that is different from the graphical information of this one here. Someone is a dirty, stinking liar. OK, boy, the language is getting strong here. Dirty, stinking, liar. Now, in this case, I'm speaking vaguely, someone. I don't have John, Samantha's concrete. It's just something, something or other.
- 5.84** I want to say that it's close to the something that's there. Someone, something, something. We've got vaguer information. But I want to seem to have strong enough information to rule out a whole lot of others. How would we say that there is something that is in the D circle, in the S circle, and in the L circle? Well, the only possibility is to say that it's in there. There's something in there. Maybe you walk into a room, you haven't identified any of the people yet, but you know something's going on really bad.
- 5.85** All right, all humans are rational and capable of self-agency. All right, this is the one that we did the circles for. Now, let's do the diagramming of the information for it. All humans are capable of rationality. Let's take that part. Well, this here would be a human being that is not capable of rationality. It would be outside of the R circle. Same here, so we're going to cross this out. Nothing here. That's empty.
- 5.86** And capable of self-agency. Well, these would be humans that we think are not capable of self-agency. We know there's nothing there, either. That gets crossed up. That then is to say, all of the humans have to be in here. If we're just speaking abstractly, maybe we're talking about things we don't really sure that humans exist or just hypothetically, we would not put an X there. That's the material implication. If we stay in the hypothetical mode, we won't make that assumption.
- 5.87** If we're talking in a context where we know we're talking about real entities, then I would put an X here. If I change this argument, though, to unicorns, suppose, all unicorns are rational and capable of—I've just made up this fictional creature. Well, we

would not use material implication in that case for reasons that, when we enter into a fictional space, the metaphysical assumptions are different. We don't want to make those assumptions when we're doing the logic, so we would just leave it with this abstract characterization. As I said, that's an interesting issue, but a little to the side for us right now.

- 5.88** Now, this, again, something worth practicing. Take individual propositions and diagram them until one becomes fluid with putting the X's where they need to go. Now for the big payoff, once we are comfortable with taking individual categorical propositions and diagramming them, we can then take entire arguments. In this case, we have two propositions and a conclusion. We're going to then say we have three categories, so we can use those same three circles and start down that exercise.
- 5.89** At the end of engaging in that exercise, we will be able to step back, look at the diagram, and it will tell us whether this is a valid argument or an invalid argument. The thing to be keeping in mind will be the formulation we put out several times before. In a valid argument, if the premises are true, the conclusion has to be true. Or if the premises are taken for granted, the conclusion follows logically.
- 5.90** What we'll be looking for is, once we take the information from the premises, if it already just follows necessarily that the conclusion is true, then when we diagram the information in the premises, that is to say, we capture visually the abstract information of the premises, we will find that the conclusion is already diagrammed for us. The conclusion in that sense is built into the information already given in the premises.
- 5.91** Or one way of putting it, another way of putting it, is to say that the conclusion is implicit in the argument's premises, and the purpose of the argument is to make explicit what's already implicit in the premises, or you've got enough information to make that move safely. Let's, with that preamble, do the same thing here.
- 5.92** So we have humans, mammals, and mortals. So we'll call those the humans. These will be all of the mammals. These will be all of the mortals. All humans are mammals. Well, what's that to say? This would be a human that's not a mammal, so there's nothing there. This would be a human that's not a mammal. It can't be anything there, so we can safely cross this out. Those two possible zones, given this information, must be empty zones.
- 5.93** All mammals are mortal. So now we look at the mammals and the mortals. We know, well, this would be a mammal that's not mortal. There can't be anything there. This would be a mammal that is not mortal. There can't be anything there either. We can cross those zones out. Those are empty zones. That's the information we get from this second premise. Now we stop, you diagram the premises and no more. We look and we ask, have we already diagrammed the conclusion?
- 5.94** Now, if we were going to diagram the conclusion, we're not going to, but we would say, all humans are mortal. We would say something like, well, this would be a human that's

immortal, and that doesn't exist. This would be a human that's immortal, and that doesn't exist. But we already know that that doesn't possible. It's already been diagrammed for us in the information that we got from the two premises. In this case, our conclusion is already diagrammed. It's represented graphically, based on this abstract information.

- 5.95** We've not made stuff up. We've not gone beyond the available evidence. We have reasoned well. This now is a valid argument. Having done this exercise, do it with several examples until it becomes obvious. You say, "Oh, yeah, that's great." Then just memorize that one. This always is a good pattern of reasoning.
- 5.96** By contrast, to show how the diagram will prove to you that something does not follow, that it's an invalid argument, let's use our whipping boy here. This is the one we've used several times. Intuitively or just looking at it, something seems wrong about it, but let's show graphically why this is going to be so. All men here. I guess I need M-E, M-O for all of the mortals, and pigs. We'll put those guys down there. Three categories, three circles, each one labeled.
- 5.97** We say, all men are mortal. Well, we've done that one before. So this would be a man that isn't mortal, outside the middle space, so there's nothing there. This would be, something that is a man, but also outside the mortal category, so there can't be anything there. Almost typically, the universal positive proposition, you're crossing out those two zones. All the men have to be over here. That's what we know.
- 5.98** Second piece of information, though, that we're given in the argument, is that all of the pigs are mortal, too. This would be a pig that is not mortal. There's nothing there. That's empty. This also would be something that's a pig. It's in the P circle and not mortal, but it's already crossed out. We just leave that one alone. We've diagrammed this one, we've diagrammed this one. We stop. If the argument is valid, our conclusion is diagrammed for us. If it's not diagrammed, then that would mean we would need more information in the diagram in order to capture the information of the premise.
- 5.99** If it's not diagrammed, that means it's an invalid argument. We say, all men are pigs. That's the asserted conclusion. We look at the diagram. Well, it could be, as far as we know; there are some men in this category here, but it also could be that there are some men in this category. This is still possible. These are men who are not pigs. That then is to say, this argument is asserting that there's nothing there, that all the men have to be down there. But we don't know that yet, given this information.
- 5.100** This argument goes beyond the information in the premises. The conclusion does not necessarily follow. This is now an invalid argument. Then the claim is, there's 256 of them. You can go through, and what you will find is, 24 of them work or 16 of them work, depending on the material implication issue. The other ones are going to be invalid, invalid, invalid. It's a useful exercise to do a lot of those because then you start

to recognize certain patterns, and things crop up on a regular basis, as well as just becoming really good at doing Venn diagram quite quickly.

- 5.101** But even so, don't feel obligated to do all of that. Do enough of these to make sure that the Venn diagram works for you and seems a reliable method. Then when necessary, dealing with something complicated or interesting or something you want to double-check, do the diagram for your own reassurance.
- 5.102** One of the great controversies in the philosophy of the logic has been over the status of logical principles. We develop Venn diagrams, we develop rules, we make distinctions between valid and invalid. In many cases, it gets very abstract very quickly, and logic shades into formal mathematics very quickly. One philosophical tradition has instituted a kind of divide between formal methods of thinking that exist only in the mind and applications in the actual world. For various philosophical reasons, the claim ends up being that these formal abstract principles are just that; they are just empty formal definitions.
- 5.103** Now, one form of this came to be popular in the early 20th century, where it came out of a philosophy of language that said language is made up of terms that we have defined, but we can define terms any way that we want. Grammar is a matter of a whole bunch of rules, but we can make up, if we want, all kinds of grammatical rules, whatever we want, including grammar and rules for chess and all sorts of games, including logic and mathematics. We start to see them just as games based on rules that we made up.
- 5.104** But they're not real. They are empty. They are just, we have decided to define terms a certain way and make combinations of symbols in certain ways. They're not actually saying anything about the way the world works. The early Wittgenstein is very close to this position and a work that he called *Tractatus: Logico-Philosophicus*. There's an arresting quotation. Of course, there's a broader context for all of this. But the point for our purpose is to say that all the propositions of logic say the same thing— that is, nothing.
- 5.105** So logic is ultimately empty. It's just a formal game that we play. It doesn't tell us anything about the world, the structure of the world, the causal nature of reality, and so on. It is a subjective game. Now, it might be a game that we really like to play. And given my constitution, I might like the idea of defining rules a certain way, and then making sure I can stay within the bounds of those rules and see how far I can go with those rules. But I'm not really saying anything about reality if I'm just using abstract S's and P's and all's and this, that; it's just words and so on.
- 5.106** Now, another anti-logic position, even more extreme, has been to say, yes, we do have logical principles and so forth, but they also are in subjective origin, not so much because we've decided to define terms, even define terms carefully or rules of grammar, but they have a different kind of subjective source. And Friedrich Nietzsche is

one of the big names behind this tradition here. Let me read this quotation from his late work, *The Will to Power*.

- 5.107** "Our subjective compulsion to believe in logic indicates merely that long before we became conscious of logic itself, we did nothing but introduce its postulates into events. Now we find them in events we can no longer help. The world seems logical to us because we have already made it logical." The idea is, the world is what it is. For all we know, it could be chaos, then we are just complete randomness and so forth. But something subjective in us doesn't like that. So what we do, even before we start to think about it, is we start to impose structure, impose order, impose logic on the world.
- 5.108** That, according to his mentor, largely his mentor Schopenhauer, makes us feel more comfortable, which is a subjective emotional state. Nietzsche is not quite so strong in that way, but nonetheless, what we are doing is imposing what we want onto the world. If we happen to want the world to be logical and orderly, then we will construct, even without realizing that we are doing it, a basically logical and orderly way of looking at it. Then the logicians come along and say, "Oh, we will just take that subjective structuring and run with it and be very silly." But really, it says more about us than it does about reality itself.
- 5.109** Then the third position here is represented by H. W. B. Joseph, early 20th century, in the Aristotelian tradition of logic—that logic is formal, but it's not only formal; that the forms of logic are abstractions on patterns in reality; that we should have a compulsion to be logical, but it's not something that just happens automatically. We know it's possible for us to have a compulsion not to be logical, and we have to choose the right choice, which is to be logical because we want our subject to match objects as they are out there.
- 5.110** In his formulation, "Unless the primary laws of thought," those are the logical principles, "were laws of things, our thought would be doomed by its very nature to misapprehend the nature of things." And he goes on to argue that, yes, we do have laws of thought, but they're not just arbitrary semantic definitions that we came up with. It's not just emotionalists tidying things up because we want things to be tidied up. Our laws of thought are reflective or mirroring or abstractions on, or drawn in some way from, the structure of reality.
- 5.111** Things are what they are. They move in various ways and in various abstract ways. That's where we get our logic from. Now, this is just to introduce another deep debate in the philosophy of logic, particularly as we get more and more abstract, as we've seen in this lecture. Where you stand on this issue becomes fundamentally important.
- 5.112** Is there a way to use logic to deal with the appeal to majority, for instance?
- 5.113** Yes, or in particular, the appeal to majority? Or, yes. OK. I think the best way is to start with oneself because we all have our individual minds, but we all are also social beings. We typically, to the extent we are social and we like other people, we want to be in

agreement with other people. If we are benevolent, we typically like to give people the benefit of the doubt initially. So the first thing would be to recognize that that's all fine and normal and healthy.

5.114 The point would be just to make sure that, when you are thinking about something that matters to you and you know your own mind, make sure you're not letting those social desires influence your thinking. Sometimes that requires just to self-identify. "Oh, I'm dealing with this issue. I really want to say yes, but I'm not sure." Make it your own habit. Then how you do that with other people, I think it depends on how close the relationship is with you, how much time you are spending with someone.

5.115 But if you are, say, the mother, then you're going to spend a lot of time with people just modeling that in your behavior. You want to get along with your kids, you want your kids to agree with you, but show them that you're being reasonable in doing so, that you're not just going to agree with them automatically, but you know how to raise gentle criticisms and provide the counter-evidence, and also that you don't expect them to do the same with respect to you and your partner. If you outnumber them, it's going to be, we want you to think for this and come to agreement, but by your own light. So I think it has to be a slow, carefully built-up set of habits.

5.116 If we go back to fallacies, I was thinking about the hot debates that are happening today and, given the abortion debate, so pro-choice or pro-life, the usual argument is, when life begins, that's usually the thing. But nowadays, there's an argument about the bodily autonomy, as far as to say that women should have pro-choice. So is that a logical fallacy, because it's not really centering on the whether or not it's a life, or is it a logical argument?

5.117 Right. Now, that's a good complex set of issues. So one way, and the abortion issue, like all controversial issues, is rife with logical fallacies all over the place. To take the appeal to emotion, for example, people on the pro-life will say, abortion is the murder of tiny, defenseless human beings. Just put that out there really strongly. Tiny, defenseless, and then human beings, and murder—so it's all strong language. And if all you have is that, then that is appeal to emotion.

5.118 On the other side, if the pro-life people will say, well, abortion is just the surgical removal of an unwanted clump of cells, or something like that, so they're going out of their way in that language to dehumanize whatever potential status the fetus has. Both of those are signs that you're dealing with people who are not interested in actually looking at the issue logically and see what the facts on the ground are.

5.119 The second part of your question, I think, is not fallacious because I think you're right to say, part of the abortion argument does involve the status of the fetus. So we do need to spend a certain amount of time looking at embryology and fetal development and figuring out what the significant milestones are, what traits are present, which ones are

activated at what point. And all of that has to feed into our decision about what is the status of the fetus, at what point does it become human person rights-bearing.

- 5.120** And all of those concepts are slightly different. It's one thing to say something has a personality, to say that it's a human personality. Dogs have personalities. So the person language, but human language. Then to be a rights-bearing being is another thing in there. What is the status of the fetus requires, I think, a lot of moral investigation and a lot of biological investigation.
- 5.121** But the other side of the argument, or the other part of the argument, it's not a side of the argument, is to say that the fetus is in the woman's body, and the woman is a human being. She is a person. She is a rights-bearing being. So what is her status? And then again, we have to do the same sorts of things—attend to her biology, attend to her moral significance. Then once we've got it worked out, the right solution is going to talk about the relationship between those two in the best resolution.
- 5.122** I don't think it's an unfair thing for one person to say, "Well, look, the fetus has a certain status." They may say a good argument for why the fetus has that status. For the other side to come back immediately and say, "Yes, but the woman also has certain status and rights as well." That's just to put two parts of the overall picture on the table for discussion, and that seems perfectly legitimate to me.
- 5.123** On the theme of fallacies, what are some factors that make someone more susceptible to fallacies?
- 5.124** Yes, that's a good question. There are dozens of other commonly committed fallacies. Sometimes, logicians will just use the label non-sequitur, borrowing from the Latin. Just because there's so many of them, you don't want to name all of them. But in many cases, just the conclusion doesn't follow from the premises. It's not even sequenced, it's a non-sequitur.
- 5.125** So your question, though, is in part a psychological question, I think. So we know that there are people who don't commit fallacies very much. They're clean in their thinking. Other people are, yeah, it's like, whoa. Yeah, so what's going on there? And maybe we don't want to know too much of what's going on in the analysis. But I would say, probably the two biggest ones are laziness. That seems to be common to many of the fallacies, because what the fallacies almost always enable you to do is, there's some complicated or important issue, is just not to have to deal with it, not to have to do the work, and so forth.
- 5.126** If you can just throw a fallacy out there, it sounds like you've said your two cents worth, you've made an argument, but it's almost like a get-out-of-jail-free card from having to do thinking. In that case, the difference would be between people who are committed to doing work, if it needs to be done, and people who are just lazy.

- 5.127** I think another bigger one, if I were just to mention another one, is that probably subjectivism is—it could be an own standalone fallacy, but subjectivism often will utilize the other fallacies in self-defense as rhetorical strategy. I want to believe what it is that I want to believe, and someone's giving me a hard time. They're presenting those pesky facts and counterarguments and criticisms and so forth. I don't want to deal with it, because I don't want seriously to consider the possibility that I'm wrong and I need to rethink certain important things in my view.
- 5.128** Depending on the kind of argument that you are dealing with or the kind of data that you are dealing with, it might be useful as a subjectivist just then to say, "Well, I don't like this person's race." So use an ad hominem to get rid of that person. Or I will be dealing with someone who seems kind of sophisticated and is used to multi-step arguments, so I will argue in a circle and maybe it will track, so you're begging the question in that case.
- 5.129** If it's someone who is under my power, who's raising difficult questions, I can threaten that kid. And that's the use of appeal to force. So maybe what's really going on is the subjectivism, but the person just has all the fallacies, so to speak, as tools in the toolkit, to be used depending on the circumstance. I think that's the best short answer we can come up with.
- 5.130** A question about, and I'm going to mispronounce his name, Wittgenstein.
- 5.131** Wittgenstein.
- 5.132** Wittgenstein, Nietzsche, and Joseph. It seems their relationship to logic is literally the relationship to truth, as in subjective. There is an objective truth, and no, it's all subjective. I mean, Nietzsche practically said that logic is like unconscious bias, that we manifest belief.
- 5.133** Exactly. Nietzsche is a Kantian, in that respect. We just have these—but a Kantian with a twist. We have these inbuilt structures, is the Kantian idea. Whatever reality is, we have no idea what that is. All we can do is think that our subconscious minds or our unconscious minds structure the data a certain way, and so it's a working-out of subjectivity.
- 5.134** Nietzsche is a couple of generations later, and he argues that these subjective structures are not constant and universal. They're more relative and subject to change, and so it might be that there are different logics or different subjectivities at work. But you're exactly right. This is the fundamental issue of objectivity versus subjectivity. What sets the terms? The small-O objectivists, Aristotle on down to H. W. B. Joseph in the 20th century, want to say, what's out there objectively in reality sets the terms, and the subject initially has to take its marching orders from reality.
- 5.135** It might be that we have distinctive methods because of our mental structure about how we process that data. We might devise tools, but nonetheless, those methods that

we use have to prove themselves as being capable of keeping us connected objectively to reality. The subjectivists are almost always skeptical; or skeptics, initially, what they find is, they think for various reasons that we can't maintain a connection between our minds and reality. There's a breakdown at some point. Our subjective side is disconnected from reality, and it just takes on a life of its own, and we don't talk about reality at all.

- 5.136** Now, the concept of truth that you mentioned right along the way, in my conceptual scheme, that is a correlative concept with the concept of fact. All of human cognition is our minds in relationship with reality. It's a relationship. If the relation is a positive relationship, then we say things like, it's an objective relationship. But then we have terms to designate the two sides of the relationship. What's the nature of reality? There we're talking about facts.
- 5.137** Then we want to get those facts inside our minds in some sense, but then we form concepts and propositions. When we use truth language, we're talking about what's going on on the subject side. So we say, you take these initial data, we're trying to figure out what the facts of the world are, and the result is, we want to have in the mind true propositions about that.
- 5.138** And so, if we say it's a fact, then in that context, we're emphasizing the object side of the relationship. If we're using the word true, we're emphasizing and stressing the subject side of the relationship. Already, we use, I think, eight or more important cognitive concepts, and all that needs to get sorted out. Very good question, too.

Lecture 6: Deduction: Hypothetical & Disjunctive Syllogisms

At a glance — Deduction: hypothetical and disjunctive syllogisms.

- 6.1 All right, let's turn to hypothetical syllogisms. Categorical syllogisms, as we have seen, have all been taking two categories, and it gets complicated because, with two categories, we can say a lot of different things about them. There are 256 possibilities, and we have Venn diagrams and rule methods to sort out the valid from the invalid.
- 6.2 Hypothetical arguments are much simpler because there's only six possible hypothetical forms. Three of them are valid and three are not valid. We can just talk through them, identify the forms, use a couple of examples. These ones are susceptible to memory. These are ones that you should make a commitment to memorizing the patterns and making it automatic, so that you recognize them, valid/invalid. The red flags go up automatically when they need to do so.
- 6.3 Now, a hypothetical proposition or a sentence is one that is governed by the if-then structure and connects with our ordinary use of hypothesis. I consider this as a possibility, and if this is possible, then we draw out some implications. Now, as with all other kinds of propositions, English is very rich and varied in how it makes its hypothetical statements. The first thing to do is to get comfortable with recognizing hypothetical statements when you see them.
- 6.4 One thing is to just, the if-then structure, if-then, start to circle it if you're dealing with something in print and you're comfortable with putting print a circle to it or write it out. But pay attention to the if-then structure. Sometimes people don't say, "If this is true, then this other thing is true." They will say, "This other thing is true, if this other thing is true." So sometimes you will get the "then" part first, and then the "if" part first.
- 6.5 Just a little piece of language, terminology. The part between the if and the then, we call it the antecedent. The part after the then, we call it the consequent. We'll come back to that because, in naming the arguments, we use this antecedent and consequent language. But we have the if, then the antecedent, whatever it is the claim is there, a then, and then a consequent.
- 6.6 Now, all of these on the board are hypothetical statements, but they don't, obviously, anyways, include the if-then structure. Sometimes we use other hypothetical pieces of language. "Should the floodwaters rise too high." We're not saying the floodwaters are going to rise too high, but we're saying if they will rise too high, then something will happen. In this case, I didn't even put the word then in there.
- 6.7 But what we have is an antecedent, which is this here. If, write it out formally, flood waters rise too high. That's a big, long phrase. Then, making it explicit, they will burst the temporary dams. The waters will burst the temporary dams. In standard form, we take this English and we will put it into that, a clear symbolized if-then structure.

- 6.8** "He must have at least a high school diploma, should it turn out that he's a chartered accountant." In this case, we have this should, which is not a moral should, but it's a hypothetical should. Should it turn out he's a chartered accountant, this is actually our antecedent here. We would say, if he really is a chartered accountant, then he must have at least a high school diploma. We would rewrite this one. If he's a CA, if that's true, then high school diploma, something like that.
- 6.9** "No pain, no gain." Catch your slogan, but we logicians, let's make it really obvious. If not pain, then not gain. I got to be sure. If you don't do painful things, then you won't get the gains from it. That's the import there. I'm starting to stiltify the language. It's a little more stilted here. Not no pain, but not pain. That's for things down the road, particularly when we turn to symbolic logic, how we deal with negations. If we'll have to say that things are unmoved, we will say not moved. Things are impossible. We will say it is not possible.
- 6.10** Because as within your math classes, when you're dealing with positives and negatives, when you're multiplying and subtracting and so on, it's useful to have that negative operator separated because then you can do various kinds of operations. What we will find is the same thing in logic. We want to be able to say, there's this and there's the not-this, and we want to detach or sometimes we will reattach the not, but I'm just flagging that for future attention.
- 6.11** "Seeing is believing." A straightforward declaration sentence. But how would we write that out formally? We're saying something like, well, if I don't see, then I'm not going to believe. That seems to make sense. So it could be a complicated double negative sentence. If I don't see something, then not believe. Or it could be to say, well, if I believe something, then it must be that I see it with my own eyes or I saw it with my own eyes.
- 6.12** And those are, logically, two equivalent ways of doing the same thing. If believe, then see; if not see, then not believe. We give them a label. We call these the contrapositive of each other. We love to label everything, break it down. But this shows that we take what's in the antecedent position here, and we move it to the consequent position and negate it. We take what was in the consequent position, move it to the antecedent position and negate it.
- 6.13** So, switch the two terms and negate both; it ends up being logically possible or logically equivalent. But we can prove that. We'll probably prove that in the next lecture when we do some more sophisticated things with hypotheticals. But this point just now is that something like this, these are equally legitimate ways of translating it.
- 6.14** "Live by the sword, die by the sword." An old cliché. The import of this is fun. If you embrace a life of violence, then you expect to die violently and probably shorter. So if live by the sword, then die by the sword. "Time flies when you're having fun." Another cliché. In this case, this is probably our antecedent. If you are having fun, then time

goes fast, or time, psychologically, seems to go fast. I'm not going to try to squeeze that in there. This would be a case where we get the consequent first, we get the antecedent second, and we would just make that reversal in writing it out formally.

- 6.15** This one from Socrates. "The unexamined life is not worth living." A kind of call to doing philosophy. Unexamined life is not worth living. Now we have a not here. We also have a negative here, an unexamined, a not-examined life. We might think about detaching that. But what is he saying? He's saying, if you not examine your life, then it's not the case that your life is worth living. If you have a not-examined life, then it's not the case that your life is worth living. Stilted, but captures, I think so.
- 6.16** We could also, in this case, say, if you have a life worth living, then you must have examined your life, making that as a necessary condition of the good life. Once again, those two are equivalent formulations, contrapositive. In this case, what's in the antecedent position goes to the consequent position and is negated. That would be not-not EL, the double negative cancels out. What's in the consequent position goes to the antecedent position and again is negated. Those are equivalent operations.
- 6.17** This one from Ecclesiastes. "He that increaseth knowledge, increaseth sorrow." King James version, so 400-year-old English. Why not? This is somewhat pessimistic. If you increase your knowledge, then you increase your sorrow. And then, "If God did not exist, it would be necessary to invent him." A little bit of humor from Voltaire. Then, if not the case that God exists, then we should invent God.
- 6.18** All right, again, some starting examples where some of them are old English, some of them are newer English, different orders, and just do as many of them as you can, for whatever your logic practice source is going to be, to become familiar with it. Now, that said, we take this exercise seriously to translate the ordinary English into standard form. Then we can see how these hypothetical statements are used in argumentation. Then we want to evaluate those arguments.
- 6.19** I mentioned that hypothetical arguments, it turns out there's only six forms that are possible. Here are four of them. We'll save the other two. These four are called mixed hypothetical arguments. We don't need to get hung up on the names. But why they are so called is that they are driven first by a hypothetical proposition. That sets the term. If Roberto worked hard, then he is tired. If there is fire, then there is oxygen.
- 6.20** But then that works with just a standard declarative sentence, and we reach a conclusion. That's just a straightforward declarative sentence. The other kinds of hypothetical arguments are going to be pure hypothetical arguments, where if this, then that, if that, then this other thing, and everything stays at the level of hypothetical propositions. But what we have then is, if Roberto worked hard, then he is tired, this proposition, and then, as we can see, this is our conclusion indicator. We know that that is the conclusion.

- 6.21** Now, does that seem like a valid argument to you? It's always worth just checking your own logical sense. If Roberto works harder, then he's tired. Roberto did work hard, therefore he is tired. Chances are good, you will nod and say, "Yeah, that seems fine." This is a valid argument form. The way it goes is, if Roberto works hard, then Roberto is tired. Roberto works hard, therefore Roberto is tired.
- 6.22** Now, the label that we put to this argument, we call this affirming the antecedent. The naming of this argument is to say, well, Roberto worked hard. This is our antecedent. But what the second premise in this argument is doing is telling you that the antecedent is in fact true. It affirms the antecedent. If RWH, then something happens, and then we affirm, yes, we do in fact have RWH. Therefore, we can draw the consequent as the conclusion.
- 6.23** So I'll put the label here, affirming the antecedent. Affirming the antecedent is valid. That's a legitimate form. Just memorize that one, after doing a certain number of examples. I might be a taxi driver. I would say, "If you give me \$20, I will drive you to the airport." The affirmation would be, "Oh, you gave me \$20, therefore I will drive you to the airport." That would also be affirming the antecedent.
- 6.24** Now, contrast that one to this second one here. If Roberto worked hard, then he is tired. Then what we have in the second premise is, Roberto has not worked hard, therefore he is not tired. If we symbolize this one, we would say, if Roberto worked hard, then Roberto is tired. Then not, detaching it. It's not the case that Roberto worked hard. Then I draw the conclusion, it's not the case that Roberto is tired.
- 6.25** Now, in terms of naming conventions, we have the antecedent and consequent. What the second premise is doing is denying the antecedent, saying that the antecedent did not obtain, it didn't happen, it's not true, or whatever. We label this argument denying the antecedent. Internal check. Does that seem like a valid argument? If it were true that one would be tired if one worked hard, or Roberto would be tired if he worked hard, but he didn't work hard, therefore he's not tired, you should have a red flag at this point or hope to get a red flag.
- 6.26** Because what this tells you is that there's one way Roberto can become tired, that is, if he worked hard. But the fact that he did not work hard doesn't exhaust the ways in which he could become tired. Maybe he didn't work hard, but maybe he went to the gym for a lot of hours. So he didn't work hard, but he did something else and that other thing. Or maybe he didn't work hard, but he's been awake for 48 hours straight, and so he's still tired. So it's possible that this is true. If he had worked hard, he would be tired. It's true that he didn't work hard, but this not to be true. And that then means this is an invalid argument. Never works.
- 6.27** Now, this is one of the ones, again, worth memorizing and practicing because this happens all the time, everywhere. You see it all over the place. When people aren't being careful, they slip into denying the antecedent. They think they've proved

something, but they hadn't. It's not just people are being lazy. Sometimes you're dealing with complicated stuff, and this seems to get you where you want to go, but it doesn't actually advance your case. So, so far, we have one valid one, one invalid one.

- 6.28** Let's carry on with this one, just changing up the kind of example to make it a little more interesting. If there is fire, then there is oxygen. So a claim from physics and combustion, you have to have oxygen in order to get fire. In this case, there's no oxygen, whatever setup we are talking about. And I make the conclusion, therefore, there is no fire. So simplify. If fire, then oxygen. That's the first claim. Then we make the claim, not oxygen. There's no oxygen there. Then I draw the conclusion, there can't be any fire, not fire.
- 6.29** Sometimes taking the larger amounts of language and simplifying it helps clarify the logic. Internal check. So if you have fire, then you have to have oxygen. Don't have any oxygen, therefore we don't have any fire. How does that compute? And that should sound good. This is a valid form, always legitimate, often used in the sciences. The sciences will make, have a hypothesis. On the basis of that hypothesis, you will make some predictions, and then you experimentally test those predictions. If the predictions do not come out the way you want, you consider the hypothesis challenged, perhaps refuted.
- 6.30** What would we call this one? What we would look at, see what's happening is, in this second premise, it is denying the consequent, saying whatever's in the consequent did not happen. That is what we call it. We call it denying the consequent. And that is a perfectly valid argument. One more possibility. You can see in this case, we affirmed the consequent. Here we denied, or sorry, we affirmed the antecedent. Here we denied the antecedent. Here we denied the consequent.
- 6.31** So the fourth possibility is going to be to affirm, sorry, is going to be to affirm the consequent. So that's what we have here. If there is fire, then there is oxygen present. There is oxygen present, same there, therefore, fire is present. Let's symbolize it a little. If fire, then oxygen. In this case here, we do have oxygen, we say, whatever our circumstance is; we conclude, therefore, we must have fire. And yes, this is affirming the consequent. Put that down here. Internal check. Does it seem like it's a valid form or invalid? It is invalid.
- 6.32** And again, one of the ones worth recognizing a pattern, because it too gets committed with frequency. Try with a number of different examples. If you insult him, he will become angry. When you see him, he's very angry. Does it follow that you insulted him? No. There's lots of reasons why he might be angry. So it can be true that if you insult him, he will be angry. It can be true that he is angry, but it doesn't follow that he is angry because you insulted him in that case. That's not a legitimate form of reasoning. So try lots of them and get comfortable with them.

- 6.33** Setting us up in part for our next session, I want to add one more set of labels here. Two of these have Latin names. Affirming the antecedent argument is also called modus ponens. If you're not into the terminology and language, don't worry about it, but it is called modus ponens. Basically, it's arguing in the ponens mode, going back to the original Latin. So this valid affirming the antecedent is called MP or modus ponens. This one over here is called modus tollens. So we are arguing in the tollens mode, according to ancient Latin. Don't pay a lot of attention to that necessarily. It just sets us up for some material we'll be covering a little bit later.
- 6.34** Yes, the two, notice the two valid forms. Affirming the antecedent gets a Latin name, denying the consequent gets a Latin name. These ones are so bad, they don't deserve a Latin name, so you don't get one. They're just bad arguments. But it is useful to have some sort of a label for them to know what we're talking about.
- 6.35** Hypothetical arguments. We've seen the four mixed forms—affirming the antecedent, denying the antecedent, affirming the consequent, denying the consequent. Those are a mixture of hypothetical propositions or hypothetical statements that drive the argument or set the context for the argument. But then we bring in other non-hypothetical information to argue on the basis of.
- 6.36** Pure hypothetical syllogisms represent the other two possible uses of hypothetical arguments, and they're called pure because every proposition or every sentence in the argument is a hypothetical argument. So you stay hypothetical mode. In this case, we have four propositions. All of them are hypotheticals. If a student is young, then he's enthusiastic. If he's enthusiastic, if he is diligent, if a student is young, in the conclusion and so forth. Each of them has an antecedent and a consequent.
- 6.37** We know what our conclusion in this case is. This is our conclusion indicator. If a student is young, he will likely succeed. This is already somewhat pre-digested, but we say, if a student is young, here we have the antecedent, then we have a consequent. He is enthusiastic. That is the consequent. The consequent of the first proposition becomes the antecedent of the next proposition. He is enthusiastic, then we have another consequent of that.
- 6.38** Then he will be diligent. That is the consequent in that proposition. It becomes the antecedent in the next. If he is diligent, new consequent, he will likely succeed or succeed likely. Then what the conclusion does is, it takes your very first antecedent and your final consequent, and says, ta-da, those are connected. So this antecedent and this consequent are linked.
- 6.39** Now, this is sometimes called informally just a chain argument. For obvious reasons, metaphorically, chain links this to this, links this to this, links to this to this, and so forth. Therefore, the very first thing is also linked to the last thing, and you can assert that that linking has occurred. More formally, we just call this one the hypothetical

sylllogism. We don't even put the pure part in because everything is hypothetical. Or we will sometimes abbreviate it as HS, the hypothetical syllogism.

- 6.40** For sake of completeness, this is valid. If this is linked to this, and this is linked to that, and that's linked to that, and that's linked to that, then the first thing is linked to the other thing. Now, you might not agree with this conclusion. You may be more pessimistic about education outcomes, but there's nothing wrong with the logic here. That would then mean you probably have doubts about some of these premises. Maybe it's not true, maybe it's a little bit overstated. If you change the premises, then you might mess up the argument, and it's no longer legitimate.
- 6.41** Now, this one here should look somewhat familiar, but let's read it through. It's another hypothetical syllogism because each of the propositions is a hypothetical. Each is an if-then. If you're a physician, then you went to college. If you're an attorney, then you went to college. Thus, if you are a physician, you are an attorney. So, if we were to symbolize this one here, we would just say, if you're a physician, then you went to college. If you are an attorney, put my N in, then you went to college. Then we want to link these two in the conclusion. Therefore, if you are a physician, then you are an attorney.
- 6.42** Now, red flags are going up at this point. This is true. This is true. But we end up with something false. There are some people who are both physicians and attorneys, but it's not a necessary connection. So this can be true. This can be true. It can be true that you're a physician, but not that you're an attorney. So this does not follow from that set of conclusions. This is an invalid argument.
- 6.43** And the reason that it's invalid is for the same reason that, when we were looking at categorical syllogisms, when we had the two terms in the premises, both in the predicate position, the analog here is that, in both cases, we have the same thing in the consequent position. And once again, it's worth noticing the formal pattern and then say, oh, wait, we have very similar things at the end. It doesn't matter whether it's the middle term or at the end, or whether it's the consequent, that's a suspect pattern of reasoning, whether you're doing hypotheticals, whether you are doing categoricals.
- 6.44** And this one is typically given the name the fallacy of the broken chain. It's the broken chain fallacy, for this kind of hypothetical argument. Called the broken chain because, as we saw in this one, we went from antecedent to consequent. Then the consequent became the antecedent. Then the consequent became the antecedent. In this case, it breaks. If you're a physician, then you went to college. But we don't get something about going to college in the next proposition in the antecedent position. We start over with attorneys, and then we come back to this issue of college. There's a break in that chain pattern, and that is illegitimate.
- 6.45** Kind of obvious in a case like this. Where again it will become less obvious is when you're dealing with much more complicated material, longer chains of reasoning, and

the language is messed up, and you've got lots of different formulations there. It's very easy if you are not careful to let the broken chain happen, and then you are lost, logically speaking.

- 6.46** Disjunctive arguments are the third kind of syllogisms that we look at. There are categorical syllogisms, hypothetical syllogisms. Disjunctives are either-or statement. We have a lot of different ways of saying either-or. Sometimes we also have, in English, built-in negations. We'll say neither-nor. That is to say, it's not either-or, and we have to pay attention to how to parse that out into formal language as well.
- 6.47** But hypothetical syllogisms, there are six of them. Categorical syllogisms, there are 256 of them. There's only three when it comes to disjunctive syllogisms. Two of them are valid, but there's a proviso that we have to talk about to show why the third one is valid. So, either the temperature or the humidity is high. The barometer says the humidity is high, so the temperature is not high. Now, this is driven by this either-or construction. We get two alternatives, and then we go on in the basis of this one here, to try to rule out something.
- 6.48** Yuriko is either a real samurai or an excellent actor. There are no more real samurai. He is, therefore, a great actor. Then, Sally is either pregnant or not pregnant. The test came back positive, thus she is not not pregnant, or not unpregnant, or thus she is pregnant. But we will keep the not in there for the clarifying of the logic reasons as well.
- 6.49** Basically, when we boil this one down, we have either the temperature is high or humidity is high. The second sentence tells us that one of these disjuncts is true, or claims that one of the disjuncts is true. It is the case that the humidity is high. I'm using the up arrow just to indicate height. If it's that one, then this argument is saying, it's not that one, and that's what the conclusion has drawn. It's not the case that the temperature is high.
- 6.50** Gut check, internal check, how does that seem to you? This one should send some red flags up. That is an invalid argument because it can be both. It could be that the temperature is high and the humidity is high. So the fact that we have affirmed one of these disjuncts doesn't mean that we can rule out the other disjunct.
- 6.51** Now, immediately contrast that one with this one. Sally is either pregnant or not pregnant. So we have two disjuncts once again. The test comes back positive, and then we do the other one. Let's do that one. She is either pregnant or not pregnant. Then, second one came back positive. That is to say, we are affirming one of the disjuncts. It's not H, P. Let me try this better. She is, in fact, pregnant. And since we have affirmed this disjunct, we think that puts us in a position to deny or get rid of or rule out the other disjunct. So we say, it's not that other one, but that other one is itself a negative formulation, not not P.

- 6.52** Now, here's the question. This one seems perfectly fine, and it is perfectly fine. This one is not fine, but they seem to have the same pattern. We have a disjunction, two possibilities. Second premise, we affirm one of the disjuncts, and in the conclusion, we rule out one of the disjuncts, the other remaining disjuncts. Why is this one good and this one is bad?
- 6.53** Here we have to add something beyond pure logic. We have to add semantics, because when we use either-or commonly in reasoning, we use the either-or formulation differently. Sometimes we use it in a way where we say, it's either this or this, and it could be both. Or we use, it's either this or it could be this, but it's not both. Those are different formulations, and we don't always make it clear and explicit in ordinary English which of those either-ors we are using.
- 6.54** To put the language to it, we will say that this is an exclusive "or". What we mean by an exclusive "or" is, we have either-or, and the truth of one excludes the possibility of the other. And that's why this isn't either-or, because if you're a woman, you are hard either-or. You're pregnant or you're not pregnant. It's not possible to be pregnant and not pregnant. So one state excludes the possibility of the other. You're either dead or you're not dead. You're dead or alive. That would be an exclusive "or". You can't be dead and alive. So the question always to ask is, is this an exclusive "or" or not?
- 6.55** Now, this we call an inclusive "or" because the truth of one of the disjuncts includes the possibility of the other disjunct also being true, or it doesn't eliminate that possibility. We have to remain inclusive in that sense, open to the possibility that both disjuncts are true. That then means, just because we find out that one of the disjuncts is true, we're not yet in a position to rule out the other one.
- 6.56** So this one over here, this is an inclusive "or". We have found out that one of the disjuncts is true, but we're not yet in a position to rule out the other one. So this is an invalid argument. This one over here, it's an exclusive "or". We bring in that semantics. Then we find we have affirmed one. That's enough for us to know that we can rule out the possibility of the other. So this is a valid pattern.
- 6.57** What about this third one? Yuriko is either a real samurai or an excellent actor. There are no more real samurai. He is, therefore, a great actor. If we do our symbolizing exercise, we either say, Yuriko is a real samurai or Yuriko is a great actor. Those are our two disjuncts. In this case here, not the case that Yuriko can be a real samurai. Therefore, Yuriko is a great actor.
- 6.58** Internal check. Does that seem good? Got two possibilities. We eliminate one. It has to be the other. This or this. It's not that one, so it must be that one. That's what our conclusion draws. That should feel good. That is logically valid. This is a—I'm going to keep my color scheme consistent here. This is a valid argument form. This one we also just typically call the DS, the disjunctive syllogism, and it's legitimate.

- 6.59 Now, if you notice structurally, the difference between this one which is valid, this one which is valid because it's exclusive, and this one which is not. In this case, our second premise is affirming one of the disjuncts, and whether that puts us in a position to eliminate the other one depends on the context. And this one here, we don't have to worry about whether it's inclusive or exclusive. If we have enough knowledge, the second premise, to rule out one of the possibilities, then we can go ahead and infer the other.
- 6.60 This is the one that's probably most commonly used in detective fiction and real detective life. So a detective starting off investigating a case might generate six possibilities. So we take a standard Agatha Christie novel. They're in some cozy country retreat, and there's six people there, and the host, and the host ends up dead. Well, immediately the detective is in a position, because it's a closed circumstance, to say it's one these six people.
- 6.61 So if we were to start formally, we would say, it's either this person or this person or this person or this person or this person or that person. Then typically what they do is, they try to find evidence that points toward one of the people. But also this argument is used a lot. You look for a way to rule somebody out. This person was killed at this time. We can guarantee that this person was somewhere else at that same time. It can't be that person. So we cross that one out, and now we're down to five. Find a reason to cross somebody else off the list. It can't be that person and so forth, until you get down to the final person.
- 6.62 Let's try three examples to illustrate different challenges in dealing with hypothetical and disjunctive arguments, determining their validity or not. Once we have learned and memorized the patterns, it's pretty straightforward, as often is the case. Wrestling with the language and getting it into a standard form and seeing what that pattern is, is at least half of the challenge.
- 6.63 In this case here, this substance is not sodium because it burns with a green flame, which it would not do if it were sodium. Immediately, what I would do is notice a few things. One is, we have this word because. That is a premise indicator. Almost always that points out, the things that follow are going to be the premises, but what immediately preceded that is likely the conclusion. What the person is trying to assert and wants us to believe is, this substance is not sodium. I would go off to the right over here. I would say the conclusion is, this is not sodium.
- 6.64 Well, why not? Another thing I would notice is, well, we have this "if" statement here. This is probably going to then be a hypothetical argument, an if-then. But we have the antecedent right at the very end. So where's the consequent? Well, we can try to reformulate this. If this substance were sodium, then it would not do something. Well, what is it that it would not do? There's a referent there. We'd say, well, it wouldn't do this thing up here. It wouldn't burn with a green flame. That seems to be the assertion.

- 6.65 But we have to do some massaging of the language to then say something like, if this were sodium, then what would not happen? It would not have a green flame when it burns. It seems all right. But then we say, this substance, whatever it is, burns with a green flame. So, read it again, see if it maintains veracity or consistency with the original English. We've done some massaging here. If this were sodium, then it would not have a green flame when it's burning. But it does have a green flame, so it's not sodium. Check. That seems fine.
- 6.66 Now, we step back and we say, well, what's going on in this argument? Hypothetical. So we look at what the second premise is doing. What the second premise is doing is denying the consequent up here. The consequent is not a green flame. This is saying the opposite of that, saying it is a green flame. So this argument form is denying the consequent. Then we consult our memories, or we check our logic sense. Is denying the consequent legitimate? Is it valid? And the answer is yes. So this is a valid argument. Good one.
- 6.67 Next, let's try this one. The electricity is on, so the circuit breaker must be working. First thing we notice, we have this "so". That's a conclusion indicator. We know what the conclusion is. So I would just draw a line, put my conclusion down here. The circuit breaker is working. Why am I supposed to believe this? Well, what comes before the conclusion indicator? The premises, the electricity is on. Electricity is on.
- 6.68 Now, here we have a challenge because, ordinarily, arguments have two premises and the conclusion, but we were only given two pieces of information here. This means we are dealing with an enthymeme. Going back to early basic argument analysis, very often people will not give you all of the necessary information. They will expect you to be able to fill in obvious things.
- 6.69 Now we have to ask ourselves, even though we're not necessarily professional electricians, what are we supposed to know about the relationship between circuit breakers and electricity? What's the information that we would add that's supposed to be true and obvious to make this argument work? One thing that we do know is that, well, if the circuit breaker gets flipped for whatever reason, there's a power surge or whatever, the electricity goes out.
- 6.70 How can we formulate that in a true statement that would work with this argument? It would say something like this. If the electric, it's not the case, say, that the electricity is on, then it might be the case that we do not have a working circuit breaker. What if we tried that? If the electricity is not on, then it might be that the circuit breaker is not doing what it's supposed to do. How does that sound?
- 6.71 What if we tried something else? What if we said, if the electricity is on, then we must have a circuit breaker that's working. I like that better because it's cleaner. It doesn't have all those negations in there. We've taken all of those out. Is it true? We should put this in brackets since we're adding our own words to the argument to indicate that. If

the electricity is on, then the circuit breaker must be working. Is that true? Yeah, normal electricity knowledge.

6.72 So, then we can step back and say, well, now we've got a full argument. What's going on in this argument? The second premise is affirming the antecedent. We say this argument is affirming the antecedent. Checking our memory, hoping that it's working in confluence with our internal logical sense, affirming the antecedent is one of the legitimate forms. So this, too, is a valid argument. Good job.

6.73 Now, one more for the challenge. Poetic language, going back to the ancient Greek poet Sappho. We'll read it first. "Death is an evil, we have the gods' word for it. They too would die, if death were a good thing." I'm sure she read it better. Now, we go to work on this. Any premise indicators—because, since, as, for this reason? No. Any conclusion indicators—thus, therefore, ergo, it follows that? No. So one challenge that we have here is trying to figure out what is the conclusion.

6.74 Now, there is an argument in here. So sometimes, of course, there isn't an argument, and that's why there are no indicator concepts. In this case, the poet decided not to use them explicitly. We do have one thing here. We have an "if". So that indicates to us that we have an if-then, we have a hypothetical statement at work. So when we reconstruct this, it's likely not to be a disjunctive or a categorical syllogism. It's going to be a hypothetical.

6.75 What if we took that and worked on that, supposing that this is a part of the argument? Now, again, we have the antecedent, the part that comes after the "if", right at the end. Is this our consequent? If we tried that again, if death were a good thing, then the gods would die. Some sort of reasoning like, the gods want all of the good things for themselves. So if death were a good thing, they would want that, too. So, if death were good, then the gods would die. Taking this, put that there.

6.76 Then we have two things left over. Death is an evil, and we have the gods' word for it. What do we do with those? Well, put it forth as a hypothesis that we have our conclusion at the beginning. What she's trying to prove to us poetically is that death is a bad thing. Death is an evil. Or since we already started with this good thing, death is not good.

6.77 So could we take this and say—we're reading contextually here, we're making an interpretive move. What she's trying to do is to say that death is not good. It's not the case that death is good. Death is an evil. Same translation. So if we were going to have a logical argument, well, so far we only have one premise, we need another premise. I could start looking at this one here and say, well, hm, what would I need to make this argument work? What could I put in there?

6.78 It has to be something that connects to death being good or the gods dying. I don't seem to have anything in the poem. Instead, what we have is this phrase here. I'll put it in question. We have the gods' word for it. We're appealing to the testimony of the

gods; or look at what the gods do, or look at what the gods don't do. I think what we're supposed to do then is fill in the blank there.

- 6.79** What do we know about the gods? The gods are immortal. They are immortal beings. Now, she doesn't say that in the poem because that's obvious, that the gods are immortal, but we are expected to infer that from here, and that's partly what makes the poem fun. She isn't just laying everything out for us. We have the gods' word for it. Look at what the gods, in fact, say and do.
- 6.80** We can then say, well, what she means is, the gods don't die. The gods are immortal. That would then, in the language of this reconstruction, it's not the case that the gods die. Now, that's a rough draft, but it looks pretty good. We might then try, let's go to the evaluative mode. What do we think about this argument? Well, it's an if-then. That means it's a hypothetical argument.
- 6.81** Then we notice what it is that the second premise is doing. It's saying, the gods don't die, or it's not the case that the gods die. That is our consequent of the first proposition, but it's the opposite of that consequent. So this is denying the consequent. And it's the same as that form over there. We can double check; that is a valid argument. So who says poets can't do logic? The best poets actually are very good with it.
- 6.82** With that, exercises and a recommendation, do as many as you can until these things start to become automatic. Another controversy about the nature of logic, now that we are getting into the intricacies of argumentation and all the stuff we are to memorize and practice and apply and make second nature, is it really worth it? And is it keeping us on the right path?
- 6.83** A point from Julian Huxley, one of the leading scientists of the 20th century, saying that, yes, the answer to that set of questions that I put out absolutely is a fundamental virtue of being a scientist and being a reasonable person. Be very careful what you let into your mind. The language he uses is here. "We should be agnostic about those things for which there is no evidence. We should not hold beliefs merely because they gratify our desires for afterlife, immortality, heaven, hell, etc."
- 6.84** At one level, toward the end, he's taking a standard set of beliefs that we all have emotional reactions to, desires for, are built into important belief systems, and making us reflect on making sure that we are believing in these things, not for subjectivist reasons, but because we have evidence for them. And we should believe in them as a matter of degree. We should be agnostic only to the extent that we have evidence. We start off from a not-believing. As evidence comes in, we consider it. And part of being logical is being careful to match your degree of belief in something to the degree of evidence—the mark of a, in his judgment, an honest, tidy mind.
- 6.85** I want to contrast to that with one of his contemporaries. This is someone who has connected to some of our earlier thinkers, read his Nietzsche, read some of the other

critics of logic as well. This is a significant historian, Carl Becker, but a very philosophical historian, setting aside logic. From his perspective, the 18th century, which was the great era when science, logic, mathematics, became explosive and institutionalized, from his philosophical perspective, that was all a big disaster.

6.86 So he writes a history that's largely critical of the Enlightenment and the 18th century. And this is one of his lines. "We now suspect that logic is something the mind has created to conceal its timidity and to keep up its courage, a hocus-pocus designed to give formal validity to the conclusions we are willing to accept if everybody else in our set will, too." So there's a kind of social conventionism at work here. My group of people, we just believe all sorts of things, and I want to fit in with my group, and if they commit to being logical, OK. If they don't, that's also perfectly fine.

6.87 But logic, notice the language that we are using here, it's just really about cowardice. It's just a bunch of hocus-pocus and so forth. We're more likely to admire or respect or like people who are not particularly committed to logic. In this same generation, we have a very deep divide. Logic is to be dismissed from the perspective of Becker, whereas for Huxley, it is a deep character virtue commitment, to make a commitment to logic.

Lecture 7: Propositional Logic

At a glance — Propositional (modern symbolic) logic.

- 7.1 This session, we're going to turn to modern symbolic logic. It's called modern because, while we've been doing logic for well over two millennia, developing very sophisticated methods for handling all kinds of logical arguments and circumstances, early modern logicians started to make connections between logic and mathematics in a more rigorous way, and some of the developments then, over the course of the modern era, including a generalization on what had been accomplished in logic up to that time, and an extension, set the stage for, in the 20th century, programming languages, the development of computers, and all of those amazing technologies.
- 7.2 There is a continuity at the same time and extension. To see that, let's start with some of the things that we know so far. On the board here, I have some hypothetical arguments and a disjunctive argument. These are four patterns of reasoning, common, and they're all valid patterns of reasoning. Having checked that out for ourselves, practiced it and so forth, we're in a position to automate it. Anytime we want to combine information in any of these patterns, we are comfortable to go ahead and do so.
- 7.3 So if we have P, then Q, it's a hypothetical statement, and we find out that we have P, then we can infer we must have Q. Now, traditionally, we would call that affirming the antecedent in English, what the second sentence or proposition here is doing, and then we draw the consequent as the conclusion. We call that antecedent. Henceforth, though, I want to start calling this argument modus ponens, or just MP for short, modus ponens.
- 7.4 The language then will be more uniform with some of the other developments in logical, modern symbolic logic, rather. That's affirming the antecedent or modus ponens, abbreviated with MP. Perfectly good. This one, we know we have our hypothetical statement here. The second sentence is denying or negating the consequent. We call it denying the consequent, and that as a matter of the conclusion means we can logically deny the antecedent. This one we call denying the consequent, but henceforth, we are going to call it modus tollens, or abbreviating it rather MT.
- 7.5 This one is the pure hypothetical form. All of the proposition constituent elements are hypotheticals. Informally, it's the chain argument. P is linked to Q, Q is linked to R, so we can link P to R, which is what we do in the conclusion. There can be as many links in the chain, as long as the pattern is consistent. This one is a pure hypothetical argument or chain argument. We are going to call it the hypothetical syllogism, henceforth, abbreviating just HS. Perfectly valid.
- 7.6 Then the fourth form I have here is the disjunctive form, either P or Q. We have two alternatives or two disjuncts. Second proposition denies or negates one of the

disjuncts. It can be either one, and that means we can affirm that the other disjunct must be the true one. So if it's either P or Q, and we know it's not P, then it must be Q, and that's a legitimate pattern as well. This is called the disjunctive syllogism, and we will just abbreviate it, shockingly, DS.

- 7.7 Now, much of the reasoning that we do is isolated arguments, where we do a modus ponens, or we do a disjunctive syllogism. But as sophisticated thinkers, we often will engage in extended arguments, and we will typically mix and match the kinds of arguments that we are putting in the extended arguments that we are doing so.
- 7.8 I have a couple of introductory examples here, moderately complicated, but the kinds of things that we do routinely on a regular basis, and we want to be able to capture that in the logic. We also want to see if we can start to axiomatize, symbolize, make everything ruthlessly simple and clear with an idea, perhaps, to programming machines, computers, and making logic and mathematics a more tightly integrated set of disciplines. Here's a family vacation planning example of reasoning.
- 7.9 We should visit either Toronto or Rochester this trip. If we go to Toronto, then we won't see Sylvia, but we have to see either Sylvia or Quincy this trip. I just found out that Quincy isn't available, so Rochester, it is. Now, in a sense, that makes perfectly good sense. We would do this in our heads; we would do it in conversation. And we would then, OK, I guess we're off to Rochester for our family vacation, all of these factors taken into account.
- 7.10 If we start to do the logical techniques that we have developed so far, we don't see any real premise indicators, but we do see one conclusion indicator. So all of this information must be the premises that support this conclusion, the decision that we are going to go to Rochester on this family trip. When we start to look at the premises, we see this one is an either-or. So at some point, we then know we've done a disjunctive syllogism or disjunctive argument, or we're doing that.
- 7.11 We have here an if-then, which indicates that we are using hypothetical reasoning at some point. It might be pure, it might be mixed with a straightforward declaration sentence or not. We have some negations, not, not, and so on. We have a combination of either-ors, this and that, not this, and if this, then that. We put it all together, more or less naturally in our reasoning then, to support this final declarative sentence. It makes sense if you think about it, and if you're in the zone, you do all this sort of thing regularly on the basis.
- 7.12 But how do we show that this is good reasoning, or if there's a mistake in the reasoning somewhere, that there is in fact a mistake in the reasoning? Now, suppose we say, look, let's look at these, and this is what, five sentences all together? Suppose we just look at the last two before the conclusion. I just found out that Quincy isn't available. So that then is to say, not Quincy. The third sentence, though, said, it's either Sylvia or

Quincy. So if we just extract that, then what we have is either Sylvia or Quincy, and not Quincy.

7.13 What we can then infer is that it must be Sylvia, and that's to do a disjunctive syllogism on the fly. So now we know Sylvia. We've got to see Sylvia this trip. Then suppose we look at this sentence here, the second sentence. If we go to Toronto, then we won't see Sylvia. That's a hypothetical statement. But if we combine that with this new thing we know—we have to see Sylvia—if we go to Toronto, we won't see Sylvia, but we are going to see Sylvia. So therefore, it makes sense to say, we can't go to Toronto.

7.14 And that is this pattern of reasoning here. We've just done modus tollens. So now we know we're not going to Toronto. Now, if we then look at the very first sentence, the very first sentence said, well, it's either Toronto or Rochester. And now, after a couple of steps of reasoning, we have eliminated Toronto as an option. So if it's either Toronto or Rochester, and we know not Toronto, then it must be Rochester. That then is to use this pattern of reasoning, the disjunctive syllogism, again. Then we are in a position to draw our conclusion.

7.15 Now, the point of that then is to say, we have initially five sentences, but more or less on the fly, or if we slow down and notice what we are doing, we are adding three more steps of the inference. We end up with eight propositions altogether, and we're pretty confident that this conclusion follows. But can we show that systematically, that in fact that follows? We can't, and that's what we want to work toward.

7.16 Here's another example, artificial party planning example. Just A, B, C, D, and then F. Just the first letter. This is not a real example. But if Alice comes to the party, then Bob will. If Carl comes, then Darlene will. If either Bob or Darlene comes, then Fred won't. Some sort of social issues, conflicts there. Either Alice or Carl is coming, therefore, Fred is definitely not coming to the party. That might seem a bit fast, but this person is putting a definitely in there. We now definitely know, given this information, Fred is not coming.

7.17 This is our conclusion indicator. This is what all of this reasoning is supporting. Again, if we then notice what we have in terms of the logical techniques we've developed so far, we have an if-then, so there's a hypothetical statement. Then we have another if-then statement here. This one's a little different because we've got a semicolon, and often when we use semicolons in our grammar, we're implicating a somewhat tight connection between two things, rather than having them as standalone sentences.

7.18 We have this other "and", this other connection. The idea is, if this, then that, and, keep this in mind very tightly, closely, if this, then that. We have an either-or, so a disjunctive sentence in there. But immediately before this either, we have an "if", so it's like a combined hypothetical disjunctive statement. That's a more sophisticated mental construction. If either of these things happens, then something else is going to happen.

- 7.19 And we have a negation built in here and another negation down there. We're looking at some of the logical operator concepts. Again, we've got a combination of if-thens and either-ors, and this one seems a little bit more sophisticated. But if we pause again, we try to work it through in our heads, we can say, we're either getting Alice or Carl. Suppose we focus our attention on this one. We don't know which one, say, but we're going to get at least one of them, either Alice or Carl.
- 7.20 Now, suppose we then say, well, if it's that one, that's the Alice part, then we're going to get Bob. If it's the other one that turns out to be the case, if it's Carl, then we're going to get Darlene. So doesn't it make sense then to say, if we're either going to get Alice or Carl, we know that, then we're also either going to get Bob or Darlene, even though we don't know which one yet? We know something definite, but we're still in hypothetical mode with some options, and that's a little bit harder to hold in your head.
- 7.21 Now, if we know that we're either going to get Bob or Darlene, but we're not sure which one, maybe both, then we get either of those ones, Fred's not coming. And so the person then says, at this point then, I'd say, well, even though we don't know which one, Alice or Bob, Carl or Darlene, we're still in that uncertainty with respect to all of that, nonetheless, we definitely know Fred, rather, is not coming. It could or should make sense, but can we show that clearly, so that we could say, explain it to a computer, that this is a valid piece of reasoning, however much the complexity is there? The answer is yes. That's what we want to work up to in this session.
- 7.22 Now, some housekeeping, notation matters. We want to, in symbolic logic, start to, more systematically and ruthlessly, symbolize things. What we want to do is, say we have this argument pattern, and we're going to just abbreviate that and assign names, modus ponens, modus tollens, hypothetical syllogism, disjunctive syllogism. There are other good patterns that we can learn as well, and we're going to work our way up to that as well.
- 7.23 But we also have some language in here, the if-then and the either-or and the word not. One of the first things we want to do is, partly just for saving ink and saving time, is to symbolize those as well. Our conventions are going to be the following. We'll say any if-then statement. We are going to replace that with a horseshoe. I would take this argument and just say P, sorry, this proposition, $P \supset Q$. So any time you see the horseshoe, it's just that.
- 7.24 The not, not, the negation, we will use the tilde sign. Very close to the mathematical negation negative sign, but in logic, the convention is mostly that we use the tilde. So in this one, we would say $P \supset Q$, and then $\sim Q$, not Q. This one, we have no more operators, so we'll just, so the final symbolized form of that argument would look like that. This argument here, the modus tollens, the inference that's drawn is therefore not P, so we abbreviate that argument that way.

- 7.25 The either-or, we'll use a hard $\vee$ symbol. That one has the nice, it's like pointing forks in the road. What we would do with this one is say $P \vee Q$. That's the symbolization. We know this not symbolizes with the tilde. It's $\sim P$. Then we bring down the conclusion, Q . This one would just be $P \supset Q$, $Q \supset R$, so $P \supset R$. We're still using the line as a convention to indicate the divide between the premises and the conclusions.
- 7.26 I want to add one more connective, as they're calling it. We call these connectives because they take constituent elements and connect them in various ways. That one is the word "and." "But" is another conjunction as well, but we use the word "and" a lot to take elements and combine them. Sometimes we will say, if you have fire, then you must have oxygen. Or we might say, if you have oxygen and a combustible, then you can get fire. That is an if-then statement, but the antecedent has a conjunction in it. You have to have oxygen and something to burn, then you can get the fire.
- 7.27 We will just use a dot, usually not like a period dot, not like the upper colon part of the colon dot, just kind of in the, in the middle. The idea then is going to be, we can take all of the propositions that we use, and we combine propositions in various ways to draw various sorts of conclusions. We will take the propositions and symbolize them. When we're thinking through all of these things abstractly, we'll use letters P , Q , R , just to symbolize all of the propositions.
- 7.28 But then the initial claim is that there are only four basic ways to combine propositions. You can speak hypothetically. If you have this one, then you get that. You can speak disjunctively. It's either this or that, or this and that, or we can negate things. Now, that is a claim, I think, for further research and thinking. We do have lots of other words that we use, but most of those end up being one of the four basic combinations in various ways.
- 7.29 So, "and", for example, is a connection, but we also have the word "but." "But" is a kind of conjunction, but it usually indicates also a negation is coming. It's really conjunction with some warning of negation at work there. Either-or, we have neither-nor, but that is either-or, and you are adding negations to both of them. I might say, if and only if, and there's that word "only" in there, and that's making some really strong claims, if and only if.
- 7.30 It's more than if this happens, then that happens. If and only if. When we work that one out, it turns out you're saying, if it's, if P , if and only if Q , then what you're saying is, if you have P , then you get Q , and if you have Q , then you get P . It's a combination of a couple of hypotheticals with a connective, and we can work all of that. The idea, though, is spend some time with the English or whatever your language is, and see if the constituent propositions have any other ways of being put together, logically, other than these four.
- 7.31 If you discover a new one, that's great. We'll find a way to work that into the logic, but we're going to work with this assumption here. We have five things, a basic

proposition, P. We might connect the P with a Q. We might do a disjunct P or Q. We might say the P is not true, the P is negated. We might speak hypothetically, if we have P, then Q.

7.32 Now, as ordinary language speakers, we know what all of these things mean, and we use them fluidly. But what we want to try to do is axiomatize. If we're going to axiomatize, take things very basic, strip them down as far as we possibly can, then what we need to do is define them in an axiomatic way. So all of those issues about concept formation, here we're doing concept formation about logical connectives, how we're going to connect things in various ways. Can we define the connectives rigorously and ruthlessly?

7.33 So this is what we devise what we call truth tables for. Truth tables for the connective. What do we mean when we are using this connective? Under what circumstances will we say that using that connective is being done truly or falsely? Now, one way in which this is axiomatizing is that we start with the claim that, for any given proposition that we are talking about, it is either true or it is false.

7.34 We go all the way back to Aristotle and what was called the law of excluded middle. Every proposition, we might not know if it is true or false, but ultimately, it either is true or it is false. We're going to use the words, the symbol T for truth, and the symbol F for false. Defining those terms, we've got our propositional variables and all of the connectors indicated there.

7.35 Now, in that case, if we know that a proposition is true or false, then I can say P. But we don't know what P is, but we do know that P is either true or false, whatever P turns out to be. When I say P and Q, I don't know what Q is, I don't know what P is. But if I make the assertion P and Q, under what circumstances would that sentence, rather, be true or false? We know there's how many possibilities? Well, it could be that P is true and Q is also true. It could be, though, that P is true and Q is false. It could be that P is false and Q is true, or it could be that they're both false.

7.36 So we have two propositional variables, and each of them has two possible states, true or false. So, there's four possibilities. So the idea of a table then is to say, let's diagram that in a table. So what we will then first do is, I'll start over here, I think, to say, well, let's take P, and let's take Q, and make a table. What do we know about P or Q?

7.37 Well, it could be that this one is true and this one is true. It could be that this one is true, but this one is false. It could be, this one is false and this one is true, or it could be that both of them are false. But since we are speaking abstractly, we've just symbolized and abstracted everything, we don't actually know what the P is and its connection to reality or not.

7.38 What we mean, though, and here we are defining carefully P and Q. Now we're defining that connective "and". If I say P and Q, I'm making a kind of commitment. Under what circumstances am I speaking truly with that compound proposition? Under

what circumstances would I be speaking falsely? Well, if it turns out that P is true and Q is true, then saying P and Q is true. So we will put the true here.

- 7.39 Suppose I'd said, Peter and Quincy are coming to the party, and Peter comes to the party, and Quincy comes to the party. Then what I said was true. Now, suppose, though, I say Peter is coming to the party and Quincy is coming to the party, but Peter comes and Quincy does not come. Then what I said was false. I said both of them are coming, but only one came. So in that circumstance, the "and" statement is false. Both of the conjuncts have to be true for the whole thing to be true.
- 7.40 And the same thing is going to hold here. If I said Peter and Quincy are coming to the party, Quincy shows up, but Peter doesn't, then what I said was false. So that is false. If I said Peter and Quincy are coming to the party, but neither one of them shows up, then my statement was false. What we are doing, working from basics all the way up, is now we have a definition, a truth table definition, of exactly what we mean when we use the "and" statement and the circumstances under which it is true.
- 7.41 Now, let's do the same thing for the disjunctive proposition. As I say, well, I've got P, I've got Q. Those are my propositional variables. What I do know is that they could both be true, could be true here, false here, false here, true here, or both of them false. The compound that we are interested in is P or Q. To say P or Q is different from saying P and Q, what's the difference graphically? Well, under what circumstances is this compound statement true? We can stick with the party example.
- 7.42 If I say, well, either Peter or Quincy is coming to the party, and they both show up, this first situation here, well, what I said was true. Suppose Peter shows up at the party, but Quincy doesn't. What I said was still true. I said either-or, I got at least one, so that's still true. Same thing here. Quincy shows up, Peter doesn't. It was true. But suppose I said, either Peter or Quincy is going to show up at the party, and neither of them shows, then in that case, what I said was false.
- 7.43 And this comparison now will indicate what makes sense to us in common speaking, that it's a lot harder for an "and" statement to be true than for an either-or statement to be true. You have more wiggle room, so to speak, with the either-or statement, but you're making a stronger claim with an "and" statement. There are fewer circumstances under which it is true.
- 7.44 OK, let's do the same thing for the negation. This one in a way is simpler. We have P, make our truth table. What do we know about P? Not much. We only know that it's either true or it's false. How do we define negation, not P? Just the opposite. If P is true, then not P is false. And if P is false, then not P is true. OK.
- 7.45 What about the fourth one? Now we have P and Q, and we are interested in—let me put one more line in here for consistency— the hypothetical statement. $P \supset Q$. Same for possibilities. If P, then Q. We'll help with our English. I suppose we say, if Peter comes to the party, then Quincy will come to the party. I make that statement. And

what happens is, Peter comes to the party and Quincy comes to the party. I said, well, if Peter comes, Quincy will come. They both came. So that is true.

- 7.46 So as I say, if Peter comes to the party, then Quincy will come to the party. Peter comes to the party, but Quincy doesn't show up. Then my hypothetical claim was false. Now, in the assignment of these next two, it's a little bit more complicated, and sometimes it's a little counterintuitive. I'm just going to say that what we have shown, we being logicians collectively, is that the right answer here is to put a T and a T there.
- 7.47 As long as, in any other circumstance, no matter what, if the antecedent is false, the corresponding consequent is true. We might then say, well, if I said, if Peter comes to the party, then Quincy is going to come to the party, and Peter doesn't show up, but Quincy does, what does that say about my initial statement? The controversy is, you might say, well, that should leave that undefined or unknown. I understand that. Think about that.
- 7.48 Or suppose neither Peter nor Quincy show up at the party. This should be undefined. Now, this will require one more lecture than we are going to get into today. But what you can show is there are various equivalences. You can show that $P \supset Q$ is equivalent to other formulations. When you take those other equivalent formulations and you do the truth table for those ones, this is the array that you get. Right now, take that as a promissory note. You can work on that on your own time or check in your logic textbooks. But this is what we are going to go with.
- 7.49 Now what we have done is, we have symbolized and essentialized the propositions, P's and Q's, maybe there will be R's. We've taken all of the ways that we can connect pieces of language together. If, then, either-or, and so forth. We've defined exactly what they mean. What can we do next with these things? That's where we want to say, what we did in discussing hypothetical syllogisms and disjunctive syllogisms before, where we went in a somewhat intuitive sense—this makes sense, this doesn't make sense, this seems valid, this seems not valid—lots of examples, and they all work to illustrate the point.
- 7.50 Now we can axiomatically, from these axioms, prove that the valid forms are in fact valid and that the invalid forms are in fact invalid. I want to keep all of this in mind and now go and work on some examples. Let's start with, as our first example, one of the forms, the modus tollens. We know what modus tollens is, is, if P, then Q. We have P, therefore Q.
- 7.51 What if we tried to make a really big truth table? In this case, we have two propositions. I'm going to draw a line here. P, and we have Q. Again, we know there's four possibilities. In this case, what we have is one proposition, $P \supset Q$. We're going to diagram the circumstances under which that would be true. We have another proposition, which is P. We'll diagram or fill in the table under what circumstances that

will be true. Then we have our conclusion, Q . We use the double line to indicate separation between premises and conclusion.

7.52 Now we have our initial possibilities, and then we have here an array of the constituent elements of the argument that makes up modus ponens. All right, let's start to fill in. For this, I'll go back to some blue. Well, these ones are easiest. P is just the same as P . I will just fill in there. P, T , rather, T, F, F . Q is just the same as Q . What do we mean with $P \supset Q$? We defined that over here. The right array for $P \supset Q$ is T, F, T, T . And then we're done filling in the table.

7.53 Now for the analysis, what does this tell us? We're interested in showing, rigorously, that this is a valid argument. Modus ponens is a legitimate form of argument. They would say, what do we mean by saying that an argument is valid? And we said things like the following all through the previous sessions. Whenever the premises are true, the conclusion has to be true. If it's possible for the premises all to be true, but you end up with a false conclusion, then that's a bad method of argument.

7.54 We take that as our test and we say, well, now we know there are cases here where all of the premises are true. We would look at this one, for example. In this case, we don't know that it's actually the correct circumstance, but both of the premises were true, and the conclusion was true. There are no circumstances where both of the premises are true, but the conclusion is false. If we have that result, that's the result we want to get. That shows that this is a valid argument.

7.55 In this case, we might end up with a false conclusion in this circumstance here. But we would say the reason for that is that we had a false premise, so we blame it on the premise. Here, we also end up with a false conclusion, but in those circumstances, it's going to be because we have a false premise as well.

7.56 Now, let me give you, by contrast, an example of the other type. Let me make a little space here. I'm going to leave those there. As we take the argument where we say, if we have P , then you get Q . We have Q , therefore we must have P . I'm going to put a big red circle around this one, just because we know from previous investigations that that's one of the invalid forms of argument. I'll say this is invalid.

7.57 Can we show that this is invalid? Do the same thing. P, Q . I'll start to do this faster now because, otherwise, you will all get bored. OK. Our four possibilities, we don't know which one obtains. $P \supset Q$, filled in according to our previous specifications over here. Q just duplicates the Q over here, but I did not do that correctly, did I? So, T . Sorry, I'm doing this. I was thinking I was doing the P for something. It's T, F, T, F . And then the P is a duplication of P over here. OK, good.

7.58 Fill it in. This is the argument, two premises and a conclusion. If it is a valid argument, whenever the premises are both true, the conclusion will also be true. In this case, premises are both true, and the conclusion is true, so we're OK. In this case, we don't pay attention because premises were false. But in this case, row 3, and I'll put a line

around it, a box around it, both of our premises are true, but we end up with a false conclusion. That shows the argument is invalid.

7.59 For all invalid arguments, there will be at least one situation in the world where the premises have a combination of truths and falsehoods, and that's the one that will mess you up. But what that tells you is that that way of combining propositions to draw a conclusion is not reliable. You can't depend upon it. And in logic, we are looking for the reliability. Now, let's do the same thing for a different argument. I'm going to save myself some writing here.

7.60 Let me add a tilde here. I'll put the $\vee$ there. Let's try it with the disjunctive syllogism. It's either P or Q, first premise, second premise, not Q. Conclusion, therefore P. Well, P or Q, under what circumstances is that one true? Well, that's our earlier specification over here. We would just fill in this array as well. As long as one of the disjuncts is true, the whole thing is true. So true, true, true, and false.

7.61 Not Q, the other premise, diagram that. That's the opposite of Q. It's going to be false, true, false, true. P is the same as our P over here. True, true, false, false. We filled in the truth table. All of the possibilities have been articulated. And if the argument is valid, whenever the premises are all true, the conclusion will be true also. If we can find a case where the premises are all true, but the conclusion is false, then that tells us this pattern doesn't work.

7.62 So in this case, we have only one circumstance when both premises are true. It's this second row of the truth table. Both premises are true, and the conclusion is true. In other cases, here we might accidentally come up with the truth, even though we've got one false premise. In these cases, we have a false conclusion, but we can blame that on the premises in those cases, or we suspect the premises in that case. So disjunctive syllogism, we've now axiomatically proven that it is a valid form of inference. We're free to use it anytime that we want.

7.63 Let's do it for the hypothetical syllogism. This is good for practice. I'm going to recommend that you do it as often as possible. For this, we're going to need to make space. Let me go with the blue. What if we now then take the hypothetical syllogism argument, which looks like this: $P \supset Q$, $Q \supset R$, therefore $P \supset R$. Now, our logical sense tells us that's a valid argument, but we want to make that rigorous.

7.64 Immediately, something is different now because we don't just have P and Q, we also have R. So we know the R could also be either true or false. We know that those truths and falses can work with all of the truth and false combinations of the P's and Q's. If they're just P and Q alone, there's four possibilities, but adding another sentence variable doubles the possibilities. It tells us we're going to need eight rows in the truth table, and our work is more complicated now.

7.65 What we will do is, we'll say, all right, let's take P and Q and R. Well, what are the possibilities? Well, it could be that all three of them are true. It could be that the P and

Q are true, but the R is false. Could be that the P is true, the Q is false, and the R is true. It could be that this one is true and both of those two are false.

- 7.66 If we hold the P constant and assume that it's true, then we have the other combinations there. Of course, it might be that the P is false and the others are true. The P is false, the Q is false, the R is true. What's my next one going to be? This one is false, this one is true, and this one is false. Then finally, it could be that all of them are false.
- 7.67 All right, that's a lot. You can already start to see how this starts to become tedious, and we start to want to think, "Let's invent computers, so that they can do a lot of this boring work." But we need to do what we did before. Let me draw a little separation line here. Now we're interested in this argument, the hypothetical syllogism. We put the first premise here, $P \supset Q$. $Q \supset R$, the second premise. Our conclusion is going to be $P \supset R$.
- 7.68 What we want to know is, whenever both premises are true, is the conclusion also true? We have to fill in now eight rows of the truth table. We go back, well, this is how we define the horseshoe, the conditional proposition. What it tells us is that basically it's true in all circumstances, except when the antecedent is true and the consequent is false. Let's look at $P \supset Q$ and do that one first. That means we're looking at these two columns in the truth table.
- 7.69 Well, in the first one, P is true and Q is true, so that is true. It's the same. In this situation, the antecedent is true, but the consequent is false. That is this circumstance here on our initial definition. That means we would put an F here. That would be false. This one is the same. That also is an F. The next four, the antecedent is false. In these cases, when the antecedent is false, both times it was true, so that's easy. Thank you for the easiness. T, T, T, T.
- 7.70 Now, let's do the second premise, $Q \supset R$. Well, T $\supset$ T, T. True $\supset$ false, that's the false one. False. False, false. False $\supset$ true. False $\supset$ false. Both of those are true. True $\supset$ true is true. False $\supset$ true is true. Make sure I'm keeping things in line here. True $\supset$ false, that's the other bad one. False. And false $\supset$ false is true.
- 7.71 Now, $P \supset R$. What we're interested in is this first column and the third column. So, true $\supset$ true takes true. True $\supset$ false takes false. True $\supset$ true, true. True $\supset$ false, false. And then again, all of the false antecedents take true, true, true, true. OK, that's boring, that's tedious, but we're doing good work because we want to be able to scale this up. The only way to scale it up is making sure at the tedious level everything is correct.
- 7.72 Now, we have proof-table-defined this argument and all of the possibilities. There's eight circumstances. Why there are eight circumstances is because we're just dealing with these things abstractly. We don't know what the P's, the Q's, and the R's are. We have to work all of the possible combinations of each of them being true or false. And that gives us this result.

- 7.73** If this is a valid form, the hypothetical syllogism, anytime the premises are true, the conclusion will also be true. If there are any circumstances where the premises are all true, and we end up with a false conclusion, that's not a reliable pattern. So true, true, and the conclusion is true. True, false, we end up with a false here. False, false, here. False, false, here. Here's another both true. This is true here, this is true, and we have a true conclusion.
- 7.74** True here, true here. We've still got truth. True, false. Still got truth, probably accidentally. And truth, truth. And we still have truth. There are four especially relevant circumstances here. This one, this one, and this one. No, and there's my fourth one. I didn't miscount. OK, good. When we are in the real world, dealing with real P's, Q's, and R's, there will be four times when the P's, Q's, and R's all end up in this array, and in every case, we'll end up with a true conclusion. That's a perfectly reliable pattern of reasoning. Go ahead and do hypothetical syllogism as much as you want.
- 7.75** Now, where we want to start economizing and developing more sophisticated methods for ourselves, but also, of course, mechanical devices that can do all these things for us is, I don't know about you, but my limit personally is with about four rows on the truth table. This becomes boring. But what if we had a longer chain argument? If P, then Q, if Q, then R, if R, then S, now we've got four sentence variables. We would need 16 rows on our truth table.
- 7.76** If we had a longer chain of reasoning, 32, 64, and pretty soon we're getting into extraordinarily tedious proofs of longer forms of argument. It's also going to be, as we saw earlier in this lecture here, sometimes in the case of party planning. If Alice comes to the party, then Bob will come, and if Carl comes to the party, then Darlene will come, and then we have Fred in there as well. So we've got five people that we are talking about and their relationship between coming to the party or not.
- 7.77** And five people, if we were to do a whole truth table for that method, for that particular combination of argument or whatever it is, it's going to be 32 rows on the truth table. I don't want to do that. What I do want to be able to do, though, is to say, if I can show that that argument is a combination of hypothetical syllogisms and disjunctive syllogisms and certain negations, and I know under what circumstances those individually are valid, then I can go ahead and prove that five-combination or five-proposition, five-people sentence without having to do the mechanical rows on the truth table.
- 7.78** But we do—and I'll just telegraph something I want to end with a little more strongly—start thinking about, this is becoming very mechanical, and the more you start to do this, it starts to feel mechanical. Same thing over and over, repetition over and over. We know as human beings that we can devise machines that can do simple mechanical things over and over again. Can we start to figure out a machine that would do trues and falses over and over in various combinations?

- 7.79 If we set the machine up, the way it works, it either flips to true or flips to false, depending on a certain rule that we set up and embed in the machine. Here we're starting to think about machine intelligence or machine application or machine working out of what we intelligently have already figured out that it wants to do. That takes us into the earliest of the calculating machines. Then as science develops, more sophisticated calculating machines of different types. They become intellectual labor-saving devices.
- 7.80 All right, so let me give you a little argument. $R \supset \sim B$. $D \vee R$, B . Suppose I give you an abstract argument, and that is what this is. I tell you these are three premises, but it's completely independent of knowing what we are talking about with respect to R , B , and what's the other one? D . I've got three things that we're talking about, and I'm connecting them in various ways.
- 7.81 Suppose I know, or I tell you, if you know this, if R , then not B . It's either D or R , and we know B . Those are the three things we have, the three pieces of information at our disposal. I want to then say that it is definitely the case that D follows. I have listed my three pieces of information, and this is a way of saying, I'm putting the conclusion off to the side. This is what we think follows, or I think follows, or that we should be able to prove.
- 7.82 Now, let's number our pieces of information. We know that one thing, we know this one thing, and we know that one thing. Now we could do a big truth table. Take R , B , D , eight rows on the truth table. We'll write out all of the premises, followed by the conclusion, fill in all of the truth tables, look for cases where the premises are all true and the conclusion is false, and see if the D situation takes us where we want to be.
- 7.83 But there's a shorter way, because we've done all of that work, and we know that certain patterns of reasoning are legitimate. We know, for example, that modus tollens is a legitimate form of argument. What is modus tollens? It says, well, if you have P , then you get Q . If you don't have Q , then you know you don't have P . That's modus tollens, or denying the consequent.
- 7.84 Is there a modus tollens that we can do with the information available here? Well, what if we looked at the first premise and the third premise? They didn't come together in this case. They were separated by other information. But we notice a connection there. They're both talking about B . This one says, if we have R , then we won't get B . But this tells us that we did get B . So what can we infer? Not B . So, I'm making an addition to this. I'm starting the argument. I now know that it is, sorry, not R .
- 7.85 That's the exact same pattern applied to this case here. If R were true, we would not have B , but we got B . So it must be that R is not true. So now we know it's not R . We're making progress. Now for notation, we want to show our work. How did we do that? So what we do is, just off to the side, I took premise one, and I took premise three, and I did what we know to be a legitimate operation. We did modus tollens.

- 7.86** Now we know not R, so we can go see, what can we use not R for? I say, well, R pops up in premise number 2. Premise 2 tells us it's either D or it's R. But we now know that it's not R, so that means it must be D. How do we know that? I took premise 2, I took premise 4. Actually, premise 4 is an inference. It's a conclusion of this earlier argument, but we're now using it as a premise for the next argument. And in that case, I did disjunctive syllogism.
- 7.87** All right, a two-step argument, but from that information, we said this does follow, and it does follow, and we can prove it in two steps. And that's enormously powerful. We did this at a comfortable pace, and it took us two or three minutes, say; it would take us a lot longer to do the truth table. So the idea is, now we know certain patterns of argument are legitimate. We can start to be creative in looking for opportunities to do so.
- 7.88** In many cases, we'll have huge amounts of information that comes at us in straight declarative form. These things are true, this is true, this is true, this is true. Sometimes it comes in hypothetical form, sometimes we are facing alternatives, but we can start combining those to extract further information that we know must be true, given that we are using legitimate methods at our disposal.
- 7.89** Now, so far, we are looking at just a handful of legitimate argumentation methods—hypothetical syllogism, disjunctive syllogism, modus tollens, modus ponens, and so forth. We can do truth tables to demonstrate more and build up our arsenal of legitimate argumentative techniques. Let me show you one right now that I will do it just semi-intuitively here. This is a fun one that is fairly useful as well.
- 7.90** Suppose I say something like, we have P, then we get Q, and if we have R, then we get S. This is a compound statement, taking two hypothetical statements and putting them together with a conjunction. So the reading of this would be, if you have P, then you get Q, and if you have R, then you get S. More sophisticated form.
- 7.91** Now, sometimes this is a hypothetical mode. We have partial information. All we know, say, is that we either have P or we have R, but we don't know which one. Doesn't it make sense, trying to hold all of this in mind, to say, well, if I get the P, that tells us I'm going to get the Q? This one tells me, if it turns out to be the R, then I'm going to get the S. I don't know yet whether I'm going to do P or R, but I do know that I'm either going to get Q or S. So, can I conclude, therefore, Q or S?
- 7.92** This is, I'll put the label to it, what we call the constructive dilemma. Often this dilemma form is, it's either this or that, and then you do a hypothetical for each of those. If you do this, then you get that, and if you do this, you get that. So you make that following conclusion. Now, what this is, is just doing two modus ponenses together or two affirming the antecedents together. If you were to ignore this part of the board, $P \supset Q$, P , therefore Q . If you ignore the left half, it's $R \supset S$, R , therefore S . But we carry down this either-or to there.

- 7.93 There's a destructive form of the dilemma. We've all heard the expression, "You are damned if you do and damned if you don't." And the conclusion of that is, the situation is hopeless, either way. But let's have a look at that one. That then is to say, if you do, then you are a deal, then you are damned. Damned if you do. And damned if you don't, then not do. Then you are damned.
- 7.94 All right. So yes, you're damned if you do and you are damned if you don't do. Now, we add to that. This is the enthymeme part of it. Well, you're either going to do it or not do it. That's the standing condition. So, either do or not do. Then we take the constructive dilemma form. Well, if I do it, then I'm damned. If I don't do it, I'm damned. I'm either damned or I'm damned. So simplification, I'm damned. It's a legitimate form of argumentation.
- 7.95 There are other fun uses of these more complicated structures in humor. You might have heard the one that goes something like, for everything in life, all you really need to fix things is either duct tape or WD-40. We use WD-40 as a lubricant for loosening things, duct tape to stick things together, and it's amazingly versatile. But the argument then will be, this is hopefully the joke, all you need is those two. If it moves, but it's not supposed to, then you use duct tape. If it doesn't move, but it's supposed to, then you use WD-40.
- 7.96 I'm exaggerating the if, and, and the nots. We could write that all out. Then you would just add, well, either it moves or it doesn't move. Those are the two possibilities. In one case, WD-40, in the other case, duct tape, you're done. But the argument is pithy because it is using this constructive dilemma form, along with some negations.
- 7.97 Now, with that in mind, we can then add constructive dilemma to our toolkit, so to speak. We could do the truth table for this. This would be 16 rows. I don't want to do it. You can do that on your own time. But we know now constructive dilemma is right up there with disjunctive syllogism, hypothetical syllogism, modus tollens, and so forth. There are lots of other ones that we can build up and practice.
- 7.98 Now, with that said, let me turn back to these two that we started with at the beginning. Let's take this one about family planning trip and start to do the work a little more economically. We look at the first premise and what does it tell us? One, Toronto or Rochester. Sentence two. If we go to Toronto, then we won't see Sylvia. If Toronto, then not Sylvia. Sentence three, where I have to see either Sylvia or Quincy this trip. Sentence four, I just found out that Quincy isn't available. New information. Conclusion, Rochester. The assertion then is that that follows as a conclusion.
- 7.99 OK, can we validate this reasoning, that it's valid reasoning? Well, let's now be more creative rather than just mechanical. Look at this not Q. Well, that's nice because then you can rule something out. Well, if it's either S or Q, but we have eliminated Q, it must be S. How do we know that? I take my third line and my fourth line, and I'm doing disjunctive syllogism. Beautiful.

- 7.100** Now I know S. Well, S pops up over here as well. This says if T, then not S, but S is definite, so it must be step 6, not T. How do we know that? Take premise 2, take premise 5, and do modus tollens, or denying the consequent. If it's not T, T also pops up in the very first premise. It's either T or it's R, but now we have eliminated T, so therefore, step 7, it must be R. What we have then is premise number 1 and premise number 6, and that is another disjunctive syllogism.
- 7.101** We have five sentences here, but we have now shown in a three-step proof that, given those five sentences, they are logically integrated. The first four provide valid support for the fifth. Let's try one more. This is the party planning example. If Alice comes to the party, then Bob will. If Alice comes to the party, then we get Bob.
- 7.102** Now, we have this semicolon in the "and" thing. This is meant from the person making this argument for this to be a whole sentence unto itself. You might find at various points that it's useful to break up parts of the sentences and use them as separate information. But if it comes to you initially as a package, honor the argument's intentions.
- 7.103** So, and if Carl comes, then Darlene will. So put those in part. And if Carl comes, then Darlene will. If either Bob or Darlene comes, then Fred won't. OK. So we have either Bob or Darlene, or then we're told that if either of those, then we don't get Fred, not Fred. We are told next, either Alice or Carl is coming. Then the person concludes, therefore, we're not going to get Fred.
- 7.104** OK. A lot of symbols. Can we work on that, given what we know, and prove, or actually fail to prove, in which case we might be dealing with an invalid argument? Let me throw a constructive dilemma out there since we just encountered that one. If we have A, then B, and if we have C, then we get D. But we know we're either going to have that A or we're going to have that C. Can we then say, by constructive dilemma, it must be that we're going to get either B or we're going to get D?
- 7.105** That, showing my work, is to take the first premise, that big monster, and combine it with number 3, and I'm using the constructive dilemma pattern of reasoning. Now, this B or D, it shows up up here, but it shows up packaged as the antecedent of a hypothetical statement. That's more complicated. But this is the antecedent. We know if you have the antecedent affirmed, you can also affirm the consequent.
- 7.106** So if it's the case we have either B or D, then we're going to not get F, and we do have either B or D. It must be the case that is not F. In that case, we are taking number 2 premise and number 4 premise, and we are doing modus ponens. So, yes, this colloquial English reasoning, we've now proved it in two steps using these legitimate methodologies.
- 7.107** The point then is, this is a remarkably powerful set of tools where we go all the way down to fundamentals. We have our categorical schemes in place, our concepts about propositions, all of the connectives. We define each of them rigorously. Then we start

to play around with combinations of them, and we know with that set of tools, we can sort out the legitimate forms of reasoning from the illegitimate forms of reasoning.

- 7.108** We can learn to recognize both, particularly the commonly used ones that are in both categories, to do increasingly complicated and creative thinking. And so logic ends up being not a merely abstract set of interesting tools that you may or may not have a good time with, but actually practically, enormously valuable.
- 7.109** Now, one of the practically, enormously valuable implications of this has, of course, been in the field of computers and more increasingly sophisticated language. And the basic insight is that what we can do is take what the logicians have developed, which is a binary between truth and falsity. We can take everything that we do in language and everything that we do in our thinking, and say all of the constituent propositions are either true or false, and the various ways of combining them, we can assign truth or false values.
- 7.110** It doesn't matter how complicated it gets, we now have the logic at our disposal to do so. But what we can also do is make very strong connections to mathematics. One of the things that mathematicians do is they use, of course, numbers for much of what they do. Comfortably, most of us learn a base 10 number system. Maybe it's because we have 10 digits, and our first counting and keeping track of things was based on that. We get up to 10, we run out, so then we start over with a new device.
- 7.111** But there are other number base systems, and the mathematical base system that seems closest to this is to say that we can map anything true or anything false onto binary, which is just a one or a zero. So whatever we're doing in all of the complicated logic, we'll just translate that into mathematical language. Anytime T is there, it'll be a one. Anytime false is there, it'll be a zero. And that also makes conceptual sense as well.
- 7.112** So what we can then do is say, all of the logical work explains and is integrated with all of the mathematical work if we translate everything into binary. And then we're interested not only in mathematics abstractly and logic abstractly, we're interested in the real world. Then scientific and technological discoveries come along that enable us to utilize certain forces of nature that are themselves rather binary.
- 7.113** For example, electricity and electrical switches. We can take an electrical switch where there's a current running through, and we can either have that switch be in the on position or in the off position. Then various combinations of electrical lines running with various combinations of off and on switches will then be machine representing logical structures or mathematical structures. So that maps onto the one-zero, which also maps onto on or off.
- 7.114** With that, we have opened the floodgates for further discoveries, further power at the hands of human beings. We know how sophisticated this can get. We can work out the logic for all of these things, and we can express that mathematically, and then we can

start to design circuits, but have the layout of those circuits. Sometimes the circuit is on or off. Sometimes we can take an electrical wire and split it, so the current can either go in this direction or it can go in that direction.

- 7.115** That's represented by the logic of or, and we know what to do with that. Or we can take a current that has electricity running through it, another current that has electricity running through it, and combine them. It's this and this, and we put them together. We have more sophisticated switches. If there is current here, then the switch turns on. That's an if-then. But if there is no current there, then the switch does not turn on.
- 7.116** Then we can start saying, OK, those are all components. And now what if we take a dozen of those and arrange them in various ways? A hundred of those, a million of those. And at that point, we are into the very sophisticated chips that we use in our computers, but we can rely on those computers because we know the logic has been worked out axiomatically, it's been mapped onto the math, and then physically instantiated in the chips. That's a beautiful thing.
- 7.117** One of the exciting things about the development of logic has been the increasing universalization of logic, so that we can take all of the kinds of reasoning that we engage, and all human beings, having the same basic physical and mental makeup, all of them can learn logic. The extent of the universalization extends beyond human beings to machines, computing devices, and so forth.
- 7.118** Now, within, though, the philosophy of logic, that is a controversial claim because there are many who want to argue that logic is not universal, that in fact there are multiple logics because human beings are not all the same, and the way they construct the reality within which they live is not all the same. So from various philosophical sectors and from various politicized philosophical sectors, you will see attacks on the universality of logic in favor of relativized, group-subjectivized accounts of logic.
- 7.119** So here are three examples from mid to late modern history. These are all names perhaps that you have heard of. Georg Hegel on the left, Karl Marx in the middle, Adolf Hitler on the right, and all three of them represent attacks on logic at the fundamental, logic as the other logicians, the universal logicians, want it to be.
- 7.120** Hegel, in his philosophy, argued that the world is evolving through various stages, and the human beings that are born into those stages are shaped and molded and come to think according to the stage in historical development their world is in. But that the German nation and the French nation and all of the nations of the world are on different developmental paths. And so as a result of that, the humans in those develop different logics, different ways of thinking about their world, such that they can't really communicate with each other at a fundamental level.
- 7.121** Even to the point of contradiction, which is the deepest no-no in the universal systems of logic, reality is not contradictory, so our thoughts can't be contradictory. Hegel

argued that reality evolves through contradictory stages and across nations, different contradictory situations as well.

- 7.122 His philosophy, for example, wants to say the law of non-contradiction is not a legitimate principle for some peoples at some points, which then, of course, means you don't have the capacity, really, to argue with them or expect them to come around to your side of the debate as appeal to facts using methods that are available to all of us in the same world.
- 7.123 Karl Marx and the next generation, instead of emphasizing the nation or the ethnic group, emphasize economic classes, arguing that human beings are born into different economic circumstances, and those are also, of course, evolving across time. But if you're born into these economic circumstances, you come to think about the world a certain way, including your logic.
- 7.124 If you're born into this class, it's the same. You have a different understanding of the world and a different logic. Those two classes cannot really understand and/or communicate with each other. Hence, things are going to be solved not through argumentation, diplomacy, but rather by means of violent method.
- 7.125 Adolf Hitler and his theoreticians represent just another variation on this poly-logist theme, which is to say that it's not your nationality, it's not your class membership economically, but rather your race. Different races construct their worlds differently, think and have actually different logics as well. For his preferred races, there's no point in trying to accommodate, understand, or work with members of other races. Again, we're into a violence with respect to sociability.
- 7.126 The point then is, this is a deep issue in the philosophy of logic. Hegel was a deep thinker, Marx was a deep thinker. Many of the theoreticians behind Adolf Hitler were PhDs in philosophy who had a lot training in what they thought was the outmoded non-Aryan logic that they wanted to reject, but they were smart people who knew what they were doing. This is another deeper issue in the philosophy of logic, just to attend to for further thinking.

Lecture 8: Induction & Open-Ended Discovery

At a glance — Induction and open-ended discovery.

- 8.1** So far in this course, we have been focusing on what we call deductive reasoning, and that's already very complicated. Working out the methodologies for handling all of the different kinds of deductive arguments takes a certain amount of time. There's a standard contrast between deductive reasoning and inductive reasoning. In this session, I want to turn to induction, which in many ways is even more complicated than deduction, but to stake out some basic territory for inductive reasoning.
- 8.2** Now, on the board here, I have four arguments, and the idea is to contrast the black ones with the ones that are in blue. We take this one here. All plants contain chlorophyll. Fuchsias are a type of plant. Fuchsias contain chlorophyll, is the conclusion there. We recognize that as a categorical argument. We're talking about all plants. Here's a particular kind of plant. We apply the generalization about all plants to this particular plant and draw the conclusion. Perfectly valid form of reasoning.
- 8.3** Here we have a disjunctive argument. He's either stupid or lazy. You know he's not stupid, and you conclude, therefore, it must be that he is being lazy. Again, we recognize this as a valid form, and it follows necessarily. If those are our two only options, you rule one out, it's got to be absolutely the other one. Now, I'm emphasizing in my tone some language that will be different when we read these other arguments here. Go to this one here.
- 8.4** We examined 280 chronic liars. We're psychologists or social psychologists, and we're doing a systematic study. 280 chronic liars, all of them suffered from low self-esteem. Our discovery. So, now, here's our conclusion. Probably all chronic liars have weak self-esteem. Now, a couple of things are interesting here, is that we started with an "all" statement about an entire class, and then started to focus on an element within that class or a type within that broader class, all plants, applying it to two particulars.
- 8.5** Our intellectual move is different here. Here we are talking initially about 280 individuals, but then we are generalizing to the entire class of people who are chronic liars, and that's a lot more than 280 individuals. Here we're going from all to a particular, or from a generalization to a particular. Here we're starting with a number of particulars and generalizing to the universal statement, and that's a different intellectual direction.
- 8.6** The other thing is that we are, say, well, it has to be that fuchsias contain chlorophyll. It follows necessarily as a result of this piece of information here. But this person is being careful and puts the word "probably" in there. From the fact that we've studied 280 chronic liars, that's pretty good. But if we're talking about potentially thousands, hundreds of thousands, millions of people who are chronic liars, do you feel super confident that your study of 280 of them applies more universally? Maybe there's a

difference between reasoning deductively, which enables us to feel certainty, it's a valid argument or it isn't, versus less certain, more probabilistic or possibilistic kinds of reasoning.

- 8.7** In this example here, a magazine did a survey. 90% of those who responded were against free love. Therefore, 90% of the general population are opposed to free love. Now, in this case, we have a particular magazine, the particular individuals who responded to the magazine, and then we're going to a more general, the general population, which is much more. How confident are we about the nature of that generalization? There is some discretion and judgment that is called for, as well as there being this issue about how we do these generalizations in the first place.
- 8.8** Now, that we do generalizations, and also lots of other animals seem to do generalizations, is important and it's powerful. So you take kids, for example, and you tell them, "Don't touch the hot stove, don't stick your hand in the fire," and so forth. But at some point, they are going to experiment, and we hope it's only one experiment. So they might touch the hot stove one time, suffer the pain, but they don't then say, "I should probably run that experiment again." No, one-time generalization, never touch the hot stove, negatively speaking.
- 8.9** We do generalizations all over the case. I'm stuck in the burn mode. You go out in the sun one or two times, you stay out too long, you get a sunburn, you learn your lesson. Never engage in too much sun exposure, whatever that's going to be. Generalization is very handy in lots of cooking. For example, you're cooking pasta. I expect you do the same thing that I do. You put the pasta in the bowl or in the pot, you boil it for a certain amount of time. Is all of the pasta in the pot cooked as much as I would like it to do?
- 8.10** You take a device and you take one piece of pasta out. You let it cool off a little bit, you taste it, that's good, and then you generalize. All of the pasta in the pot is cooked. And so that again is going from particular. I don't have to test each piece of pasta individually to do so. The question, though, about why we do that and under what circumstances we can feel that we are doing so legitimately, there's an immediate philosophical puzzle that we can raise because we can say, "Well, look, the pasta in the pot, they are all the same."
- 8.11** They're the same product. They're subject to the same manufacturing process. We're putting them in the same conditions. Water, the water is uniform all around the pot. If all of the same antecedent factors are the same, then we should have the same effect on all of the individuals. I only need to test one piece of pasta and then I can make the assumption.
- 8.12** What makes that assumption legitimate is the broader assumption that we are living in a cause-and-effect world. If you have the same cause, you'll get the same effect, or if you have the same set of causes, you will get the same effect as well, because the world as a whole is cause and effect. Now, the interesting question philosophically

comes in, because that right there is a generalization that justifies our saying the same causal processes are working in this case, this case, and so forth, deductively.

- 8.13** But that very big generalization about the world as a whole, where did we get that from? The idea that all of reality is cause and effect? I might then say, well, I was looking at reality over and doing with this part, and it was all cause and effect operating here. I was looking at reality over here, and it was cause and effect in various ways, and it's also cause and effect over here. I've checked all of these areas, and therefore I'm generalizing.
- 8.14** But that's exactly the thing that we have in question here. Under what circumstances is it legitimate to generalize, and how do we do so? How do we get to great big whopping generalizations, like all of reality is cause and effect, when we know reality is a pretty big place and we have not experienced all of reality? At the same time, we do feel quite confident, fire burning, pasta cooking, and lots of other examples.
- 8.15** But where we have to start being cautionary is that we know lots of times we do generalizations, and the generalization is just way off bat. I might, for example, maybe a thousand years ago, I walk under a ladder, and then within a few steps, I trip and I fall off the curb and twist my ankle, and I say, "Oh, cause and effect. Walking under a ladder causes bad luck." There I've got a superstition, and I'm generalizing to a certain causal understanding of the universe, but that's just wrong.
- 8.16** And lots of times we know we might meet individuals who are members of certain ethnic groups or whatever. I go to Manhattan and I ask someone for directions, and he gives me a certain hand sign and says he's not interested. I say, "Whoa, New Yorkers are so rude." And so immediately from this example, I'm going up to the big generalization. And it might, of course, be that that generalization, I've just soaked it up from osmosis listening to other people and their experiences, but that's not very systematic at all.
- 8.17** Of course, we can go the other way. Canadians have a reputation for being nice and polite. You meet a Canadian who comes to visit and say, "Oh, boy, that's such nice people. Those Canadians are just so wonderful." Well, let me tell you, many are, some aren't. So that's an inductive generalization that remains to be put to the test.
- 8.18** More amusingly, there's a famous example from Bertrand Russell, who was, wrote on, a famous philosopher from about 100 years ago now, who gave the example of the chicken. And the chicken in the barnyard, and every morning the rooster crows, the chickens come out. The farmer comes by with a bucket of feed and throws the feed down and the chickens, happily—if it happens the next day, happens the next day.
- 8.19** So you imagine from the chicken's perspective, say, "Wow, this farmer is such a great guy, every day without fail, blah, blah, blah." And it's now been dozens and dozens of days. "The farmer is looking after us. He's feeding us." And then, of course, one morning, sun comes up, rooster crows, the chickens come out, farmer grabs the guy by

the neck, takes him over to the chopping block, whack. So that generalization, if we can imagine the chickens are engaging in generalization, that the farmer is benevolent and looking after you, that that is going to extend, is obviously faulty.

8.20 So generalizations are enormously powerful to us in our thinking, but at the same time they are fraught with danger and there's lots of false generalizations. You think about what scientists are doing for most of their career. They're looking at particular examples, trying to find decisive experiments to run, to generalize to a whole class of whatever it is that they are studying, and to come up something they want to call a law or a principle. We then use those laws. In many cases, we stake our lives on the basis of those scientific discoveries and the experiments that they've been done at the time. It's extraordinarily fraught.

8.21 Not only in science per se, one of the most famous examples of the power of generalizations comes from Adam Smith, writing in 1776, when he published *On the Wealth of Nations*. There's a famous example that he opens with, of a pin factory. Now human beings have been around for tens of thousands of years, working, making stuff, and we're always concerned with, "How can I make stuff more efficiently, lower cost, and increase my productivity and so on?"

8.22 But Smith, as a social scientist, happened to go and study a pin factory that makes pins that we use in clothing and so forth. He had studied or looked at a certain number of pin factories, and he was blown away by how productive this pin factory was. Because up until the early modern world, the standard was that one person would have an area of expertise and would do all of the functions in that area of expertise.

8.23 If you want to make pins, then you buy your metals, and then you hammer it out and make it into wires and you cut it, and then you shape it into long strands. Then you take some snips and you cut it to length, and then you put it in a device, and you tap it on the end with a hammer. So you have a little bit of a cap at the end of your pin. You sharpen the edges, then you put it in the box and you wrap up the box. And so if you're really good in practice, you can make a certain number of pins per day. Maybe you can make 500 pins in a given day.

8.24 But he went to this factory, and it turned out this factory was cranking out 50,000 pins a day, and it only had about 10 people working at the factory. But 10 independent craftsmen, each of them, if they were very productive, would only be making about 5,000 pins per day. And so what he noticed was that they had divided the labor. They had one person who was responsible for shaping the metal, pounding out the metal. Then the next guy would cut it, the next guy would shape it, the next guy would cut it to length, the next guy would sharpen, the next guy would tap on the pin.

8.25 And for whatever reason, this is what he was interested in, that method of division of labor made each of them more efficient. Now, a lot of it has to do with transition times. If you are doing this part of the operation and you're going to do the next part of the

operation, well, then you have to put down your tools for that part of the operation, get your tools, practice a little bit until you get into a flow, then you put the tools down. That might explain a lot of it.

8.26 But Smith wants to say, they have divided the labor into parts, and then each individual person is specializing in that part of the labor, and that's adding up to enormous productivity gains in this particular case. Now, this is pins. Does it also work for other things, not just pins? What if we're talking about clothing manufacturing, even to stay within that territory as well? And so he said, yes, it does, in fact. And so at a certain point, he's ready to generalize through this principle of division of labor.

8.27 And the idea then is, this is an enormously powerful generalization for those of us in business. If we're interested in business productivity, whatever it is that I'm doing 100 years, 200 years later, I can take that generalization, apply it to my business, and chances are good, I will be more productive as well. That is social science using generalizations and so on. How do we do this?

8.28 Now, we know there are all of the fraught cases, all of the superstitions that typically, and all the stereotypes and so on. At the same time, there are very good generalizations that we've engaged in. Now, the challenge is that when we think about evaluating inductive arguments, we use criteria that are different from the criteria we use in deductive arguments.

8.29 One thing we say in the case of deductive arguments is, say the conclusion follows necessarily from the premises. If it's going to be a valid argument, the conclusion follows necessarily. But it doesn't seem the case that the conclusion follows necessarily from an inductive argument. Instead of using necessity language, we start switching to possibility language, probability language, likely, maybe, and we're more cautious, particularly in cases like this about many of our generations.

8.30 We can't be looking for immediately. We might get to certainty, but we can't be expecting certainty in all inductive arguments. Sometimes it's a process where we generalize cautiously, then we generalize more certainly, we have studied, and so forth. The other thing we sometimes do is to say, well, in a deductive argument, really the conclusion is contained in the premises, and all we're doing is making explicit the information that's contained in the premises.

8.31 We saw that, for example, when we were doing the Venn diagrams for categorical syllogisms. We take the information that we get in the premises and the categorical syllogism, and we diagram it. Then we stop and we look and we see already we have diagrammed our conclusion, if it's a valid argument. In a way, the conclusion was already in the premises and we're just making explicit the information that's already there.

8.32 In this case, we might say, in the first premise, we're talking about all plants, and then we're already applying that in a particular case to fuchsias. But fuchsias are a plant, so

already implicitly we've been talking about fuchsias because they're in the category that we established initially, and we're just drawing out information. Now, that then is to say, we're not expanding the range of our knowledge, we're not amplifying anything, we're not going beyond the available evidence.

8.33 But all of that does not apply in the case of inductive arguments, because that is exactly what it seems like we're doing in the case of the inductive arguments. We're going beyond the initial 280 people that we studied here to all of them. It is an amplification. It is a generalization, is the word that we use, or even a universalization. In this case here, we're saying, well, 90% of those who responded, we didn't even have a full response rate. We're against free love, so therefore we are going beyond, we're amplifying and making a claim about the general audience. We need different evaluative standards.

8.34 Now, what has been developed over the course of time as logicians and philosophical scientists especially, who are studying the methods that they are using, is to say, what we have are five rules of thumb. Initially, we will call them rules of thumb for evaluating inductive arguments. Sometimes it's going to be a matter of more or less, depending on how well you satisfy these criteria.

8.35 So, one of the obvious criteria is to ask, how big is your sample? So the more instances you have initially in your sample, the more credence will reside. So you go to New York, you've never been to New York, you've never heard about New York, you don't have any preconceptions about New Yorkers. You meet one, and maybe the person is rude. You ask for directions, and he blows you off. All right, that's one. To generalize to that, all New Yorkers are rude. Based on one, that's too much.

8.36 On the other hand, if you had five New Yorkers in a row, that's more. But still, probably not enough for all New Yorkers are rude. Maybe it was rush hour. Maybe it was a super hot day. Maybe everybody was crowded in the subway. So there can be other factors at work. But still, five rude New Yorkers in a row, that's better than one. So that's one thing to check. Partly this is, don't generalize too hastily or too quickly. If you have more instances available, look for them and bring them into your survey sample.

8.37 Another one close is the diversity of your sample. In the case of pasta in a pot, it's hard to get diverse samples because they are all the same. But when you're dealing with more complicated things than pasta, say, like human beings or plants that are actually growing out in the wood, you know there's a lot of diversity among plants, compared to strands of pasta. So chances are good, before you make generalizations about plants, you'll need to consciously check a number of different types of plants.

8.38 And then in the case of human beings who are even more complicated than plants and much more variable, you have to pay attention to all of the ways in which human beings are diverse amongst themselves. It might be relevant to whatever it is that you are talking about and get samples there. If you're at a very basic biological level, and

you sense that humans have all of the same kinds of cells, whatever cells are in the skin, for example, it's all pretty much the same. That might be one thing. On the other hand, if you're talking about human beings' psychology, psychology is a lot more variable among human beings than their skin cells, then your project has to be a lot more diverse.

- 8.39** A third thing is a self-check. That one of the things that we know that we have a problem within our reasoning is looking for things that support the thing that we want to believe or the thing that we want to investigate, and not to as easily pay attention to things that might go against our hypothesis. So the rule of thumb in this case is to go out of your way to try to find examples that don't fit the pattern. Try to find cases where there's an element in the group, but it doesn't have the same trace, and see if you can disprove your generalization.
- 8.40** This is a psychological check. Basically, it's a variation on the diversity point, but you're looking for a diverse factor that might, instead of just looking for diverse factors that might confirm and still fit your hypothesis. So, a standard example here is, we might take something relatively simple like water and want to know what the temperature water boils at. We might have 10 pots of water, and we might boil them, say, in pots that are made out of different metals.
- 8.41** We have the thermometers in there and we find, well, all 10 of them, it doesn't matter what kind of pot they are in, they all boil at 212 degrees Fahrenheit or 100 degrees Celsius. Then we're ready to generalize. Water boils at—but then it might be that we have set ourselves to the project of looking for diverse circumstances and find that, if we were, for example, already up in the mountains, that it might not be 212 degrees. The temperature might be a little bit different.
- 8.42** If we went down sea level, the temperature of boiling seems to be the same, if we went even up higher in elevation. If we were consciously setting our minds to looking for that as a possible factor, as we know, for example, that we are affected by, say, changes in elevations, maybe the water is affected by changes in elevations, but we're looking for something that might disconfirm our example.
- 8.43** Another important one is to make sure whatever it is that you study, that it fits with all of your previous background knowledge. Partly, this is the thing that helps you in figuring out what number of diversity factors might be relevant. But it's also important to say, on any important thing, before I get too much into the generalization range, to make sure that this general hypothesis fits with other important things that I know by analogy.
- 8.44** For example, this is in the history of sciences, the history of zoology, actually, for a long time, people in the West thought that swans were white, and that's a reasonable thing to believe. Because if you are in England or you're in Western Europe and you are among the first people to start doing systematic biology, all of the swans that you

encounter are white. It might be the case that all of the cardinals that you have encountered, the males are reddish and the females are a more mixture of olive greens and browns and tans and so forth.

- 8.45** And so even though you've studied, say, all of North American cardinals or you've studied all European swans, you're comfortable with a generalization that swans are white. But even before you travel to other parts of the world—and there are other parts of the world where the swans are not white, there are native black swans in various parts of the world—but you can reason and say, well, one of the things I do know about birds is that birds are a kind of animal species. And animal species like dogs and cows and horses and so forth are much more varied in their external coloring.
- 8.46** Human beings, I also know, are much more varied in their external covering. So before I get too locked into this idea that all swans are white, their external covering of the feathers, that also might be subject, or something that is subject to high variation depending upon where one is in the world, by analogy to human beings and other animals. So make that a formal part of one's check.
- 8.47** It's also in a social context where this part is useful. So you run your ideas by someone else who has different experience, different background knowledge. Often they can point out things that you need to pay attention to. And then the next one, and this is where the science really gets down to business, is to say, if you can find underlying causal mechanisms that explain whatever it is that you are talking about.
- 8.48** So you have the pasta in the pot, and you can do the chemical analysis of the pasta. You can see, say at the microscopic level, that the pasta's molecular structures are all the same and all identical, then you know the underlying process and how it's interacting with the heated water, and you're in a safer position to generalize about surface matters like texture and how soft or hard it is.
- 8.49** If you can, as we all know, fish need to stay in water in order to stay alive, we can learn that, we can be quite comfortable in that generalization, but if we can start doing the science and figure out that the way fish process their oxygen, the way that we process the oxygen, the mechanisms are different underlying. We know what our lung structure is like. We know what their gill and underlying causal structures are like. They are different. They work differently. Therefore, we can explain the effect, fish needing to be in water, by appeal to a deeper mechanism. That makes the inductive generalization stronger.
- 8.50** Now, let me run a couple of examples by you and just ask for your reaction on them. Suppose I gave you this one. The barber did a good job cutting Rick's, Bill's, and Larry's hair. Those are my premises. Good job on Bill, good job on Rick, good job on Larry. He will do a good job on mine. That's the generalization. He's a good barber. I can plug myself into that mix. We will get the same effect.

- 8.51** So, what would we think in this case? Well, we would say, Rick, Bill, Larry, and I, we are all human beings. Those are all male-sounding names. So same general style of haircut. I'm a male, I'm going to get the same general style of haircut. Any other factors that we should think about? Well, is it that they all go first thing in the morning? Maybe there are morning people and I'm an afternoon person. He's good in the morning, but as he gets tired, he gets worse and worse. Maybe I should think about this as a factor.
- 8.52** Is it that Rick, Bill, and Larry are all friends of his? But he doesn't really know me, and maybe he pays extra attention with friends, and maybe he's not quite so good with strangers. We would think through all of that. We'd say, well, this is only three people. Is that a big enough sample? And so forth. But those are the kinds of questions you would ask. I would say this is a weak generalization, but there's probably something. If it's only one haircut, maybe it's worth the risk.
- 8.53** Out of 20 letters to the editor, 18 complained of the inconvenience caused by the strike. The public is clearly antagonized by the strike. We got 20 letters to the editor, 18 were complaining about the strike. Therefore, the general public is antagonized by the strike. How strong is that? Well, it's 9 out of 10, or 90%. That's pretty high.
- 8.54** On the other hand, we might say there's a difference between people who write letters to the editor and people who don't write letters to the editor. People who are really worked up about something are more likely to write a letter to an editor. Maybe the majority of people are not particularly excited by the issue, so they're not writing. I'm drawing only from a much smaller pool of people who are worked up about it.
- 8.55** We might also bring in some background knowledge. We also know that if things are going fine, people don't typically say anything. But when something is going wrong, that's when people start to complain, and the squeaky wheel gets the grease. It might be that we have a disproportionate sample toward not only people who are worked up about the issue, but people who are on the negative side of that issue. So I would say this is a weak generalization. You've got something to go on here. Clearly, there's enough people maybe to pay attention to, but this generalization seems to be much too strong.
- 8.56** Buster, assuming he's a young guy, maybe he's in his teens, should prove to be an excellent batter. He plays just like Babe Ruth, one of the most famous baseball players of all time. He has the same stance. He stays low and has a real good eye. He should do really well in baseball. OK, so some optimistic thinking about the prospects of this individual. In this case, what would we say?
- 8.57** Well, we're making a comparison between someone who was a great baseball player and someone who is a young and developing baseball player. We are pointing out some similarities— stance, his eye-hand coordination, and so forth. But this one, I would say, seems to be weak because, given our background knowledge about human beings, there's a great deal of variation.

- 8.58** If we have more specialized knowledge about baseball, we know that to become a very good hitter, much less the best hitter in the history, that there are dozens and dozens of factors, not only a dozen physical factors, lots of psychological factors, and we don't have any information about any of those. So predicting that this teenager is going to turn into Babe Ruth or something close is relatively weak.
- 8.59** All right, now suppose we are getting past wanting to be accomplished thinkers as we go about our lives. We're going to be engaging with particulars, learning all sorts of new things, informing all kinds of generalizations ourselves. Sometimes we're going to be hearing generalizations from other people, and we want to know whether we should accept them provisionally or wholeheartedly, so we know how to think through them for ourselves and for the ones we're hearing from other people.
- 8.60** Is it a big sample? Is it a diverse sample? Did they go out of their way to find disconfirming examples? Are they looking for underlying causal mechanisms, et cetera? Then in a rough and ready but sophisticated way, we're ready to take on the complicated world and navigate it with all sorts of generalizations. If we upgrade and we want to do that process more systematically, we want to be scientists, social scientists of various sorts, then there have been developed methods that are called inductive methods.
- 8.61** More historically, credit-wise, they are sometimes called Mill's methods, because John Stuart Mill, the great 1800s British philosopher who wrote significantly on logic, was among the first to systematically identify four inductive methods that scientists can and should regularly use, depending on what they're studying, in order to make more rigorous their approaches and their findings.
- 8.62** Let's take this first case, and I've got a schematic of it here, and it's just this upper left quadrant on the board. You are a scientist, and what you have is, all right, I'm studying four different cases. I'm looking for a cause and effect explanation here. I noticed beforehand, we had A, B, C, and G, and as a result of that, E happened. In the second case, we had A, B, D, and G, and E also happened. In this case, A, C, F, and G happened. Then we also got the effect. In this case, it was B, C, F, and G, and we also got the same effect.
- 8.63** Having gathered all of that data, put it in this table to structure the data, we start to analyze. One of the things we are thinking is, well, we're trying to find the cause of the effect. A principle of cause and effect says you will get the same cause leading to the same effect. Something that isn't present can't be a cause. The thing that is the cause has to be present in all of the cases.
- 8.64** We can start going through and sorting, and we say, well, in this case, C was not present. We had D. In this case, D was not present. So C can't be the cause, D can't be the cause. In this case, we had F. In this case, we also had F, but we didn't have F in these cases, but we still got the effect, so it can't be that F is the cause as well. So the

only thing that is in common in all of these cases, or the only thing that is, that agrees in all cases, is the presence of G. Every time we got the effect, G was present, and that's the only thing that was present every single time. Then we can draw a conclusion from these four cases. G was the cause of E.

- 8.65** Now, a homey example would be, suppose you and—how many people would it be? Three friends. You went to a party, you all ate and drank a bunch of stuff, and then the next morning, everybody has a terrible stomachache, all four of you. And so you start to do the postmortem, so to speak. You're still alive, so postmortem is a bit harsh in this case. But you say, "You, what did you eat and drink? You, what did you eat and drink?" And someone says, "Well, I had a burger, I had a Diet Coke, I had some salad, and I had some lemonade." And then someone says, "Well, I had a burger and some Diet Coke and some pretzels." And then the third person had a different array of things.
- 8.66** But it turns out, lemonade is the only thing that all four of you had. Everybody had different combinations of everything. Then it's quite reasonable to say it has to be the lemonade. Now, I said that pretty strongly. It has to be the lemonade. Now, this is where we might start and say this is good science. We're being systematic and we're probably isolating something important, but it might be something else.
- 8.67** It could be that you all went on a little expedition in the same car, and someone had left something rotten in the car, and there was some sort of mold in the air that you all breathed in. If we're going to be really serious about it, we would have to control for that variable as well. But presumably then we would just add that to the list, and we would see another kind of test, to see if we can differentiate the lemonade from, say, that mold. But this is a perfectly strong, powerful, commonly used method.
- 8.68** This is, as Mill labeled it, the method of agreement. It's one of the basic elements of scientific method to learn. This one here, two cases, in any case it's simpler. In initial conditions, you have A, B, C, and F, and you consciously make sure people in this group are exposed to F, whatever that is, and you get the effect. In the second case, you make sure that people are exposed to not all of those factors, you explicitly withhold F, so they're not exposed to it, and you don't get the effect.
- 8.69** Then on the basis of that, it's reasonable to conclude, F is the cause. F is present, it happens. F is not present, it doesn't happen. This is the standard for initial rounds of drug testing, for example, among many things. You have a new experimental drug. You want to know, does it work? You get a certain number of volunteers, say, 200 people. You divide them into two groups, and hopefully you can divide them into, the groups are more or less homogeneously.
- 8.70** If you want to be super controlled, you will say, let's all get the same amount of sleep, have the same amount of food, drink the same sorts of things. Those might be the A, B, and C. They both slept sufficiently. They ate the same breakfast. They had the same amount of coffee. One control group, you give them your experimental drug. The other

group, you give them a pill that's a placebo pill. It doesn't have any active ingredients whatsoever.

- 8.71** In one case, you get the effect. Hopefully, it's the effect that you want. In the other case, it doesn't make any difference. Therefore, you can include positively in this case, there is something to your new pill. Now, how big do your human groups need to be? How many of these variables do you need to be controlling? That is an art, as well as a science. This John Stuart Mill labeled the method of difference. As in the method of agreement, you're looking for the thing that agrees or is common in all cases. In this case, you're looking for the one differentiating factor, and we label it so accordingly.
- 8.72** Now, there are other cases in science, though, where the factors, the causal factors and the effects, we're not able to isolate them independently from each other. We have to figure out methods in order to be able to test for one, even while the other ones might be changing. Sometimes we can hold them constant in varying degrees, but we can't separate them out. It's not, we will give you the pill or we won't give you the pill. We will drink the lemonade or we won't drink the lemonade. Those are either-or, off-on sorts of things.
- 8.73** An example to think through this would be something we know in reality, and that is that pressure, temperature, and volume, say, when we are studying physics. I can take a flask or a container of some sort, and suppose I have a sealed container with a certain amount of oxygen or some gas that is inside that container. I know I have a volume, I know I have it at a certain pressure, and the whole thing might be at a certain temperature. But we want to know what's the relationship between pressure and volume and temperature.
- 8.74** We can't just say, well, let's just take the volume away. We'll have things that have no volume and we'll test them against things that have volume, or things that have no pressure and things that have some pressure and so forth. We can't get things down to really absolute zero. I think we can get pretty close, but we just can't say, let's have no temperature and compare that to things with temperature. Instead, what we know is that these things vary with respect to each other. In order to come up with laws like Boyle's law, we have to then hold what we can hold constant and notice the variations in the others.
- 8.75** One thing we might then do is to say, well, we have this container. Let's keep the size of the container the same. That then is to say, we are going to hold the volume constant. We're not going to change that. We have a gas inside the volume, and right now it is at a certain temperature. We know what the volume is, and maybe we can measure the pressure that it is at. Now, one thing we can control the variation on is temperature. I might take a heat source and apply it to the container. I'm going to add heat into this container, and then the volume is going to stay the same, but my pressure measuring is going to change.

- 8.76** Then I can then say, as my temperature changes, say at this rate, the pressure is going to change at this rate. It might be also that we have an experimental device, where we have our volume is expandable or not, say, like a balloon or whatever. We might then be able to say, well, we have a certain volume of gas in there at a certain pressure. What happens as we increase heat? Well, the volume of the balloon will expand. The volume will change with the temperature.
- 8.77** If we can measure each of these things carefully, then we can form precise mathematical laws about the relationship, without being able to do either of these methods over here. In this case, it might be pressure, temperature, and volume, and we get a certain effect. But we say, as we increase this factor, whatever the effect is that we are measuring, it also increases. Or it might be, as we lessen this factor, the effect that we're trying to measure also lessens. And so Mill called this one, this has the most fun British name, the method of concomitant variations. It is variations, but the variations are concomitant with each other. That is to say, they match each other. Concomitant.
- 8.78** And a fourth one that Mill emphasizes, and it's one that's very commonly used, also involves not being able to separate out factors and measure them independently. In many cases, we're dealing with compounds, and they are separate, but not necessarily separable in the case of experiment. We can use what he called, and I'll give you the name first, the method of residue. That's to say, whatever's leftover, the leftover, and trying to identify the leftover.
- 8.79** Now, my favorite homey example of this, this is something that I used to do, is suppose you want to know how much your dog weighs. You have a little tiny bathroom scale and you have a big dog, and you say to your dog, "Please get on the scale and stay motionless for 30 seconds, so that it can stabilize and I can find out how much you weigh." You know that experiment is never going to succeed because your dog is not going to stand still on the scale for that amount of time.
- 8.80** What you can do, though, is you can scoop up your dog and then you, with your dog, go and stand on the scale. Everything is stable for a certain amount of time. You look down and you say, "Oh, 235 pounds," just to make up a number. Then you step off the scale. Now what I know is, this is me, this is my clothes, this is my dog, and I get $A + B + C$ is the entire weight altogether. Suppose that, for sake of example, ends up being 235.
- 8.81** Then what I can do is, I can put the dog over there, come over, step on the scale, weigh myself, and suppose it comes out to 175 or whatever. Then I can step off, I can take my clothes off, I can weigh them or I can weigh myself, and I can say something like, "Well, I know I am, say, 175. Maybe my clothes are five, if I'm wearing heavy shoes or whatever. Therefore the residue," and here I do the math, "is my dog weighs, what would it be, 50 pounds, 45 pounds? Something like that."

- 8.82** So that is a scientific method. It applies obviously to beyond just measuring dogs and so forth. One very interesting example from the history of science is the discovery of Pluto as a planet. So for many centuries on up until the 1600s, we thought there was only seven bodies up in the sky, sun, moon, and a certain number of planets. I don't know that they had even gotten to Saturn at that point, maybe. But certainly, Pluto, Uranus came along 1700s, 1800s. And then finally, Pluto was discovered.
- 8.83** But the discovery of Pluto was not by direct observation with a powerful telescope. Instead, the scientists were using the method of residue. So what they were doing is, they were able to figure out where Uranus was and what its gravitational attraction was on the various other planets, where Neptune was and what its gravitational attraction was, and then the various other planets and all of the effects on each of them.
- 8.84** You can't ask the planets to stop moving, so you can measure these effects. They're using a lot of concomitant variations in order to tease out the contribution of the gravitational attractions of each planet on the others. Then what they noticed with one of the planets, I can't remember right now whether it was Neptune or Uranus, I think it was Neptune, that if they had taken into account all of the gravitational forces exerted by all of the other planets, then Neptune should be here.
- 8.85** We will predict that Neptune, as it continues its path, will go here. But it turned out that Neptune was there, just a little bit off. But they were very sure about the calculations they had done for all of the other planets. And so the inference was, well, there must be something else out there exerting a certain amount of gravitational attraction, something that we haven't seen yet. And it has to be probably in this position, and it probably has to be about this big in order to alter what we think Neptune's planet should be.
- 8.86** So at that point, they had the telescope technologies. They could train them on that part of the sky and wait. Sure enough, Pluto pops into the picture and we have a discovery of a new planet. That is a lot of concomitant variations in the background. In this case, it's the residue. It's what's left over in the math, and we need to figure out that there's something there that accounts for that.
- 8.87** This is an introduction to inductive logic, and it's very complicated and very open-ended. As we are reaching to the end of this eighth session, this introduction to logic course, I want to do one more cumulative example, but rehearse, before we get to that cumulative example, how far we have come so far just in eight lectures and how much more interesting stuff there is to do.
- 8.88** We started off talking about basic argument analysis, sorting premises and conclusions, filling in missing pieces, attuning ourselves to indicators, and all of that requires working with our ordinary English or whatever our language's facility, to become very good at the grammar and the parsing of sentences. We talked a lot about the

importance of definition and concept formation, and forming hierarchies of concepts in our mind, and being able to position things well and clearly.

8.89 And then the precision of definition in doing so, being absolutely essential to anything that we do later, building on, we've decided that we are using these words as meaning certain things. But there's a lot more work still to be done with concept formation, how we do this, and psychology is the discipline that is going to work out more understanding of how we do these amazing abstractions that we do with the sophistication that we do. So logic has a lot of work to do, not only with linguistics, but ongoing with psychology.

8.90 Prior to all of that, the initial data seems to come from our senses. We are bodily beings, and learning how to use our senses well so that we can notice the similarities and notice the differences that are going to feed into our conceptual structure; and hopefully in an ongoing way, let us finetune or change when we need to our conceptual structures. Logic has to work hand in hand with the empirical evidence that comes along.

8.91 As we started then to look at deductive arguments with increasing abstraction, increasing sophistication, we start to develop all of these methods for dealing with categorical argumentations. Are we now, with better graphical methods, able to have better diagramming techniques, so that with categorical syllogisms right now, we know how to diagram and model on a two-dimensional surface, three-term arguments? But what about four-term arguments and five-term arguments? And so, will the new graphical programs and sophistication enable us to do better with even more complicated such arguments?

8.92 And then, as we saw in the last session, the link between hypothetical arguments and disjunctive arguments and the various combinations of those and truth tables and all of the exciting things that come out of that with respect to integrations with mathematics, integrations with machines that can do our calculations for us. There also are ways, as we progress further down in that road, to integrate in categorical syllogisms into a more broader symbolic logic framework. All of that is cutting-edge and still a work in progress.

8.93 The example that I want to work through in closing, though, is an example of how all of that fits together. To my mind, it is a beautiful example from the history of medicine, when something absolutely important was discovered, but it was only discovered because the person responsible, his name was Ignaz Semmelweis, he was a medical doctor working at University of Vienna Hospital. But for him to figure out this important medical discovery, he had to be doing everything and everything very well that we have been studying in this course so far.

8.94 Let's go back to the 1840s. We are well into the modern world, well into the scientific world. Medicine also is being revolutionized, starting in the 1700s on into the 1800s.

1840s, though, we do not yet have germ theory. The idea that there are little tiny microorganisms, that some of them are good for us and some of them are bad for us, that was all perhaps wild abstract speculation, but we did not have that germ theory. We also did not have any antiseptic theory. That's something that's going to be developed in the next generation.

- 8.95** Ignaz Semmelweis was originally from what we now think of as Hungary, but he had come to the University of Vienna as a young medical doctor, and Vienna was one of the great centers of medicine, medical research, and the associated sciences at the time, and so he was in a very prestigious position. This is a picture of Vienna Hospital in the 1840s, and Semmelweis was working with maternal issues.
- 8.96** Women who arrived at Vienna Hospital, pregnant, who sometimes stayed at Vienna Hospital to give birth, and then they would be in recovery from birth for a certain amount of time. As it happened at Vienna Hospital, you can see it's a very large establishment. Vienna is a pretty large city. They had two different divisions for maternity. One was in one part of the hospital and another was in another part of the hospital. They called them Division One and Division Two.
- 8.97** In these two divisions, women would come to Vienna Hospital. Some would be assigned to the first division, some would be assigned to the second division. Semmelweis shows up. He's making his rounds, checking in on the women in the first division, some of them are still pregnant, close to delivery, some have given birth and are in recovery, and same thing happens over in the other division as well.
- 8.98** As he's going around, he's an alert guy; he's learning on the job and practicing, but he notices something. And he notices that it seems like in the first division, women who have given birth are dying at a much higher rate than women are dying after having given birth in this division. Now, in one sense, this is not particularly remarkable because giving birth to children had been the number one killer of women throughout human history, prior to the modern world and the medical innovations.
- 8.99** But nonetheless, even so, Semmelweis was alert enough. He was observing and he was keeping track. There's a woman who died. Here's another woman who died in the first division. There's a woman who died in the second division. This woman, fine, this woman fine, and so on. In a rough and ready way, he is observing the difference between life and death, rather obviously, but a whole bunch of other things. But he has a category scheme. First division, second division, and he's starting to categorize deaths according to that geographical location.
- 8.100** Now, one might say, "Well, women die in childbirth. That's just the way things go. I'm a busy guy, got a lot of things to do," but Semmelweis cared. He took some extra time to check. And so he made a point of consulting his own records. He consulted the hospital's general records, including from before he was there, and he did the math. And what he found was, this is 1840-something, say '41, 1842, 1843, going back a

certain way. Then we count up all of the deaths in the first division. We put the numbers in there. Second division, we put all of the numbers in there.

- 8.101** So he's making an effort to do all of the previous observations of women dying. All of the record-keeping has been categorically organized. He can use it with his own, and he comes up with a table. It turns out that his initial sense was right, that in fact, women in the first division were dying at almost four times the rate of women dying in the second division. That would not have happened without lots of observation, lots of record-keeping, lots of good categorization, and someone who cares enough to pay attention to all of that, to come and do so. And then the basic math that's involved in calculating those numbers.
- 8.102** So he now knows there is a problem here. Now, again, I might say, "Well, it's a mystery. Who knows why? The world is chaos," etc. Or I could say, "No, there's cause and effect operative in the world. There has to be something going on in the first division that is not happening in the second division, and I think I can figure this out. So I'm going to," and he takes this on then as a research project.
- 8.103** Now, he starts then to consider hypotheses. What could it possibly be? Well, childbirth is complicated. People are complicated. The number of people who are involved in the hospital, there's all sorts of factors. What could it possibly be? So one of his initial hypotheses then was to say, maybe it has something to do with whether the women voluntarily are coming into the hospital or whether they are brought in against their will, as is sometimes the case. So that might be a relevant factor, something to do with their psychological state. Are they there under coercion or not?
- 8.104** Then he goes through the records and see the terms under which the women were admitted. If it was that, then I should find a difference in the two divisions. Notice I'm making a prediction. Then when I examine the records, I don't get that predicted result. It doesn't seem to matter whether they came voluntarily or whether they were brought under some duress. Therefore, that hypothesis is false. Now, we now know that is denying the consequent. That's modus tollens, and he more or less effortlessly is doing so.
- 8.105** Then he says, well, maybe it has something to do with overcrowding. It turns out, when he was looking around, there seemed like a lot more women were packed into the first division than were in the second division. So now we have a new hypothesis. Overpopulation is the cause of the childbirth fever, the death.
- 8.106** Now, I should mention that when he was looking at the deaths more closely, he started to notice some other factors. It wasn't just that they were dying. They seemed to have certain common symptoms, many, many, many of them involving certain blotches on their skin, certain kinds of anemia. He noticed that, as a result of paying more careful attention to the women, particularly now that he knows that he has a problem on his

hands. So, increased observation to noticing something relatively subtle that might have escaped the gaze of other physicians as well.

- 8.107** Then he starts to say, well, OK, if it's not the overcrowding, because that one then gets disproved by the data, another *modus tollens*, what could it be? Now, months are going by, more months are going by, because it takes time to come up with a new idea, to consult all the records, to do the calculations, and so forth. One thing he notices is that the attending physicians in the first delivery seemed to prefer that the women deliver their children while lying on their backs, whereas in the second division, the attending physicians seemed to prefer women in a side position. You put the woman on her side, raise her leg, and you deliver the baby that way.
- 8.108** So then he's thinking, maybe it has something to do with this position of delivery. And so let's run the experiment, and we gather all of the data that's now hopefully in the, in our records. How many women in the first division over here did back delivery, how many did side delivery, and what was the death rate for each of them? Same thing for the second division. Crunch all of the numbers, and it turns out, not the delivery position. It makes absolutely no difference. Another *modus tollens*, another rejected hypothesis.
- 8.109** Now, a year has gone by, two years has gone by, and what you might think is, well, a lot of people would say, "I'm out of ideas, I give up." But Semmelweis does not. He wants to solve this problem. There has to be something. There has to be an explanation. Now, he's starting to get desperate because a couple of years have gone by and he doesn't seem to be any closer to getting the explanation or the causal explanation.
- 8.110** So, one day he's hanging out in the first division, scratching his head, wondering what he can do next. And he notices that a priest walks in one part of the first division, crosses through the first division and out the door and into another part of the hospital. And so he starts to wonder about this priest. He says, "Could it be that the priest is the problem?" Because he goes and talks to the priest and he investigates the priest, and the priest is a good guy. "Why is this guy asking me all of these suspicious questions?"
- 8.111** It does turn out that the priest has his office close to the first division in the hospital, and that whenever he is called to do last rites anywhere in the hospital, the easiest way for him to get anywhere in the hospital is just to cross through the first division, and then he has access to pretty much everything. But he very rarely goes to the second division. So Semmelweis thinks about this, and he says, well, maybe it's not primarily a physical cause. It's not birth position. It's not overcrowding. Maybe it's a psychological cause. Maybe it's something like this.
- 8.112** Women who are giving birth are stressed. They're physically tired, they're physically exhausted. Then they go through the process of birth. That just is draining, physically very difficult. Their emotions are all up and down, new life, and so forth. At the same time, they are psychologically drained. There are often physical things that have

happened, tearings that have happened, and they need time to recover from all of that. What we have is women who are really at the outer reaches of what a human being can endure. That seems like a pretty good description of much of childbirth historically.

- 8.113** Then what happens in the first division is this priest comes in. From the psychological perspective of someone who's maybe this close to death, you see a priest and you know that someone is dying, and this priest is going to give last rites. Maybe you add, maybe this person is going to heaven, but maybe they're going to hell. And all of these increasingly stressful thoughts are happening inside the women, and it's just that little bit too much that pushes them over the edge.
- 8.114** And so we have a new hypothesis. The priest is the cause of the death in combination with, of course, all of the other stuff that's going on. It's the straw that broke the camel's back, metaphorically speaking. All right, well, it's out there, but we're desperate and we really want a cause, so we check this out.
- 8.115** We go to the priest, and Semmelweis says to the priest, "Look, I'm working on this hypothesis. Don't look at me like that, but would you please do me a favor. Whenever, for the next little while you are called to give last rites anywhere in the hospital, take the long way around. Please just avoid the first division. Also, stay out of the second division as much as you possibly can." We're trying to control this variable, the priest. The priest is a good guy. He says, "Yes, absolutely."
- 8.116** For the next month, the next two months, the priest goes the long route, and Semmelweis is monitoring very carefully the death rates in the two wards. If it's the priest, then the death rate in the first division should go down because we've taken out the causal factor. And unfortunately, the death rate stays the same. It's still four times as many women dying as in the second division. Another *modus tollens*, it's not the priest. We have to eliminate that hypothesis.
- 8.117** Semmelweis is very close to being out of ideas. He doesn't know what to do. Then a lucky accident occurs. Lucky for Semmelweis, that is, lucky for women and people who care about women. Not so lucky for Semmelweis' colleague, another medical doctor named Kolletschka. Dr. Kolletschka was working with students. This is a hospital, but it's also a teaching hospital working with the University of Vienna medical school.
- 8.118** What typically happens then is, credentialed physicians will, of course, lecture to their students, but they will also take the students on rounds through various parts of the hospital, so the students can learn what the practice of real medicine looks like. Professor Kolletschka was working with a group of students, and as part of their routines, they had anatomy lab. In the anatomy lab, what they would do is they would have corpses, and they would do dissections on the corpses. This is the way that students can learn anatomy, physiology, and all sorts of things.
- 8.119** What had happened was that, in the course of doing dissection, one of the students was holding a scalpel and doing various things, and Professor Kolletschka wanted to

point to something inside the body, but the timing was bad. He pointed just as the student's scalpel was going to do something, and he got a cut on his pointing finger right there. Now, OK, you get cuts, these things happen, but the next day, Professor Kolletschka was sick, and then the day after that, he was very sick.

- 8.120** And Semmelweis comes to see his colleague who is sick, and he sees that Kolletschka has exactly the same symptoms as the women who have been dying at very high rates, particularly in the first ward, the discolorings, the various kinds of discomforts, and so on. And so this is an observation point within a certain categorical scheme that Semmelweis is operating in turn. He sees those symptoms in the women. He sees the exact same symptoms in Kolletschka, even though Kolletschka is a male, even though Kolletschka has not given birth recently, and probably never will give birth.
- 8.121** Different categorical schemes, but nonetheless, strong overlap on the symptoms, and he observes those and is willing then to form a category adjustment. Whatever kills Kolletschka—and it did kill him the next day, he was done—is the same thing that kills the women. We know what killed Kolletschka. We put it in archaic language. We would say it was the cadaveric material or some blood on skin tissues, or whatever on the scalpel from doing the dissection had punctured his skin, gotten into his blood stream, and gone into the rest of his body and done whatever nasty things that thing does to people's bodies.
- 8.122** Now, that's what killed Kolletschka. How might it be that that is also killing the pregnant women? Because that's the anatomy lab. Here we have the women. Then we find out, and this is where it gets dark, that after anatomy lab, it was the practice of Professor Kolletschka and the other teaching professors to go to other parts of the hospital. But no germ theory, no antiseptic theory. They would perfunctorily wash their hands, but not in a very serious fashion. Then they would go and be handling patients in the other parts of their rounds.
- 8.123** In this case, we know they're going into the maternity ward, because it turns out the anatomy lab is pretty close to the first division, rather far afield from the second division. In many cases, they would go from anatomy lab into the first division, and the professors and the students would be handling the women who had recently just given birth. They have cadaveric material on their hands and fingers. The women have abrasions and lesions and so forth. That then gets introduced into their bloodstream. So now we have also a causal transmission measure. We know now what caused puerperal fever.
- 8.124** Now, Semmelweis, this is a great scientific discovery, and as a result of that, we then know what we can start to do. We can introduce more rigorous hygiene regimes and thereby save thousands and thousands of women who otherwise ordinarily would die. Now, that's a great value, but my point as a logic teacher is to say, this would not have happened without someone raised in an environment in which commitment to logic and all of the constituent elements are praised, practiced, and internalized, not only as

part of his intellectual furniture, but as part of his moral character as a professional practicing medicine person.

- 8.125** Excellent sensory skills, use of categorization schemes, precise definitions, whether we're willing to alter the conceptual schemes when new data comes along, a commitment to generating abstract hypotheses and predictions from them, and doing logical and mathematical operations on them to sort the good ones from the bad ones. Then in his case, not simply to do this as a matter of a five-minute exercise on the whiteboard or even a month's work or a year's work, but to stick with it, consistently applying those methods for several years, then and only then we get the ultimate discovery.
- 8.126** Now, this is Semmelweis, and he is a great hero in the history of medicine, from my perspective, not only as a human being, but especially as a logic professor. But there's an additional challenge that I want to add to it here. It's not enough that only individuals isolated here and there be able to do this. A sad part of the story is that Professor Semmelweis, then having engaged in this stupendous amount of research yielding these results, had a very difficult time, almost impossible time, getting anything changed at Vienna Hospital.
- 8.127** He would go to the other physicians, explain his data, if they would listen to him, and suggest that they change their procedures. And many of them would say, "I'm not interested in doing so. I read my textbook, I listened to my professors, I'm going to carry on them. Who are you, Semmelweis, to tell me that we need to do something different germ theory, I've heard about that, but that's just weird speculation. I'm not in that particular scientific school, and I'm not really open to discussion on it."
- 8.128** In some cases, it was an issue of age. Semmelweis is relatively young or an early middle career at this point. I may be an older doctor, and I have a certain reputation, and who is this young whippersnapper to come along and tell me? Certainly, Semmelweis doesn't have power on the hospital boards that make the policy decisions and so on. Those are all occupied by other people. Sometimes there's religious prejudice, and sometimes there's political prejudice, and sometimes there's personal animosities. And so many of those people just don't like Semmelweis for various reasons, and so they're not willing to pay attention to his argument.
- 8.129** So the personal commitment matters, but also part of the lesson of logic is a social. The more we have more people who are able to do these things, appreciate them, and have internalized them, the better we are going to succeed. Wonderful intellectual heights, I think, still to scale, and logic will help us do so, but also wonderful improvements to life that logic, as part of the package, will also enable. We'll stop there.
- 8.130** When we are trying to find cause and effect, there's a lot of studies that have come out that have been debunked now because they took correlation as causation. We know

that correlation doesn't always equal causation. Is that a fault in our methods, or is that merely human error or maybe a moral issue with the scientists?

- 8.131** Sometimes correlation is causation, and sometimes correlation is not causation. That's exactly what we need to figure out. That's precisely why we're developing these methods and using them in increasingly sophisticated ways to do so. I think the quickest thing I can say in response to that is that the problem is going to be human error. That the person is too quick to think that they've used the methods, and so they generalize too hastily.
- 8.132** In many cases, it might just be that they want to get to a conclusion, or they already have had that hypothesis in mind and they're somewhat invested in it, so they do the other hard work rather hastily. But I don't think it's even in a majority of cases in scientists, it's that, because I think scientists, by and large, have good training and good ethics for the most part. I think the more likely thing just is that reality is more complicated than any of us knows at any given time.
- 8.133** In many cases, there are other factors that we've just not become aware of. We are operating, sometimes even in a pretty sophisticated situation. I think there are eight variables, and that's already a lot to isolate. I'm controlling seven of the eight variables. But just given our current state of knowledge, I don't know about 9, 10, and 11. Those will be discovered later for more research, better tools, and so on.
- 8.134** So I think the way to do it is to always have some intellectual objectivity. I don't like the word intellectual modesty. I think scientists can be proud of their accomplishments and their commitments. Being a scientist is nothing to be modest about or humble about. But nonetheless, built into that is an awareness that, however much you know, reality often has proved to be more complicated. So calibrate your commitment to the results, to the in-depth of your study of the antecedent factors, and always be aware of the historical record, that sometimes reality still surprises us in various ways.